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INSTITUTO SUPERIOR TÉCNICO

CURRICULUM VITAE

Alexandre José Malheiro Bernardino

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Contents

Summary	3
Highlights	3
1 Presentation	4
1.1 Personal Data	4
1.2 Languages	4
1.3 Researcher Data	5
1.4 Contact Information	5
1.5 Education and Academic Degrees	5
1.6 Professional Career	6
1.7 Research Career	6
2 Pedagogical Activity	7
2.1 Courses	7
2.2 Innovation and Pedagogical Materials	7
2.3 Supervision	8
3 Scientific Activity	9
3.1 Research Lines	9
Bioinspired and Learning Robots	10
Vision for Decision and Control	10
Object Grasping and Manipulation	11
Human-Robot Interaction	11
3.2 Creation And Reinforcement of Research Infrastructures	12
3.3 Scientific Publications	13
The Ten Most Representative Works	14
International Collaborations	21
3.4 Scientific Project Grants	22
3.5 Recognition and Visibility	23
4 External Engagement	23
4.1 Technology Transfer and Services to the Industry	23
4.2 Services to the Scientific Community	24
4.3 Services to Society	24
5 Academic and Research Service	24
5.1 Management	24
5.2 Tutoring	24
5.3 Academic Juries	24
6 Teaching Activities	25
6.1 Courses	25
6.1.1 PhD Level Courses (3rd Cycle)	25
6.1.2 MSc Level Courses (2 nd Cycle)	25
6.1.3 BSc Level Courses (1 st Cycle)	28
6.1.4 Tutorial Courses (1 st and 2 nd Cycle)	30
6.2 Innovation	31
6.3 Pedagogical Materials	33
7 Supervision Activities	38

7.1	Concluded PhD Theses	38
7.2	Ongoing PhD Theses	43
7.3	Supervised and Co-Supervised MSc Theses	45
7.4	PostDoc Researchers	59
7.5	Research Assistants	59
7.6	Visiting Students	62
8	Publications	62
8.1	Papers in Peer-Reviewed International Scientific Journals	63
8.2	Papers In Peer-Reviewed International Conference Proceedings	71
8.3	Chapters in Edited Books	89
8.4	Other Papers and Abstracts in Peer-Reviewed International Meetings	90
8.5	Papers in Peer-Reviewed National Journals or Conference Proceedings	93
8.6	Theses	96
9	Impact Factors	96
9.1	Citations	96
9.2	Journal Impact Factors	100
10	Project Grants	104
10.1	Project Grants as Global Coordinator	104
10.2	Project Grants As Coordinator of Participating Institution	108
10.3	Project Grants As Team Member	112
11	Recognition And Visibility	123
11.1	Prizes	123
11.2	Invited Seminars	124
11.2.1	International	124
11.2.2	National	125
11.3	Scientific and Professional Societies	127
12	External Engagement	127
12.1	Technology Transfer, Patents and Utility Models	127
12.2	Services to Industry	127
12.3	Organization of International Conferences and Workshops	128
12.4	Member of Program, Scientific or Technical Committees	129
12.5	Editorial Boards	131
12.6	Evaluation of Academic Project Proposals	131
12.7	Reviewing Activity	131
12.7.1	Journals	131
12.7.2	Conferences	132
12.8	Communications to the Public	134
13	Academic and Research Service	135
13.1	Management Positions	135
13.2	Tutoring	136
13.3	Juries of Academic Degrees Examinations	136
13.3.1	PhD Thesis Committees	136
13.3.2	MSc Thesis Committees	140
13.4	Committees of Research and Faculty Positions	144
13.5	Juries of Academic Competitions and Awards	144
		144

Summary

Alexandre Bernardino holds a PhD in Electrical and Computer Engineering from *Instituto Superior Técnico* (IST), *University of Lisbon* (UL) (2004). He is currently an Associate Professor with tenure at the scientific area of *Systems, Decision and Control* (ACSDC) of the *Electrical and Computer Engineering Department* (DEEC). He is a researcher of the *Institute for Systems and Robotics of IST* (ISR/IST) and integrated member of the *Laboratory of Robotics and Systems Engineering* (LARSyS).

Alexandre Bernardino started his pedagogical activity in 1995 and since then has taught **21 different disciplinary curricular units** of the BSc, MSc, and PhD degrees at IST/UL, on the broad areas of Robotics, Control, Signal Processing and Artificial Intelligence. He has supervised and co-supervised **9 postdoctoral fellows** and graduated **22 PhD students** and more than **210 MSc students**. He has hosted **23 visiting students** in their internships at ISR and supervised **84 research fellowships** integrated in research projects.

The research interests of Alexandre Bernardino span the areas of Bioinspired and Learning Robots, Vision for Decision and Control, Object Grasping and Manipulation, and Human-Robot Interaction. He has published numerous peer-review articles in top international publications: **112 in journals**, **207 in conferences**, and **11 in book chapters**. He has **463 co-authors** in scientific publications.

He participated in more than **40 national and international research projects**, being the global coordinator in **8 of them**. He was the Technical Manager of the EU Project Handle, and the local coordinator of the IST team participation on the ERC Project ORIENT led by Prof. John Van Opstal of the Radboud University, Nijmegen.

Beyond the significant pedagogical and scientific achievements, Alexandre Bernardino has contributed to numerous managerial, professional, knowledge transfer and outreach initiatives. He is a current member of the **executive board of ISR**, **coordinator of the LARSyS Thematic Line INTERACTION**, **adjunct coordinator of the scientific area** of Systems, Decision and Control, and **adjunct coordinator of the Computer and Robot Vision Lab** (VisLab) of IST. He was member of the **executive board of the ECE department** from 2010 to 2016, **chair of the IEEE Portuguese RAS Chapter** from 2014 to 2019, and **joint coordinator of the Gender Balance interest group at IST from 2020 to 2026**. He was the **founder of a spin-off company**, and author of the disclosure of **3 inventions**. He has co-supervised several students in **theses with industry** and has established several **contracts and protocols with industrial partners**.

Highlights

As of 26 May 2026

- Ph.D. in ECE IST (since 2004)
- Faculty member at IST (since 1996)
- Associate Professor of ECE (since 2014)
- Researcher at ISR-Lisboa (since 1994)
- Integrated Member of the Associated Laboratory LARSyS (since 2001)
- Chair of the IEEE RAS Portuguese Chapter (2014-2019)
- Member of the Executive Board of the ECE Dept. (2010-2016)
- Adjunct Coordinator of the Computer and Robot Vision Lab (since 2004)

- Coordinator of the Thematic Line “INTERACTION” at LARSyS (since 2013)
- Principal Investigator in **8 projects** of programmes FCT-PT, CMU-PT, EU-FP7 (local) H2020 (local), QREN-PT
- I10-index: **187** (Google)
- H-index: **48/35** (Google/Scopus)
- Citations: **10415/6288** (Google/Scopus)
- **9** supervised post-doctoral researchers.
- **84** supervised research assistants.
- **22** concluded Ph.D. students
- **10** ongoing Ph.D. students
- **210+** concluded MSc students
- **23** supervised visiting students
- **112** papers in intl. peer reviewed journals
- **207** papers in intl. peer-reviewed conferences.
- **41** papers in workshops and symposia
- **33** papers in national publications
- **11** publications in book chapters
- **463** co-authors (Scopus)

1 Presentation

1.1 Personal Data

Full Name: Alexandre José Malheiro Bernardino

Date of Birth: 29 May 1971

Birthplace: Lisbon, Portugal

Nationality: Portuguese

1.2 Languages

Portuguese: Native

English: Fluent

Italian: Proficient

French: Conversational

Spanish: Conversational

1.3 Researcher Data

ORCID ID: 0000-0003-3991-1269

Researcher ID: G-1316-2010

Scopus Author ID: 7003407125

Google Scholar: <https://scholar.google.pt/citations?user=auFmh9MAAAAJ&hl=en>

Research Gate: Alexandre-Bernardino

Ciência ID: 1118-49F0-B28C

1.4 Contact Information

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1.5 Education and Academic Degrees

2004: PhD in Electrical and Computer Engineering

Institution: IST/UTL

Scientific Area: Systems, Decision and Control

Title of the Thesis: *Binocular Head Control with Foveal Vision: Methods and Applications.*

Supervisor: Prof. José Santos-Victor

Classification: Approved

1997: MSc in Electrical and Computer Engineering

Institution: IST/UTL

Scientific Area: Systems, Decision and Control

Title of the Thesis: *Binocular Tracking of Moving Targets using Log-Polar Images*

Supervisor: Prof. José Santos-Victor

Classification: Approved

1994: Licentiate (5 Years) in Electrical and Computer Engineering

Institution: IST/UTL

Scientific Area: Systems, Decision and Control

Título da Tese: *A Biological Approach to Active Vision*

Supervisors: Prof. Carlos Pinto-Ferreira and Prof. José Santos-Victor

Classification: 18/20

1.6 Professional Career

1. Associate Professor with Tenure at the Dept. of Electrical and Computer Engineering

Institution: *Instituto Superior Técnico, Universidade de Lisboa* (IST/UL)

Period: Since December 2014

Activity: Electrical and Computer Engineering; Systems, Decision and Control

Evaluation: RADIST evaluation for 2013/2015 and 2016/2018: Excellent

2. Assistant Professor with Tenure at the Dept. of Electrical and Computer Engineering

Institution: *Instituto Superior Técnico, Universidade de Lisboa* (IST/UL)

Period: April 2009 – December 2014

Activity: Electrical and Computer Engineering; Systems, Decision and Control

Evaluation: RADIST evaluation for 2010/2012: Excellent

3. Assistant Professor at the Dept. of Electrical and Computer Engineering

Institution: *Instituto Superior Técnico, Universidade de Lisboa* (IST/UL)

Period: April 2004 – April 2009

Activity: Electrical and Computer Engineering; Systems, Decision and Control

Evaluation: RADIST evaluation for 2004/2007 and 2008/2009: Excellent

4. Teaching Assistant at the Dept. of Electrical and Computer Engineering

Institution: *Instituto Superior Técnico, Universidade de Lisboa* (IST/UL)

Period: March 1997 – April 2004

Activity: Electrical and Computer Engineering; Systems, Decision and Control

5. Intern Teaching Assistant at the Dept. of Electrical and Computer Engineering

Institution: *Instituto Superior Técnico, Universidade de Lisboa* (IST/UL)

Period: March 1996 – March 1997

Activity: Electrical and Computer Engineering; Systems, Decision and Control

6. Invited Teaching Assistant at the Dept. of Electrical and Computer Engineering

Institution: *Instituto Superior Técnico, Universidade de Lisboa* (IST/UL)

Period: September 1995 – March 1996

Activity: Electrical and Computer Engineering; Systems, Decision and Control.

1.7 Research Career

1. Senior Researcher

Institution: Institute for Systems and Robotics, Lisboa (ISR-Lisboa)

Roles: Research and Development (since 1994), Adjunct Coordinator of the Computer and Robot Vision Lab (since 2004), Member of the Scientific Board (2013-2022), Member of the Executive Board (since 2023).

2. Integrated Member at Associated Laboratory

Institution: Laboratory of Robotics and Engineering Systems (LARSyS)

Roles: Coordinator of Thematic Line “INTERACTION” (since 2013).

3. Visiting Scientist

Institution: *Institut Systèmes Intelligents et de Robotique (ISIR), Université Pierre-et-Marie-Curie (UPMC)*, Paris, France

Period: May 2010 – June 2010

Activity: Research on Robot Manipulation in the HANDLE project (EU-FP7)

4. Visiting Researcher

Institution: Laboratory for Integrated Advanced Robotics (LIRA-Lab) of the Department of Communication, Computer and Systems Science (DIST) of the University of Genova, Italy.

Period: January 2000 – September 2000

Activity: Research on Active Robot Vision.

5. Research Grant Holder (JNICT)

Institution: *Institute for Systems and Robotics, Lisboa*

Period: January 1995 – March 1996

Activity: Research on Computer and Robot Vision

2 Pedagogical Activity

My pedagogical activity has involved the coordination and teaching of courses at IST and the supervision of doctoral and master students, in the broad disciplinary areas of Electrical and Computer Engineering, Aerospace Engineering, and Computer Science.

2.1 Courses

I have lectured more than **21 different non-tutorial curricular units** in the 1st, 2nd and 3rd cycles, most on the core of the Integrated Master Programme in Electrical and Computer Engineering, on the disciplinary areas of Control (Control Theory, Modeling and Simulation, Systems Theory, among others), Robotics (Automation, Autonomous Systems, Robotics, among others), and Signals (Signal Processing, Circuit Theory, Machine Learning, Algorithms, Computer Vision, Neural Networks, among others).

In the last years I have been responsible for the MSc courses of Distributed Real Time Control Systems (~80 students) and Machine Learning (~270 students). In the last two years I have become responsible for the PhD course of Estimation and Classification (~10 students). I was granted three times the IST Excellent Teaching award, issued by the IST Course Unit Quality System (based on students' surveys), on the course of Distributed Real Time Control Systems (2016/2017, 2020/2021 and 2023/2024). The full list of courses is provided in Appendix I.1, including the ranks given by students to my teaching performance, according to the student's surveys organized by IST pedagogical council program QUC (IST Course Unit Quality System).

2.2 Innovation and Pedagogical Materials

My teaching activities are characterized by continued efforts to introduce new contents in the courses I have lectured, both in the theoretical and practical components, with emphasis in

experimental and laboratory resources. Full lists of my innovation actions and pedagogical materials are presented in Appendices I.2 and I.3.

I started my teaching career in 1995 as a Teaching Assistant associated to the laboratorial component of courses of scientific area of Systems Decision and Control, where I always thrived for the introduction of new concepts, projects and equipment. In 1996 I prepared, together with the **Neural Networks** course coordinator, Prof. Luís Borges de Almeida, a set of original lab projects involving new software for the design and training of neural networks. In 1998 I prepared, together with the **Computer Vision** course coordinator, Prof. José Santos-Victor, an original lab project for 3D reconstruction based on cast shadows. In 2002 I prepared, together with the **Modeling, Identification and Control** course coordinator, Prof. João Lemos, a new project for the control of a ball-in-beam educational kit. In the follow up of this project, during 2003-2007 I collaborated in several proposals for the improvement of the control labs with Quanser educational kits and I/O acquisition boards to be used in the teaching of control topics in several courses, corresponding to an investment around 50kEur.

In 2003, I got the coordination of the course on **Modeling, Identification and Control** (2003-2011), where I developed exciting lab projects for the Quanser educational kits ball-in-beam and flexible-bar. Simultaneously, I prepared, together with the Optimal and Adaptive Control course coordinator, Prof. João Lemos, a lab project involving the implementation of the adaptive controller MUSMAR in Simulink Real-Time, for the control of a DC motor. In 2003 I prepared, together with the **Automation and Industrial Processes** course coordinator, Prof. Paulo Oliveira, a lab project involving a new automation cell composed of conveyor belts and robot manipulators.

Since 2007, I have been coordinating the 2nd cycle course on **Distributed Real Time Control Systems** (with interruption in 2010/2011, 2011/2012, and 2014/2015). This is a projects-based course where students develop a practical control system involving knowledge on control theory, system modeling and identification, distributed optimization and control, real-time concurrent systems, and control communication networks. I extensively reformulated the course by introducing decentralized control and distributed optimization topics. Also, I designed a new project that students can implement with accessible materials (basic electronic, communication and computational elements) and take home for allowing autonomous work. In 2013 I prepared a take-home kit that allows students to develop their project anywhere/anytime with real hardware. Since then, students can manage their own time and dedication to the project, which promotes their learning skills. During the years of the COVID19 pandemics, the project developed without disruptions, as it was already fully prepared for in-home execution.

During the year of 2009/2010 I coordinated the **Machine Learning** course, during the sabbatical leave of Prof. Luis Borges de Almeida. As I was involved from the beginning of the course in the preparation of laboratory projects, and taught the laboratory classes of the previous year, I was invited by the usual course coordinator to replace him on his leave.

The idea of take-home projects initiated by me on **Distributed Real Time Control Systems** course, and the disruptions in the teaching of project-based and lab-based courses that emerged from the pandemic crisis, were on the genesis of three Pedagogical Innovation Projects in the years 2021 and 2022, that allowed not only the students of **Distributed Real Time Control Systems** but also of **Control Theory**, and **Autonomous Systems**, to take the hardware and instrumentation of their projects home and improve their autonomy, creativity, and time management skills.

2.3 Supervision

I have supervised and co-supervised **22 concluded PhD students**, **10 ongoing PhD students**, and more than **210 MSc students** (concluded).

Many of my PhD students are joint supervisions with international partners in dual-degree

programmes: 3 in the IST-CMU programme, 3 in the IST-EPFL programme, and 1 in co-tutelle with UPMC, Paris, France. My supervised PhD student Pedro Vicente was awarded **Best National PhD Thesis in Robotics Award of 2022**, Portuguese Society of Robotics, and the **2nd Place of the Best Iberian PhD Thesis in Robotics 2022**, Spanish Society for Research and Development in Robotics and the Portuguese Robotics Society. My supervised PhD student Rui Figueiredo was awarded the **Honorable Mention for the Best National PhD Thesis Award of 2019/2020**, Portuguese Association of Pattern Recognition. My co-supervised PhD student Ricardo Cabral was awarded the **IBM Prize 2014**.

Most of my MSc students are from the courses on the responsibility of the Department of Electrical and Computer Engineering at Técnico (MEEC, MEE, MEAer) but I also have supervised students from courses in other Departments (Physics, Mathematics, Computer Science, Biomedical Engineering, Civil Engineering), other national universities (FCT/UNL) and other international universities (HTW Berlin, Germany), (TU/e, Netherlands). One of my MSc students (Filipe Veiga) was awarded the **Best MSc Thesis Award** by the Portuguese Society of Robotics (SPR) in 2012. Another of my MSc students (João Castelo-Branco) was awarded one of the **Outstanding Master Thesis diplomas** of 2026 by DEEC.

Beyond supervision of students in formal educational programmes, I supervised the work of **9 post-doctoral researchers, 84 research assistants, 23 visiting researchers**.

The full list of my MSc students, PhD students, post-doctoral researchers, research assistants and visiting researchers is provided in Appendix II.

3 Scientific Activity

My research activity has been developed in the broad area of robotics perception and control, with applications to relevant societal problems. I study the fundamental problem of how intelligent systems (natural and artificial) can make **use of its sensors, actuators, and computing resources to learn and adapt to unstructured and dynamic environments**. Then, I try to apply my findings to improve current robotics technologies and have a better understanding of natural intelligent systems. My work has been applied in several areas of societal impact such as **Health & Care, Environmental Monitoring** (Wildfire and Maritime scenarios), **Humanoid Robotics, Security and Surveillance**.

My research spans multiple disciplines at the intersection of **computer vision, robotics, machine learning, cognitive systems, optimization, and control**. Most of my research lines are carried-out in collaboration with colleagues of the **Computer and Robot Vision Laboratory** (Prof. Santos-Victor, Prof. José Gaspar, Prof. Salvador-Marques, Prof. Catarina Barata, Prof. Bruno Damas, Prof. Nuno Pessanha-Santos, Dr. Plinio Moreno). Additionally, I have created links with many researchers worldwide, both on the engineering and the natural sciences. As a member of ISR-Lisboa, I have joined the Associated Laboratory **LarSyS (Laboratory of Robotics and Engineering Systems)**, where I coordinate the Thematic Line “**INTERACTION - Cognitive Robots and Human Experience**”.

3.1 Research Lines

In the following paragraphs I describe in more detail in my core research activities. I will refer to related research projects, PhD students, and relevant collaborations. The actual PhD supervision teams and project’s details are described in the Sections 2.3 and 3.4.

Bioinspired and Learning Robots

One of my main research lines is on “Bioinspired and Learning Robots”. I have completed my PhD in 2004, on the topic of “Binocular Head Control with Foveal Vision”, with inspiration on the mechanics and physiology of the human visual system. In 2006, I joined the **ROBOTCUB Project**, whose purpose was the development of a standard robotic platform for research on developmental and cognitive robotics and involved a multidisciplinary research team from the engineering sciences and the human sciences (neuroscience and psychophysics). In that pioneering project, I have developed researched work on humanoid technologies, visual attention, active vision, object tracking, and learning object affordances. I had the opportunity to participate in the joint effort of the development of **the iCub Robot**, one of the most complex humanoid robots worldwide and a unique platform in Portugal. A fascinating topic on those years, prior to deep learning technology, was object recognition. In my first PhD student co-supervision (Dr. Plinio Moreno, with Prof. José Santos-Victor), we researched on biologically inspired computer vision for object recognition. On the follow-up of the ROBOTCUB project I participated in projects **MIRROR**, **CONTACT** and **ROBOSOM**, also focused humanoid technologies through synergies from Robotics and Neurosciences. Also, I have been awarded two national projects, **BIOLOOK** (2007-2010) and **BIOMORPH** (2014-2015). In those projects I have researched eye-head coordination algorithms and visual prediction mechanisms for motion estimation (PhD of Dr. Nuno Moutinho, with Prof. José Gaspar) and studied the development of an artificial fovea through the optimization of sensorimotor learning principles (PhD of Dr. Jonas Ruesch, with Prof. José Santos-Victor). Integrated in the IST-EPFL doctoral program on **Robotics, Brain and Cognition** (RBCog), I engaged in collaborations with Prof. Auke Ijspeert from the Biorobotics Laboratory and EPFL, Lausanne, Switzerland, in bioinspired locomotion behaviors in quadrupeds, and modular robots, that led to the graduation of 2 PhD students (Dr. Alexandre Tuleu, Dr. Mehmet Mutlu). Also in the RBCog Doctoral Program, I collaborated with Prof. Helder Araújo (Univ. Coimbra) on space-variant vision mechanisms for resource constrained systems (Ph.D of Dr. Rui Figueiredo), and collaborated with Prof. Lorenzo Jamone (QMUL) and Prof. José Santos-Victor on humanoid robot self-supervised learning of body schema (PhD of Dr. Pedro Vicente) which was awarded the prize of best 2021 national PhD thesis in Robotics by the Portuguese Society of Robotics. More recently, I have been collaborating in the Horizon 2020 project **ORIENT**, an ERC initiative of Prof. John van Opstal from the Donders Institute for Brain, Cognition and Behaviour, Radboud University, Nijmegen, Netherlands. In this project the ISR/IST team developed a biomimetic robot eye to study the principles underlying the optimization of the fast ocular movements called saccades.

Vision for Decision and Control

Another of my main research lines is on computer vision technologies for decision and control, such as motion estimation, activity recognition, object detection, classification, and tracking. While still working on my PhD, I joined the **NARVAL** Project, where I assumed an active role in the development of visual navigation behaviors for underwater autonomous systems. These are fundamental skills for the autonomy of advanced robotics and surveillance systems. In the sequence, I worked on probabilistic methods for motion analysis for autonomous vehicles (PhD of Dr. Hugo Silva) in collaboration with Prof. Eduardo Silva (ISEP-IPP). I established collaborations with the Video Surveillance Industry through the innovation projects **HDA** (ANI) and **DICORE2S** (EU-FP7) that led to novel methods in the detection of pedestrians (Dr. Matteo Taiana, with Prof. Jacinto Nascimento), multiple-hypothesis tracking in camera networks (Dr. Dario Figueira, with Prof. Jacinto Nascimento), and gait-based person re-identification (Dr. Athira Nambiar, with Profs. Jacinto Nascimento and José Santos-Victor). In the context of the CMU-Portugal Program, I have co-supervised Ph.D. students in the development of novel image classification algorithms using matrix completion methods (PhD thesis of Dr. Ricardo

Cabral, IBM Prize 2014) in collaboration with Prof. João Paulo Costeira (ISR/IST) and Prof. Fernando de La Torre (CMU), and on activity recognition from videos with Deep Learning methods (PhD Thesis of João Antunes Martins) in collaboration with Prof. Dan Siewiorek (CMU) and Prof. Asim Smailagic (CMU). I have made collaborations with the Portuguese Navy and Air Force through national research projects on maritime surveillance (**SEAGULL**, 2013-2015), airborne image analysis of maritime and forest environments (**VOAMAS**, 2019-2023), and wildfire monitoring (**FIREFRONT**, 2019-2023) where I supervised PhD students in 3D model-based tracking of Drones (thesis of Dr. Pessanha-Santos) in collaboration with Prof. Vitor Lobo (Navy Academy), and 2D tracking of vessels in maritime environments (Dr. Gonçalo Cruz) in collaboration with Prof. António Pedro Aguiar (FEUP). More recently I have established collaborations with Prof. Alessandro Saccon (TUE, Eindhoven), through the co-supervision of MSc students, to further research 3D model-based tracking algorithms for robotic manipulation tasks.

Object Grasping and Manipulation

Object grasping and manipulation are fundamental robotic skills for industry and are quite complex to deploy in unstructured environment. Object grasping and manipulation rely on physical contacts with objects that involve impacts with multiple surfaces in simultaneity, being extremely complex to model and simulate, particularly with human-like robot hands. Object grasping and manipulation relate to computer vision as the initial phases of grasping can be guided by vision, but as soon as fingers contact the objects, tactile and force sensing become essential and, afterwards, state estimation and control can fuse vision, force, and tactile sensing. After my participations in the **ROBOTCUP** and **MIRROR** Projects, where I researched visual skills involved in robotic manipulation, I participated actively in the resubmission of a project proposal for research in dexterous robotic manipulation (**HANDLE** project), in which I assumed the role of Technical Manager. In collaboration with the Principal Investigator, Prof. Véronique Perdereaux from the ISIR lab of UPMC, Paris, I researched human dexterity and developed methods for the high-level control of in-hand manipulation (PhD of Dr. Urbain Prieur). In comparison to visual sensors, which are becoming close to human resolution, tactile sensors for robot hands are still many orders of magnitude behind.

In the context of the individual Marie Curie grant of Dr. Lorenzo Jamone, the **LIMOMAN** project developed magnetic tactile sensors for the instrumentation of dexterous robotic hands and several methods were developed for tactile based control of robotic grasps. To further develop the tactile sensing technology, we established collaborations with Prof. Susana Freitas (IST, INESC-MN) and co-supervised two PhD students (Dr. Pedro Ribeiro and Dr. Miguel Neto) that developed highly sensitive magneto-resistive sensors and applications to robotics.

Human-Robot Interaction

My research in Human-Robot Interaction attempts to deploy in robots many of the technologies developed in my other research lines, and evaluate how such systems can be used, valued, and enjoyed by humans. In traditional robotics applications, robots perform their tasks autonomously and independently from humans, most often in separate physical spaces due to safety constraints. As robots become more versatile, robust, and safe, they can be used in applications involving close interaction with humans, e.g., domestic use, collaborative manufacturing, and services. However, to be accepted by humans, beyond competence and reliability in executing the required tasks, such robots must provide a range of other features that are much related to human psychology: usability, pleasant appearances and motions, natural communication and behaviors, sense of safety, shared agency, and control, among others. In my research, I have addressed many of these dimensions. In the Poeticon++ project I have studied how object's affordances and goal directed actions can be used by robots to understand human intentions and

provide more natural non-verbal communication channels (PhD of Giovanni Saponaro). In the **AHA** and **IntelligentCare** projects I have studied how interactive robots and virtual/augmented reality systems can be used in conjunction with therapists to aid in physical and cognitive rehabilitation processes (PhD of Min Hun Lee, PhD of Ana Rita C3cias, ongoing). In the PhD work of Jo3o Avelino, in co-supervision of Prof. Leonel Garcia-Marques (Fac. Psychology, UL) and Prof. Rodrigo Ventura (ISR/IST) we have implemented human-like salutation and engagement behaviors and illustrated their validity in human-robot interactions “in the wild” (unconstrained environments). In **ShiftHRI** project, in collaboration with Prof. Tiago Guerreiro (FCUL) and Prof. Jodi Forlizzi (CMU) we are studying how telepresence and social robots can be used to provide valuable services to older people (PhD of Hugo Sim3o, ongoing). In the **HAVATAR** project we are studying how telepresence and service robots can be used in a hospital context, providing value to customers and patients’ relatives.

I was the principal investigator of projects funded by the Portuguese Science Foundation, by the Portuguese Innovation Agency, and by special programs of collaboration of Portuguese and International Universities (CMU-PT). Also, I was technical manager and local leader of projects of programmes EU-FP7 and HORIZON 2020.

3.2 Creation And Reinforcement of Research Infrastructures

I am the deputy coordinator of the Computer and Robot Vision Lab (Vislab) of the Institute for Systems and Robotics (ISR), partly responsible for the acquisition, maintenance, and organization of general-purpose lab resources (computers, cameras, accessories, consumables, etc). In the context of national and international research projects I have been directly involved in the construction and acquisition of specialized platforms and equipment that has reinforced the Vislab infrastructure for top level research in Computer and Robot Vision:

1. The **iCub robot** is one of the most advanced humanoid platforms in the world, and unique in Portugal, as an outcome of the RobotCub project. This platform has been used in top-tier research projects.
2. The **NAO** and **Darwin-OP** humanoid robots are small humanoid platforms that have simple operation and programming interfaces to allow rapid prototyping of demonstrations and user studies.
3. The **CamNet** is a distributed camera network installed in the 3 floors of ISR (approved by the National Data Protection Commission) that has been used in video surveillance research projects. A dataset has been acquired, that counts with more than 300 users worldwide. This was an outcome of the HDA project in collaboration with the national industry.
4. The **Tobii EyeTracker** is an eye-tracker device to measure the instantaneous gaze direction of a human subject. It was acquired in the context of the Biolook project and has been used for several psychophysics studies on human visual cognition.
5. The **Vizzy** robot is a unique humanoid social robot jointly developed through several integrative efforts. I have participated in the development of several hardware and software components.
6. The **Optitrack** high speed motion capture system allows the precise acquisition of trajectories and has been used to benchmark several robot and computer vision methods.
7. The **BioRob** robot is a robotic arm composed of elastic actuators that is capable of highly dynamic motions and was used to the study dynamic motion primitives and real-time trajectory control.

8. The **Medusa** robot head is a humanoid robot head with 4 degrees-of-freedom where I have worked for the great part of my MSc and PhD and have contributed to the development of the hardware, control electronics, and software.
9. The **EuroHead** is a high precision robotic stereo head for metrology application, for which I have contributed for the specification and acquisition.
10. **PEPE** is an augmented reality platform for physical and cognitive stimulation composed of a mobile base, an ultra-wide throw projector, a computer and a motion sensor that was developed in the AHA project. It has spawned other projects and has raised interest for commercial exploitation. A national patent has been granted in 2024, and international patents submitted.
11. **Temi** and **SanBot** are commercial teleoperation robots with a high degree of autonomy that have been acquired in the context of the AHA and HAVATAR projects for research in robotics tele-autonomy.
12. The RBCog lab is a research infrastructure included in the National Roadmap of Research Infrastructures where I participate. I have contributed for its development through specifications. It includes many of the resources listed above and others like the **Baxter Robot**, a **Kinova Gen 3** robot arm, and manipulator end-effectors for several applications.

The available infrastructure has been key in advanced training at the level of MSc and PhD projects, as well as promoting research with external partners that have led to long term collaborations and joint project proposals.

3.3 Scientific Publications

I have a very diversified publication profile and publish both in journals of my core research fields of **robotics, computer vision, control, and artificial intelligence** (e.g. *Science Robotics*, *IEEE Transactions on Cognitive and Developmental Systems*, *IEEE Transactions on Image Processing*, *Neurocomputing*, *Pattern Recognition*, *Neural Networks*, *The International Journal of Robotics Research*, *Autonomous Robots*, *Pattern Analysis and Machine Intelligence*, *Journal of Field Robotics*, *IEEE RA-L*, *IEEE Transactions on Systems, Man and Cybernetics*, *Computer Vision and Image Understanding*), and in journals of applications of robotics and AI to: **health and rehabilitation** (*International Journal of Clinical and Health Psychology*, *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, *Journal of Neuroengineering and Rehabilitation*); **sensing and instrumentation** (*IEEE Internet of Things Journal*, *Computers and Electronics in Agriculture*, *International Journal of Wildland Fire*, *Expert Systems with Applications*, *Experiments in Fluids*, *IEEE Transactions on Circuits and Systems for Video Technology*, *Remote Sensing*, *IEEE Transactions on Geoscience and Remote Sensing*, *IEEE Transactions on Magnetics*, *Mechanism and Machine Theory*); and **human-machine interaction** (*International Journal of Social Robotics*, *User Modeling and User-Adapted Interaction*, *Proceedings of the ACM in Human-Computer Interaction*, *ACM Computing Surveys*).

Overall, I have published **404 peer reviewed articles (112 international journal papers, 11 book chapters, 207 papers in international conferences, 41 papers in international workshops and symposia, 33 papers in national publications)**. According to Scopus, I have **463 co-authors** in scientific papers, **6288 citations** and an **H-index of 35**, as of 26 May 2026.

The full list of my publications is provided in Appendix III. The impact of my publications is analysed in Appendix IV through the publication citations and journal impact factors provided by several indexing services: Clarivate's Web of Science (WoS), SCImago, Elsevier's Scopus, and Google Scholar (GS).

The Ten Most Representative Works

In this section I list and briefly describe the 10 works I consider most representative about my research and that are having a significant impact in the research community as measured by the number of citations in WoS/GS as of May 27th 2026. I have selected 10 works that were published in 10 different journals of the first quartile (Q1) according to WoS Journal Citation Report (JCR) of 2022. Each of the articles is classified into one or more of my main research domains: Bioinspired and Learning Robots, Vision for Decision and Control, Object Grasping and Manipulation, Human-Robot Interaction. For each article, a couple of other related papers is included.

The iCub Robot

Authors: G. Metta, L. Natale, F. Nori, G. Sandini, D. Vernon, L. Fadiga, C. von Hofsten, J. Santos-Victor, A. Bernardino, L. Montesano

Year: 2010

Title: “The iCub Humanoid Robot: An Open-Systems Platform for Research”

Journal: *Neural Networks*, 23(8-9), 1125–1134.

Domain: Bioinspired and Learning Robots

IF: 7.8

Citations: 469/773

DOI: <https://doi.org/10.1016/j.neunet.2010.08.010>

Description: This paper presents the humanoid robot iCub, in its most relevant aspects. The iCub robot was the main output of the EU project RobotCub. The iCub Robot is one of the most sophisticated humanoid robots in the world. It has a child-like morphology and was specially designed for research in developmental robotics. Currently there are about 50 copies of this robot in the world, and one is in Portugal (Vislab/IST/IST). The robot has been used in several PhD and MSc theses at Vislab, and on “spin-off” projects like the Poeticon++ project in which Vislab participated. The iCub robot has been fundamental in the creation of a community associated to the research of humanoid robots, cognitive systems, and robot learning. The open-source nature of the platform (hardware and software) promotes the exchange of experiences among several research groups. A large part of the paper presents results obtained by the Vislab team, that had a significant impact in the project, like the construction of the robot head, the design of the cognitive architecture, and the development of robot vision and learning algorithms.

Other Related Papers:

- J. Ruesch, M. Lopes, A. Bernardino, J. Hörnstein, J. Santos-Victor, R. Pfeifer (2008), “Multimodal Saliency-Based Bottom-Up Attention A Framework for the Humanoid Robot iCub”, *IEEE International Conference on Robotics and Automation (ICRA 2008)*, pp 962-967. <https://doi.org/10.1109/ROBOT.2008.4543329>
- R. Beira, M. Lopes, M. Praça, J. Santos-Victor, A. Bernardino, G. Metta, F. Becchi, R. Saltarén (2006), “Design of the Robot-Cub (iCub) Head”, *IEEE International Conference on Robotics and Automation (ICRA)*, pp 94-100. <https://doi.org/10.1109/ROBOT.2006.1641167>

Object’s Affordances

Authors: L. Montesano, M. Lopes, A. Bernardino, and J. Santos-Victor

Year: 2008

Title: “Learning Object Affordances: From Sensory Motor Maps to Imitation”

Journal: *IEEE Transactions on Robotics*, 24(1), 15-26.

Domain: Bioinspired and Learning Robots, Object Grasping and Manipulation

IF: 7.8

Citations: 281/548

DOI: <https://doi.org/10.1109/TRO.2007.914848>

Description: According to several theories on cognition, the way humans perceive objects is related more to their utility (what they afford to - affordances) rather than their identity. This contrasts with the classical view of object’s representations (3D reconstruction, recognition) in which object’s models are created without taking the task into account. Affordance is a very powerful concept for general purpose and adaptive robots because it allows to consider object manipulation as a pre-categorical skill: robots do not need to recognize the semantic class of an object to manipulate it, and thus, can experiment it for the execution of tasks even if not seen previously. This paper shows, through a set of experiments with a humanoid robot, that pre-categorical cues like shape features are sufficient to manipulate the objects in simple tasks. The robot executes several actions on the objects and learns what are the effects of those actions in the environment conditioned to the shape features of the objects. A probabilistic graphical model that encodes the joint probability distribution of object features, actions and effects can be learned from data using structure learning techniques. This model can then be used to support several cognitive skills: action planning, action prediction, object selection, action recognition and imitation.

Other Related Papers:

- L. Jamone, E. Ugur, A. Cangelosi, L. Fadiga, A. Bernardino, J. Piater and J. Santos-Victor (2018), “Affordances in Psychology, Neuroscience and Robotics: A Survey”, *IEEE Transactions on Cognitive and Developmental Systems*, 10(1), pp 4-25. <https://doi.org/10.1109/TCDS.2016.2594134>
- L. Montesano, M. Lopes, A. Bernardino, J. Santos-Victor (2007), “Modeling Affordances using Bayesian networks”, *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2007)*, pp 4102-4107. <https://doi.org/10.1109/IROS.2007.4399511>

Learning Language Associations of Robotic Actions

Authors: G. Salvi, L. Montesano, A. Bernardino, J. Santos-Victor

Year: 2012

Title: “Language Bootstrapping: Learning Word Meanings From Perception–Action Association”

Journal: *IEEE Transactions on Systems, Man, and Cybernetics, Part B: Cybernetics* 42(3), 660-671.

Domain: Bioinspired and Learning Robots, Human-Robot Interaction

IF: 6.2

Citations: 22/63

DOI: <https://doi.org/10.1109/TSMCB.2011.2172420>

Description: Before the emergence of Large Language Models (e.g. ChatGPT) and Visual-Language Learning (e.g. image captioning), this paper has shown empirically that a robot

agent can learn to associate words to the entities of a robotic task: objects, actions and effects. Learning is obtained by exploiting the co-occurrences between the spoken words of a narration of the task, and the simultaneously observed characteristics of the task measured by other perceptual modalities. In combination with object's affordance models (see previous work), the robot can learn the concepts of verbs associated to the robot actions, names for objects, and adjectives to characteristics of the objects and the actions. The model is learned in the form of a Bayesian Network to represent the inherent probabilistic nature of the problem. Once learned, the model can be used to instruct the robot to perform actions related to the objects present in the environment. Through probabilistic reasoning and contextual cues the robot can resolve ambiguous cases and incomplete instructions. This model had a key role in the cognitive architecture of the Poeticon++ project, in which the iCub robot interacted with humans in natural language to execute manipulation tasks.

Other Related Papers:

- V. Kunic, G. Salvi, A. Bernardino, L. Montesano, J. Santos-Victor (2009), "Affordance Based Word-To-Meaning Association", *IEEE International Conference on Robotics and Automation (ICRA 2009)*, pp 806–811. <https://dl.acm.org/doi/10.5555/1703435.1703566>
- Antunes, L. Jamone, G. Saponaro, A. Bernardino, R. Ventura (2016), "From Human Instructions to Robot Actions: Formulation of Goals, Affordances and Probabilistic Planning", *IEEE International Conference on Robotics and Automation (ICRA 2016)*, pp 5449-5454. <http://doi.org/10.1109/ICRA.2016.7487757>

Foveal Vision for Robots

Authors: V. Javier Traver, A. Bernardino

Year: 2010

Title: "A review of log-polar imaging for visual perception in robotics: A survey"

Journal: *Robotics and Autonomous Systems*, 58(4), 378-398.

Domain: Bioinspired and Learning Robots, Vision for Decision and Control

IF: 4.6

Citations: 134/210

DOI: <https://doi.org/10.1016/j.robot.2009.10.002>

Description: Foveal Vision mimics the distribution of photoreceptors in the human retina and is characterized by a high density in the center and a decaying resolution towards the periphery. Coupled with eye movements, this representation allows to have simultaneously a high-resolution area and a wide field of view with a limited number of pixels, which are important characteristics of visual systems for robots with limited computational resources.

The log-polar mapping is a popular mathematical representation of this type of images, having interesting computational properties such as equivariance to centered rotations and scaling. This paper reviews the main representations, computational methods, and properties of log-polar images and describes a large set of works that applied log-polar image processing in robot vision, highlighting their advantages in computation time and accuracy with respect to the conventional images. The paper presents in a unified way a set of original works from myself at Vislab where I show the advantages of foveal vision in robot eye-head control, particularly in tracking, segmentation, and depth perception of central objects.

Other Related Papers:

- A. Bernardino, J. Santos-Victor (1999), “Binocular Tracking: Integrating Perception and Control”, *IEEE Transactions on Robotics and Automation*, (15)6, 1080-1094. <https://doi.org/10.1109/70.817671>
- A. Bernardino, J. Santos-Victor (1998) “Visual Behaviours for Binocular Tracking”, *Robotics and Autonomous Systems*, (25)3-4, 137-146. [https://doi.org/10.1016/S0921-8890\(98\)00043-8](https://doi.org/10.1016/S0921-8890(98)00043-8)

Convex Methods for Learning Object Detectors in Images

Authors: R. Cabral, F. de La Torre, J. Costeira, A. Bernardino

Year: 2015

Title: “Matrix Completion for Weakly-supervised Multi-label Image Classification”

Journal: *IEEE Trans. on Pattern Analysis and Machine Intelligence*, 37(1), pp 121-135

Domain: Vision for Decision and Control

IF: 23.6

Citations: 149/429

DOI: <https://doi.org/10.1109/TPAMI.2014.2343234>

Description: The most common methodology to learn object detectors from image data is through supervised learning. This methodology requires a dataset with annotated bounding boxes around the objects to detect, which is typically done by laborious manual work on large image sets. To reduce the amount of manual work, many authors have proposed methods in which only the presence or absence of objects in an image is provided (weak labels). Then, the method must learn from the co-occurrence of object features in many images with coherent labels. However, existing methods used combinatorial approaches that make the learning process very slow. Instead, we proposed a convex method, that exploits the feature-based histogram representation of images as the linear combination of the histograms of the composing objects. With this formulation, the problem can be solved efficiently through matrix-completion with rank regularization, that is robust to noise in the images and in the labeling process, as well as to moderate amounts of missing labels and data. Experimental validation on several data sets shows that our method outperforms state-of-the-art classification algorithms, while effectively capturing each class appearance.

Other Related Papers:

- R. Cabral, F. De la Torre, J. Costeira, A. Bernardino (2013), “Unifying Nuclear Norm and Bilinear Factorization Approaches for Low-rank Matrix Decomposition”, *IEEE International Conference on Computer Vision (ICCV 2013)*, pp 2488-2495. <https://doi.org/10.1109/ICCV.2013.309>
- R. Cabral, F. De la Torre, J. Costeira, A. Bernardino (2011), “Matrix Completion for Multi-label Image Classification”, *International Conference on Neural Information Processing Systems (NIPS 2011)*, pp 190-198. <https://dl.acm.org/doi/10.5555/2986459.2986481>

Computer Vision in Transportation Systems and Autonomous Vehicles

Authors: J. Melo, A. Naftel, A. Bernardino, J. Santos-Victor

Year: 2006

Title: “Detection and Classification of Highway Lanes Using Vehicle Motion Trajectories”

Journal: *IEEE Transactions on Intelligent Transportation Systems*, 7(2), 188-200.

Domain: Vision for Decision and Control

IF: 8.5

Citations: 112/184

DOI: <https://doi.org/10.1109/TITS.2006.874706>

Description: The automation of vehicles and road infrastructure has witnessed a large penetration of computer vision and machine learning methods in the last decade. We have applied some methods from robot vision in applications to transportation systems, in particular the detection and tracking of vehicles and persons. In the first paper of this line of work, we have used low-level tracking methods to compute vehicle trajectories, that, accompanied by clustering and indexing/retrieval methods, were shown to: automatically identify the lanes of traffic, compute statistics of traffic in highways, and detect anomalous trajectories. Experimental results are presented to show the real-time application of this approach to multiple views obtained by an uncalibrated pan-tilt-zoom traffic camera monitoring the junction of two busy intersecting highways. In follow-up papers, we have worked on ways to improve object detection in autonomous driving applications.

Other Related Works:

- D. Ribeiro, J. Nascimento, A. Bernardino, G. Carneiro (2017), “Improving the performance of pedestrian detectors using convolutional learning”, *Pattern Recognition*, 61, pp 641–649. <https://doi.org/10.1016/j.patcog.2016.05.027>
- M. Taiana, J. Nascimento, A. Bernardino (2015), “On the purity of training and testing data for learning: The case of pedestrian detection”, *Neurocomputing*, 150A, pp 214–226. <https://doi.org/10.1016/j.neucom.2014.09.055>

Airborne Image Analysis for Maritime Surveillance

Authors: R. Ribeiro, G. Cruz, J. Matos, A. Bernardino

Year: 2019

Title: “A Data Set for Airborne Maritime Surveillance Environments”

Journal: *IEEE Trans. on Circuits and Syst. for Video Technology*, 29(9), pp 2720–2732

Domain: Vision for Decision and Control

IF: 8.4

Citations: 82/110

DOI: <https://doi.org/10.1109/TCSVT.2017.2775524>

Description: Maritime surveillance is a key activity for many countries. It is important to assure the safe and secure use of the oceans for transportation and trade. It allows the control of fisheries to guarantee the protection of resources and ecosystems. Maritime surveillance also ensures that environmental regulations are applied, preventing oil spill and bilge dumping that have a severe impact on fauna, flora and also coast human populations. In the last decade, Unmanned Aerial Vehicles (UAVs) have seen a huge increase not only in its deployment but also in its capabilities. Right now, UAVs offer promising technologies to aid remote sensing and oceanic surveillance. While traditional aircraft are equipped with heavy radars, UAVs normally have only light weight passive electro-optical sensors. This paper presents a data set with surveillance imagery over the sea captured by a small size UAV. Additionally, using standard evaluation frameworks, we establish a baseline of results for automated image and video analysis methods, using algorithms developed by the authors, which are better adapted to the maritime environment.

Other Related Works:

- G. Cruz, A. Bernardino (2019), “Learning Temporal Features for Detection on Maritime Airborne Video Sequences using Convolutional LSTM”, *IEEE Transactions on Geoscience and Remote Sensing*, 57(9), pp 6565-6576. <https://doi.org/10.1109/TGRS.2019.2907277>
- G. Cruz, A. Bernardino (2016), “Aerial detection in maritime scenarios using convolutional neural networks”, *International Conference on Advanced Concepts for Intelligent Vision Systems (ACIVS 2016)*, Lecture Notes in Computer Science, Springer, vol 10016, pp 373-384. https://doi.org/10.1007/978-3-319-48680-2_33

Human Recognition and Tracking

Authors: A. Nambiar, A. Bernardino, J. Nascimento

Year: 2019

Title: “Gait-Based Person Re-identification: A Survey”,

Journal: *ACM Computing Surveys (CSUR)* 52(2), 33, pp 1-34.

Domain: Vision for Decision and Control, Human-Robot Interaction

IF: 16.6

Citations: 90/146

DOI: <https://doi.org/10.1145/3243043>

Description: Person Re-Identification is the ability to recognize the same person at two different instances of time. It has important applications in surveillance systems and human-robot interaction. In our work we have developed several methodologies for person re-identification with computer vision using appearance, shape, facial, and gait features. In this paper we perform an extensive survey of gait-based person reidentification. Gait features are attractive because they are persistent across extended periods of time and are hard to fake, allowing long-term re-identification. However, the acquisition of good-quality measures of a person’s motion patterns in unconstrained environments has proved very challenging in practice. Existing technology (video cameras) suffers from changes in viewpoint, daylight, clothing, accessories, and other variations in the person’s appearance. We identify several relevant dimensions of the problem and provide a taxonomic analysis of the current state of the art. We present some of our own work in this field, discussing the levels of performance achievable with the current technology. Finally, we give a perspective of the most challenging and promising directions of research for the future.

Other Related Works:

- D. Figueira, L. Bazzani, H. Q. Minh, M. Cristani, A. Bernardino, V. Murino (2013), “Semi-supervised Multi-feature Learning for Person Re-identification”, *IEEE International Conference on Advanced Video and Signal Based Surveillance*, pp 111-116. <https://doi.org/10.1109/AVSS.2013.6636625>
- Nambiar, A. Bernardino, J. Nascimento (2015), “Shape Context for soft biometrics in person re-identification and database retrieval”, *Pattern Recognition Letters*, 68(2), pp 297–305. <https://doi.org/10.1016/j.patrec.2015.07.001>

3D Model-Based Object Tracking

Authors: N. Pessanha-Santos, V. Lobo, A. Bernardino

Year: 2020

Title: “Two-stage 3D model-based UAV pose estimation: A comparison of methods for optimization”

Journal: *Journal of Field Robotics*, 37(4), pp 580-605.

Domain: Vision for Decision and Control

IF: 8.3

Citations: 12/22

DOI: <https://doi.org/10.1002/rob.21933>

Description: 3D Model-Based Object Tracking is a methodology that exploits available 3D models of objects to detect and track their 6D pose (position and orientation) with respect to a camera sensor. It has important applications in robot navigation and control, and object grasping and manipulation. There are two main approaches to the problem: inverse and direct methods. Inverse methods try to solve the problem of transforming the 2D measurements obtained in the image to the 6D pose of the object. Direct methods, instead, compute the likelihood of the 6D object pose given the 2D data. Direct methods tend to be more robust to moderate amounts of occlusion and variability in the scene (illumination, clutter, background), but face two main problems: (i) the true likelihood function is unknown, so approximations must be carefully designed for the problem at hand, and (ii) even approximate likelihood functions are highly irregular, so sampling methods are required for its optimization which brings computational complexity issues. In this paper, we propose and compare several likelihood functions and optimization methods to the problem of 3D model-based tracking. The framework is applied to the outdoor pose estimation of a fixed-wing UAV for autonomous landing in a Fast Patrol Boat (FPB).

Other Related Works:

- N. Pessanha-Santos, V. Lobo, A. Bernardino (2020), “Directional statistics for 3D Model-Based UAV Tracking”, *IEEE Access*, 8, pp 33884-33897. <https://www.doi.org/10.1109/ACCESS.2020.2973970>
- N. Pessanha-Santos, V. Lobo and A. Bernardino (2019), “Unmanned Aerial Vehicle Tracking Using a Particle Filter Based Approach”, *IEEE Underwater Technology (UT 2019)*, pp 1-10. <https://doi.org/10.1109/UT.2019.8734465>

Robots to Study Biological Systems

Authors: A. John, C. Aleluia, A. J. Van Opstal, A. Bernardino

Year: 2021

Title: “Modelling 3D saccade generation by feedforward optimal control”

Journal: *PLoS Computational Biology* 17(5), e1008975, pp 1-35.

Domain: Bioinspired and Learning Robots

IF: 4.3

Citations: 11/20

DOI: <https://doi.org/10.1371/journal.pcbi.1008975>

Description: Using robotic models to test theories on the behavior of humans or animals can help understand many aspects of intelligence and control in natural systems. This work resulted from a collaboration between a team of roboticist and a team of neuroscientists, both studying the human oculomotor system from different perspective. The paper focus on how model-based

optimal control principles can explain stereotyped human oculomotor behaviors (Listing’s Law, Saccade dynamics), through simulations in a realistic model of the human eye with a cable-driven actuation system. We verified the contribution of different terms in the cost optimization functional to realistic 3D saccade behavior, and identified four essential terms: total energy expenditure by the motors, movement duration, gaze accuracy, and the total static force exerted by the muscles during fixation. Our findings suggest that Listing’s law, as well as the saccade dynamics and their trajectories, may all emerge from the same common mechanism that aims to optimize speed-accuracy trade-off for saccades, while minimizing the total muscle force during eccentric fixation. Other papers in this line of research have focused on learning aspects and studied the visual systems of other animal species.

Other Related Works:

- R. J. Alitappeh, A. John, B. Dias, A. J. van Opstal and A. Bernardino (2024) "Emergence of human oculomotor behavior in a cable-driven biomimetic robotic eye using optimal control," in *IEEE Transactions on Cognitive and Developmental Systems*, 6(4):1546-1560. <https://doi.org/10.1109/TCDS.2024.3376072>
- X. Liu, M. D. Loring, L. Zunino, K. E. Fouke, F. A. Longchamp, A. Bernardino, A. J. Ijspeert, E. A. Naumann (2025) "Artificial Embodied Circuits Uncover Neural Architectures of Vertebrate Visuomotor Behaviors", *Science Robotics*, 10, <https://doi.org/10.1126/scirobotics.adv4408>

International Collaborations

Among my 460+ co-authors in scientific publications, most of them are international. In this section I list 10 of my closer and more established international collaborations.

1. **Lorenzo Jamone, 29 co-authored works, 3 joint PhD students**, Associate Professor in Robotics & AI at the Department of Computer Science of University College London (UCL), **UK**.
2. **Asim Smailagic, 15 co-authored works, 3 joint PhD students**, Research Professor at CMU, Director of Laboratory for Interactive Computer Systems and Wearable Computers, **US**.
3. **Daniel Siewiorek, 14 co-authored works, 2 joint PhD students**, Buhl University Professor at CMU, Human-Computer Interaction Institute, **US**.
4. **Auke Jan Ijspeert, 11 co-authored works, 2 joint PhD students**, full professor at EPFL (the Swiss Federal Institute of Technology at Lausanne), head of the Biorobotics Laboratory (BioRob), **Switzerland**.
5. **Cecilia Laschi, 10 co-authored works**, full professor at the National University of Singapore, College of Design and Engineering, **Singapore**.
6. **A. J. Van Opstal, 9 co-authored works, 1 joint PhD student**, professor at Donders Centre for Neuroscience and Donders Institute for Brain, Cognition and Behaviour, **Netherlands**.
7. **Paolo Dario, 9 co-authored works**, professor at Scuola Superiore Sant’Anna (SSSA), Pisa, and Founding Coordinator of the Center of Micro-BioRobotics@SSSA of the Italian Institute of Technology (IIT) **Italy**.
8. **Fernando de La Torre, 8 co-authored works, 2 joint PhD students**, Research Associate Professor at CMU, Robotics Institute, **US**.
9. **Giampiero Salvi, 6 co-authored works, 1 joint PhD student**, professor at the Norwegian University of Science and Technology, **Norway**, and Associate Professor with KTH Royal Institute of Technology, **Sweden**.

10. **Jodi Forlizzi, 5 co-authored works, 1 joint PhD student**, Herbert A. Simon Professor in Computer Science and HCII at CMU, Human-Computer Interaction Institute, **US**.

3.4 Scientific Project Grants

I have participated in more than **40 scientific research projects** (national and international) involving a budget of more than **7.5M€ at IST**. While participating as **global coordinator** (8 projects) or **coordinator at the participating institution** (8 projects), I have secured a budget of more than **2.5M€ for IST**. A complete description of my projects is provided in Appendix V.

I highlight my participations in the following projects:

- I was the principal investigator of the CMU-PT project **AHA** (2014-2018) where I initiated strong academic links with four of my top international collaborators: Prof. Asim Smailagic, Prof. Dan Siewiorek, Prof. Fernando de la Torre, and Prof. Jodi Forlizzi. In this project we developed a novel robotic assistance platform (**PEPE**) designed to support healthy lifestyle, sustain active aging, and support those with motor deficits. **PEPE** has a **national and a European patent**. The platform is being used in a dozen of institutions in Portugal and has been exploited more recently to develop projects related to cognitive stimulation of senior adults.
- I was the principal investigator of the national project **HAVATAR** (2021-2024), that deepened collaborations with an important partner in the health sector, **Hospital da Luz**, GLSMED Learning Health, S.A. and studied the application of technologies in the health sector, namely the use of **teleoperation robots** to allow remote visits in hospitals, motivated by the pandemics context. It also continued the development of virtual and robotic coaches initiated in the AHA projects for **stroke rehabilitation**.
- I was the principal investigator of the national projects **FIREFRONT** (2019-2023) and **VOAMAIS** (2019-2023), and local PI of the **SEAGULL** (2013-2015) and **ELEVAR** (2017-2019) projects, that developed **airborne computer vision** and robotics technology (deployed in drones) for the **monitoring of critical infrastructures** (bridges, buildings, dams), **wildfires in the national forest**, and for the **surveillance of maritime events** like illegal naval traffic and pollution spots. These projects developed strong links with the **Naval School** and the **Air Force Academy**, as well as with national technological companies like **Tekever**, **UAVision**, and **Critical Software**.
- I was the local PI of the EU project **ORIENT** (2019-2022) in which ISR-IST participated as a beneficiary of the **ERC Grant** of the neuroscientist Prof. John Van Opstal, *Stichting Radboud Universiteit, Nijmegen* (NL). In this project Prof. John Van Opstal studied the Head-Eye orienting responses of animals in rapid object identification. Given the current limitations in the use of animal studies in Europe, Prof. John Van Opstal invited ISR-IST to develop robotic analogues of a human eye to test several theories on rapid eye movement control (saccades). We developed the **first cable driven mechanical eye with 6 degrees of freedom**, mimicking the main biomechanical aspects of the human eye, that was used usefully to **test optimal control and reinforcement learning theories** in the emergence of saccade eye movements.
- I was the Technical Manager of the EU project **HANDLE** (2009-2013) closely supporting the PI in daily technical decisions on the project. We studied how humans perform the manipulation of objects and used this knowledge to replicate grasping and skilled **in-hand movements with an anthropomorphic artificial hand**. We used the most advanced robot hand in the world, the **Shadow Robot Hand**, and produced pioneering works in dexterous manipulation using hand synergies and objects' affordances. In this project the PI and I jointly supervised a PhD student in a co-tutelle agreement between IST and UPMC, Paris.

- I was one of the key team members in the EU Project **ROBOTCUB** (2004-2010) that resulted in the construction of the humanoid robot iCub. The iCub is the most complex humanoid robot worldwide (in terms of degrees of freedom) and is an open platform to study developmental robotics, i.e. how humans develop **advanced sensorimotor coordination, learning, perception, interaction, planning, and execution skills**. Vislab has the only copy of the iCub in Portugal, as part of the RBCog lab, which has been used to support the graduation of many PhD students. It created a community of users and spin-off projects that paved the way for the growth of research in the **humanoid robotics** field.
- I was the responsible for the cross-platform computational architecture of the EU project **NARVAL** (1996-2001). I highlight this project as it was my first important contact with a research and development project while I was a PhD student. The project was globally coordinated by Prof. Santos-Victor of Vislab ISR/IST and developed **computer vision technologies for the control of underwater autonomous vehicles**. I developed **algorithms for tracking planar surfaces underwater** for stabilization and docking of AUVs. Also, I developed a **distributed software architecture** for the integration of several computational modules and interoperability with the software of other partners.

3.5 Recognition and Visibility

My scientific activities have been very well appreciated in the research community. I had **7 distinctions related to publications** (best paper, best poster, honourable mentions, and alike), and **6 distinctions related to my students' theses** (best thesis, outstanding thesis). I was a **finalist** (Top 5 among 85 participants) of the **Smart Ageing Prize**, AAL Programme, AAL Europe – ICT for Ageing Well, 2018. I am a **Senior Member of IEEE Robotics and Automation Society** and member of **ELLIS Society**, associated with the Lisbon Unit (**LUMLIS**). I am regularly invited to seminars, keynote talks, and plenary session in international and national events to present the results of my research, totalling more than **40 events**. Complete lists of my awards, invited talks, and society memberships is provided in Appendix VI.

4 External Engagement

4.1 Technology Transfer and Services to the Industry

I have made some efforts to transfer some of my know-how and the results of some projects to the industry. I am author of 3 intellectual property disclosures:

- **drawing** of the Portable Exergame Platform (PEPE) that resulted from the AHA project, registered in European Intellectual Property Office;
- **patent** of the PEPE platform granted in Portugal and in Europe;
- **software** of the DICORE2S project, registered in Portugal (IGAC), composed of an electronic coupon redemption system using a database, a smartphone app, and pattern recognition techniques.

The intellectual property of the DICORE2S software was transferred integrally to the company Observit, Lda, with an income for IST of **50k€**. The **drawing and patent of PEPE were licensed** to the company IDMind - Engenharia de Sistemas, Lda, for Exclusive Use of the Intellectual Property Rights. I provided consulting services for companies, fundraising, and collaboration protocols that resulted in about **60K** income for IST. Also, I have established a few formal agreements with companies for collaboration in co-supervision and placements of **MSc and PhD students in industrial context**. Finally, I was **evaluator of projects** for the national innovation agency (ADI/ANI) for **9 times**.

More details in my technology transfer actions and services for industry and innovation are presented in Appendix (VII.1 and VII.2).

4.2 Services to the Scientific Community

My services to the scientific community are related to organization of scientific events and participations in reviewing and editorial activities for international journals and conferences. I have participated in the organization of 11 international conferences and workshops including 3 general (co-)chairs and 3 program (co-)chairs (Appendix VII.3). I was member of the program, scientific, or technical committees of more than 40 international conferences (Appendix VII.4). I am currently Associate Editor of the “IEEE Robotics and Automation Letters” and have participated in other 5 editorial activities in international journals and special issues of journals (Appendix VII.5). I was evaluator of two EU project proposals in 2024 and 5 ELLIS PhD proposals (Appendix VII.6). I have been reviewer for more than 30 international journals and more than 50 conferences (Appendix VII.7). I am frequently Associate Editor of major robotics conferences as ICRA and IROS.

4.3 Services to Society

I am regularly invited to write or speak in public dissemination channels for the general audience, typically to talk about my projects and professional activities. I have made 19 communications to the public including 13 presences in media (newspapers, magazines, tv). The full list of my communications to the public are presented in Appendix VII.8.

5 Academic and Research Service

5.1 Management

I am regularly involved in management activities both at the academic level (university / department) and at the research level (research institute / lab). At the academic level I am currently the **adjunct coordinator of the Scientific Area** of Systems Decision and Control of DEEC/IST and **coordinator of the Dual Degree Master Programme** between IST and KTH, on Control, Robotics and Artificial Intelligence. At the research level I am a member of the **executive board of the Institute for Systems and Robotics**, **adjunct coordinator of the Computer and Robot Vision Lab**, and **coordinator of a thematic line of LARSyS**, the Associated Laboratory of Robotics and Engineering Systems. My full list of academic and research management positions is shown in Appendix VIII.1.

5.2 Tutoring

The tutoring program at IST¹ aims at monitoring the students’ academic path for the early identification of cases of academic failure or high academic performance. In both cases special training and guidance may be required. Overall, the program aims at contributing to improve the support of students and quality of teaching at IST. Since its start at the academic year 2003/04, I have been the tutor of **more than 100 students**. The distribution of my tutoring effort along the years is shown in Appendix VIII.2.

5.3 Academic Juries

I have been member of the juries of more than 70 PhD thesis committees (Appendix VIII.3.a) and of more than 70 MSc thesis committees (Appendix VIII.3.b). I was member of the evaluating

¹<https://nda.tecnico.ulisboa.pt/en/nda/tutoria/>

committee for a position of Associate Professor (FEUP), and member of 5 tenure evaluation committees of professors (IST, SSSA Italy) and researchers (IIT Italy), see Appendix VIII.4. I have been member of the evaluation committees of several prizes and awards at IST: the Maria de Lourdes Pintasilgo Prize for distinct female IST alumnae, the FEEDZAI and TagEnergy grants for distinct female students in the areas of machine learning and energy, and other. The full list is in Appendix VIII.5.

6 Teaching Activities

6.1 Courses

In this section I list all the official curricular units in which I have been involved at IST. They are grouped in PhD level courses, MSc level courses, BSc level courses, and Tutorial Courses. The former three are conventional courses with group lectures and practice/labs. The latter is of tutorial nature with meetings with students (individually or in group) to guide their preparation of the MSc thesis projects. For the former two cases, I include, whenever available, the ranks given by students to my teaching performance, according to the student's surveys organized by IST pedagogical council program QUC (IST Course Unit Quality System).

6.1.1 PhD Level Courses (3rd Cycle)

1. Estimation and Classification

2025/2026, 1st Semester, PDEEC, Teaching

2024/2025, 1st Semester, PDEEC, Teaching

6.1.2 MSc Level Courses (2nd Cycle)

2. Distributed Real Time Control Systems

2025/2026, 2nd Semester, MEEC21, Coordination and Teaching

Students Rating: NA (lectures and labs)

2024/2025, 2nd Semester, MEEC21, Coordination and Teaching

Students Rating: 8.75/9 (lectures), 9/9 (labs)

2023/2024, 2nd Semester, MEEC21, Coordination and Teaching

Students Rating: 8.62/9 (lectures), 8.5/9 (labs)

2022/2023, 2nd Semester, MEEC21, Coordination and Teaching

Students Rating: NA (lectures and labs)

2021/2022, 2nd Semester, MEEC21, Coordination and Teaching

Students Rating: 8.69/9 (lectures and labs)

2020/2021, 1st Semester, MEAer, MEEC, Coordination and Teaching

Students Rating: 8.75/9 (lectures), 9/9 (labs)

2019/2020, 1st Semester, MEAer, MEEC, Coordination and Teaching

Students Rating: 8.12/9 (lectures), 8.38/9 (labs)

2018/2019, 1st Semester, MEAer, MEEC, Coordination and Teaching

Students Rating: 8.12/9 (lectures), 8.62 (labs)

2017/2018, 1st Semester, MEAer, MEEC, Coordination and Teaching

Students Rating: 7.25/9 (lectures), 7.88 (labs)

2016/2017, 1st Semester, MEAer, MEEC, Coordination and Teaching

Students Rating: 8.88/9 (lectures), 9/9 (labs)

2015/2016, 1st Semester, MEAer, MEEC, Coordination and Teaching

Students Rating: 6.88/9 (lectures)

2013/2014, 1st Semester, MEAer, MEEC, Coordination and Teaching

Students Rating: 8.25/9 (lectures)

2012/2013, 1st Semester, MEAer, MEEC, Coordination and Teaching

Students Rating: 8.12/9 (lectures)

2009/2010, 1st Semester, MEAer, MEEC, Coordination and Teaching

Students Rating: 7.38/9 (lectures)

2008/2009, 1st Semester, MEAer, MEEC, Coordination and Teaching

Students Rating: NA (lectures)

2007/2008, 1st Semester, MEAer, MEEC, Coordination and Teaching

Students Rating: NA (lectures)

3. Modelling, Identification, and Digital Control

2013/2014, 1st Semester, MEEC, Teaching

Students Rating: 7.62/9 (labs)

2012/2013, 1st Semester, MEEC, Teaching

Students Rating: 7.5/9 (labs)

2011/2012, 1st Semester, MEEC, Teaching

Students Rating: 8.56/9 (labs)

2010/2011, 1st Semester, MEEC, Coordination and Teaching

Students Rating: 6.6/9 (lectures), 6.6/9 (labs)

2006/2007, 1st Semester, LEEC-pB, Coordination and Teaching

Students Rating: 3.83/5 (lectures)

2005/2006, 1st Semester, LEEC-pB, MEEC-pB, Coordination and Teaching

Students Rating: 3.91/5 (Lectures), 4.04/5 (Labs)

2004/2005, 1st Semester, LEEC-pB, Coordination and Teaching

Students Rating: 4.05/5 (Lectures), 4.09/5 (Labs)

2003/2004, 1st Semester, LEEC-pB, Coordination and Teaching

Students Rating: NA (lectures, labs)

2002/2003, 1st Semester, LEEC-pB, Teaching

Students Rating: NA (labs)

4. Machine Learning

2025/2026, 1st Semester, Multiple courses, Coordination and Teaching

Students Rating: NA (lectures)

2024/2025, 1st Semester, Multiple courses, Coordination and Teaching

Students Rating: 7.88/9 (lectures)

2023/2024, 1st Semester, Multiple courses, Coordination and Teaching

Students Rating: 7.88/9 (lectures)

2009/2010, 1st Semester, MEBiom, MEEC, Coordination and Teaching

Students Rating: 5.06/9 (lectures)

2008/2009, 1st Semester, MEBiom, MEEC, Teaching

Students Rating: 6.94/9 (labs)

5. Autonomous Systems

2021/2022, 2nd Semester, LEEC21, MEAer21, MEEC21, Min-IA, Teaching

Students Rating: 8.75/9 (lectures and labs)

6. Computer Controlled Systems

2014/2015, 1st Semester, MEAer, MEEC, Teaching

Students Rating: 8.07/9 (labs)

7. Optimal and Adaptive Control

2012/2013, 1st Semester, MEAer, Teaching

Students Rating: 8.81/9 (labs)

2011/2012, 1st Semester, MEAer, Teaching

Students Rating: 7.75/9 (labs)

2010/2011, 1st Semester, MEAer, Teaching

Students Rating: NA (labs)

2004/2005, 2nd Semester, LEA-pB, Teaching

Students Rating: NA (labs)

8. Robotics

2011/2012, 2nd Semester, MEEC, MEIC-A, MEBiom, Teaching

Students Rating: NA (labs)

1996/1997, 1st Semester, LEIC-pB, Teaching

Students Rating: 3.70/5 (labs)

9. State-Space Control

2006/2007, 2nd Semester, LEEC-pB, Teaching
Students Rating: 4.02/5 (practice), 4.06/5 (labs)
2005/2006, 2nd Semester, LEEC-pB, Teaching
Students Rating: 4.10/5 (practice), 4.03/5 (labs)
2004/2005, 2nd Semester, LEEC-pB, Teaching
Students Rating: 3.58/5 (labs)

10. Automation and Industrial Processes

2002/2003, 2nd Semester, LEEC-pB, Teaching
Students Rating: 4.07/5 (labs)

11. Computer Vision

2001/2002, 2nd Semester, LEIC-pB, Teaching
Students Rating: NA (lectures, labs)
1997/1998, 2nd Semester, LEIC-pB, Teaching
Students Rating: NA (labs)

12. Neural Networks

1997/1998, 2nd Semester, LEEC-pB, Teaching
Students Rating: NA (labs)
1996/1997, 2nd Semester, LEEC-pB, Teaching
Students Rating: 4.43/5 (labs)
1995/1996, 2nd Semester, LEEC-pB, Teaching
Students Rating: 4.43/5 (labs)

6.1.3 BSc Level Courses (1st Cycle)

13. Modeling and Simulation

2023/2024, 1st Semester, LEEC, Teaching
Students Rating: 6.25/9 (lectures)
2022/2023, 1st Semester, LEEC, Teaching
Students Rating: NA (lectures), 7.62/9 (labs)
2019/2020, 2nd Semester, MEEC, Teaching
Students Rating: 8.12/9 (labs)
2018/2019, 2nd Semester, MEEC, Teaching
Students Rating: 7.25/9 (labs)
2017/2018, 2nd Semester, MEEC, Teaching
Students Rating: 6.81/9 (labs)
2010/2011, 2nd Semester, MEEC, Teaching

Students Rating: 8.12/9 (labs)

2009/2010, 2nd Semester, MEEC, Teaching

Students Rating: 7.28/9 (labs)

14. Signals and Systems

2019/2020, 1st Semester, MEBiom, MEEC, Teaching

Students Rating: 6.88/9 (practice)

2008/2009, 2nd Semester, MEEC, Teaching

Students Rating: 6.9/9 (practice)

2003/2004, 2nd Semester, LEEC-pB, Teaching

Students Rating: 4.01/5 (practice)

2001/2002, 2nd Semester, LEEC-pB, Teaching

Students Rating: NA (practice)

15. Portfolio MEEC

2018/2019, 1st Semester, MEEC, Teaching

Students Rating: 8.25/9 (labs)

2017/2018, 1st Semester, MEEC, Teaching

Students Rating: 7.75/9 (labs)

16. Algorithms and Data Structures

2017/2018, 2nd Semester, MEEC, Teaching

Students Rating: 8.12/9 (labs)

17. Introduction to Control

2014/2015, 1st Semester, MEAer, Teaching

Students Rating: 6.5/9 (practice)

18. Control

2003/2004, 2nd Semester, LEA-pB, LEN-pB, Teaching

Students Rating: NA (labs)

2002/2003, 2nd Semester, LEA-pB, LEN-pB, Teaching

Students Rating: NA (practice)

1998/1999, 1st Semester, LEEC-pB, Teaching

Students Rating: 3.09/5 (labs)

1997/1998, 1st Semester, LEEC-pB, Teaching

Students Rating: 3.81/5 (labs)

1995/1996, 1st Semester, LEEC-pB, Teaching

Students Rating: NA (labs)

19. Circuits Theory

2002/2003, 1st Semestre, LEIC-pB, Teaching

Students Rating: 4.14/5 (practice)

20. Signal Processing

2002/2003, 1st Semester, LEA-pB, Teaching

Students Rating: 3.44/5 (practice)

21. Signal Theory

1996/1997, 2nd Semester, LEEC-pB, Teaching

Students Rating: 3.97/5 (practice)

6.1.4 Tutorial Courses (1st and 2nd Cycle)

22. 2nd Cycle Integrated Project in Electrical and Computer Engineering

2025/2026, 1st Semester, MEEC21, Teaching

2024/2025, 2nd Semester, MEEC21, Teaching

2024/2025, 1st Semester, MEEC21, Teaching

2023/2024, 1st Semester, MEEC21, Teaching

2022/2023, 2nd Semester, MEEC21, Teaching

2022/2023, 1st Semester, MEEC21, Teaching

2021/2022, 2nd Semester, MEEC21, Teaching

2021/2022, 1st Semester, MEEC21, Teaching

23. 2nd Cycle Integrated Project in Computer Science and Engineering

2024/2025, 1st Semester, MEIC-A, MEIC-T, Teaching

2023/2024, 2nd Semester, MEIC-A, MEIC-T, Teaching

2023/2024, 1st Semester, MEIC-A, MEIC-T, Teaching

2022/2023, 2nd Semester, MEIC-A, MEIC-T, Teaching

2021/2022, 1st Semester, MEIC-A, MEIC-T, Teaching

2020/2021, 2nd Semester, MEIC-A, MEIC-T, Teaching

24. 2nd Cycle Integrated Project in Biomedical Engineering

2023/2024, 1st Semester, MEBiom21, Teaching

25. Introduction to the Research in Electrical and Computer Engineering

2020/2021, 2nd Semester, MEEC, Teaching

2020/2021, 1st Semester, MEEC, Teaching

2019/2020, 2nd Semester, MEEC, Teaching

2019/2020, 1st Semester, MEEC, Teaching

2018/2019, 2nd Semester, MEEC, Teaching

2018/2019, 1st Semester, MEEC, Teaching
2017/2018, 2nd Semester, MEEC, Teaching
2017/2018, 1st Semester, MEEC, Teaching
2016/2017, 2nd Semester, MEEC, Teaching
2016/2017, 1st Semester, MEEC, Teaching
2015/2016, 2nd Semester, MEEC, Teaching
2015/2016, 1st Semester, MEEC, Teaching
2014/2015, 1st Semester, MEEC, Teaching
2013/2014, 2nd Semester, MEEC, Teaching
2013/2014, 1st Semester, MEEC, Teaching
2012/2013, 1st Semester, MEEC, Teaching

26. Master Project in Information and Software Engineering

2020/2021, 1st Semester, MEIC-A, MEIC-T, Teaching
2018/2019, 2nd Semester, MEIC-A, MEIC-T, Teaching
2018/2019, 1st Semester, MEIC-A, MEIC-T, Teaching

27. Project in Electronics Engineering

2018/2019, 1st Semester, MEE, Teaching

28. Introduction to Biomedical Engineering

2017/2018, 1st Semester, MEBiom, Teaching

Students Rating: NA (project)

2016/2017, 1st Semester, MEBiom, Teaching

Students Rating: NA (project)

2015/2016, 1st Semester, MEBiom, Teaching

Students Rating: NA (project)

29. 1st Cycle Integrated Project in Biomedical Engineering

2023/2024, 2nd Semester, LEBiom, Teaching

6.2 Innovation

The following list highlights my most relevant contributions to innovation, with emphasis in experimental and laboratory resources. Most of the points are associated to pedagogical materials that are listed in the next section.

1. In 1996 I prepared, with the supervision of Prof. Luis Borges de Almeida a set of 10 original assignments for the laboratories of the Neural Networks course.
2. In 1998 I prepared, with the supervision of Prof. José Santos-Victor, a new project for the course of Computer Vision, on 3D reconstruction from cast shadows.

3. In 2002 I prepared, with the supervision of Prof. João Miranda Lemos, a new project for the course of Modeling, Identification and Digital Control, on computer-based control of a ball-in-a-beam didactic kit.
4. In 2003 I prepared, with the supervision of Prof. Paulo Oliveira, a new project for the course in Automation of Industrial Processes, composed of a large automation cell with 3 conveyor belts and 2 robot manipulators.
5. In 2003, I prepared, with the supervision of Prof. João Miranda Lemos, a Matlab/Simulink implementation of the adaptive controller MUSMAR for the labs of the course of Optimal and Adaptive Control.
6. From 2003-2007 I participated with Prof. João Miranda Lemos in several proposals for the improvement of the experimental teaching of several courses of the Scientific Area on Systems, Decision, and Control, through the acquisition of didactic kits (total investment exceeding 50keur) and the preparation of laboratory projects for real-time control of mechanical systems. These kits have been used by several hundreds of students on control-based courses in Electrical and Computer Engineering and Aerospace Engineering since their first use in 2005.
7. In 2004, I prepared with Prof. João Miranda Lemos the first project based on the newly acquired didactic kits on applications to the control of a DC motor, for the course of Optimal and Adaptive Control.
8. In 2005, I adapted from C++ to Matlab/Simulink the project of the course Modelling, Identification and Digital Control. This adaptation allowed a better programming interface for the students that led them to progress faster in the project and a bigger diversity of experiments.
9. In 2006 I introduced new didactic kits and lab manuals for the course of Modelling, Identification and Digital Control, for the control of a flexible bar. Also, I prepared with Prof. João Miranda Lemos a new project for the course on State-Space Control on designing optimal controllers and observers for the flexible bar kit.
10. In 2008, with Prof. José Gaspar, I introduced, through a robotics middleware, new abstractions to the multitask programming component of the project of Distributed Real Time Control Systems.
11. In 2009, with Prof. José Gaspar, I introduced a more versatile microcontroller (Arduino) and a more accessible dynamical system (DC motor) in the project of Distributed Real Time Control Systems, which facilitated the accessibility of the students to the experimental equipment and the programming of the control algorithms.
12. In 2013 I introduced, for the first time in the Distributed Real Time Control Systems course decentralized control and distributed optimization techniques (distributed simplex), which enlarged the theoretical scope of the course.
13. In 2013 I created a new project for the course of Distributed Real Time Control Systems (Decentralized Control of a Distributed Illumination System) composed of 8 instrumented luminaires connected to an ethernet network, in which was possible to apply decentralized control algorithms for the first time.
14. In 2014, I created, for the first time, individual take home kits for the students of Distributed Real Time Control Systems. Each kit allowed students to develop a part of the work at home (a control node of the distributed control system), with more contact time with the experimental setup a better time management. Then, to develop the distributed

control, students organized in groups of 2 or 3 and created a network using microcontroller serial communications (SPI or I2C) to implement the decentralized control algorithms.

15. In 2016, I introduced the Consensus Algorithm for Distributed Optimization in the course of Distributed Real Time Control Systems. The Consensus Algorithm with ADMM presents an alternative to the distributed simplex algorithm used in the previous years.
16. In 2019, I have introduced CAN-BUS modules to the Distributed Real-Time Control Systems Project. As the CAN-BUS is a control network widely used in industry, the introduction of these modules improves the practical relevance of the course project.
17. In 2021, I submitted a Pedagogical Innovation Project (PIP) to acquire microcontrollers in sufficient quantity to provide each student with a full kit for the Distributed Real Time Control System. Due to the pandemics, student had reluctance to gather physically in groups to develop the networked component of the project. With a full kit, each student had available the whole experimental setup of the project, and collaboration could be done remotely. In addition, were acquire more modern microcontrollers (Raspberry pi pico) with two cores, which made opportune the teaching of concurrency patterns in multi-core scenarios. The project was approved, with complementary funds of the department (DEECProj) and the scientific area of Systems, Decision and Control, to acquire support materials (tools, instruments) for students. The equipment was first used with success in the spring semester of 2021/2022.
18. In 2022, given the success of the idea of take-home labs, I submitted two more PIP applications for take-home kits for the projects and lab assignments of the courses of Control (with Prof. Paulo Oliveira and Prof. Rita Cunha) and Autonomous Systems (with Prof. Rodrigo Ventura and Prof. Alberto Vale). The projects were approved with additional funds from the scientific area of Systems, Decision and Control, and the kits were first used in the spring semester of 2022/2023.

6.3 Pedagogical Materials

In this section I list the pedagogical materials of which I am the main author and/or the main contributor. Many of these materials are co-authored with colleagues.

1. Distributed Real Time Control Systems
 - a. Slides series in the following topics (evolved from a set of slides by Prof. Carlos Silvestre):
 - i. Introduction to Distributed Real Time Control Systems
 - ii. Microcontrollers Hardware and Software
 - iii. Topics in C++
 - iv. PID Control
 - v. Digital Control
 - vi. Real-Time Control
 - vii. Control Networks
 - viii. Concurrency
 - ix. Decentralized Control
 - x. Distributed Algorithms

- xi. Asynchronous Programming
- xii. Soft skills: Reporting guidelines
- b. Project on distributed illumination control (with contributions from Prof. João Paulo Neto, Prof. João Pedro Gomes, Prof. José Gaspar, Dr. Ricardo Ribeiro) involving the following topics:
 - i. Microcontroller Programming (Arduino, Raspberry Pi Pico)
 - ii. Assembly of electronic components and in small scale model of a distributed illumination system.
 - iii. Interface with Digital and Analog Inputs/Outputs.
 - iv. Actuation of a dimmable light source (LED) with PWM signals.
 - v. Measuring Lux with an LDR.
 - vi. Implementing real-time control loops with timers and interrupts.
 - vii. Feedforward and feedback control with PID controllers.
 - viii. State-machines
 - ix. Multi-task programming
 - x. Communications in buses with arbitration (CAN-BUS, I2C).
 - xi. Distributed algorithms for multi-node coordination.
 - xii. Decentralized control with competitive algorithms (Feedforward from accessible disturbance).
 - xiii. Decentralized control with cooperative algorithms (optimization based – Simplex, Consensus with ADMM).
 - xiv. PC interfaces to microcontroller networks.
 - xv. Asynchronous Programming with Serial and Socket based communications.
- c. Project on distributed command, monitoring, and control of a DC motor, with a PC and an Arduino (with Prof. José Gaspar and Prof. Rodrigo Ventura).
 - i. Physical Modelling of the DC motor plant.
 - ii. Identification of the physical parameters from step response.
 - iii. Simulation of the plant in Simulink.
 - iv. Controller design of a DC motor and implementation with an Arduino.
 - v. Arduino-to-PC communications using Ethernet Shield.
 - vi. Implementation of software modules in ROS/YARP to perform distributed computation and communications between a Control PC and a monitoring PC.
- 2. Modelling, Identification, and Digital Control
 - a. Slides series in the following topics (evolved from a set of slides by Prof. João Miranda Lemos):
 - i. Introduction to Computer Controlled Systems
 - ii. Models of Discrete Systems

- iii. Models of Sampled Systems
 - iv. Discretization of Continuous Controllers
 - v. Non-parametric Identification: correlation analysis and spectral analysis
 - vi. Parametric Identification: AR, ARMA, ARX and ARMAX structures.
 - vii. Controller design with polynomial methods
 - viii. Linear Prediction Models
 - ix. Prediction Error Minimization and Maximum Likelihood estimation
 - x. Minimum Variance Control
 - b. Lab Assignment on Identification and Digital Control of a Servo-Motor with a Flexible Bar (with Prof. João Miranda Lemos and Prof. Margarida Silveira):
 - i. Use Simulink Real-Time Workshop to interface a DC motor with analog position and velocity feedback.
 - ii. DC motor sensor calibration
 - iii. DC motor velocity and position Control
 - iv. Parameter Identification via Time Response
 - v. PID control with anti-windup.
 - vi. Ziegler-Nichols controller tuning method.
 - vii. Signals for Identification: Chirp and PRBS.
 - viii. Non-parametric identification of the system: SPA and CRA
 - ix. Parametric Identification: ARX and ARMAX.
 - x. Polynomial Control.
 - c. Lab Assignment on Identification and Digital Control of the Ball-in-a-Beam (with Prof. João Miranda Lemos):
 - i. Physical modelling and Matlab simulation.
 - ii. Parametric Identification from Step Response.
 - iii. Determination of Control
 - iv. Parametric Identification from data: ARX and ARMAX models.
 - v. Polynomial Control with two degrees of freedom.
 - vi. Cascaded Control
 - vii. Adaptive Control
- ### 3. Machine Learning
- a. Slides series in the following topics (evolved by a set of slides from Prof. Luís Borges de Almeida and Prof. Fernando Mira da Silva):
 - i. Introduction and Fundamentals of Machine Learning.
 - ii. Basics of Function Approximation.
 - iii. Regression vs Classification.

- iv. Neural Networks: Adaline, MLP
 - v. Backpropagation
 - vi. Data partitioning: Training, Validation, Testing
 - vii. Recurrent Networks
 - viii. Bayes Decision Theory
 - ix. Support Vector Machines
4. Computer Controlled Systems / State-Space Control / Optimal and Adaptive Control
- a. Lab Assignment on Identification and Computer Control of a Flexible Robot Arm Joint (with Prof. João Miranda Lemos and Prof. Margarida Silveira):
 - i. Use Simulink Real-Time Workshop to Interface with the Dynamical System (send actuation commands and read sensor data).
 - ii. Calibrate system sensors.
 - iii. Methods to prepare data for identification: sampling, filtering and detrending.
 - iv. Non-parametric identification: Spectral Analysis from Chirp Signals.
 - v. Parametric Identification: ARMAX model estimation with PRBS signals.
 - vi. LQ controller design.
 - vii. LQE estimator design.
 - viii. LQG controller design.
 - ix. Integral Control and Anti-Windup.
5. Optimal and Adaptive Control
- a. Lab Assignment on applications to the control of a DC motor (with Prof. João Miranda Lemos).
 - i. Introduction to real-time computer control with Matlab/Simulink
 - ii. Proportional Control
 - iii. Sampling and Filtering
 - iv. System Identification with Step Response
 - v. System Identification with Least Squares
 - vi. Pole Placement
 - vii. Recursive Identification
 - viii. Optimal Control and Minimum Variance Control
 - ix. Adaptive control with MUSMAR
6. Computer Vision
- a. Update of the set of slides supporting the lectures (with Prof. Santos-Victor).
 - b. Lab assignment on 3D Scanner from casting shadows on objects (with Prof. José Santos-Victor):

- i. System calibration with checkerboard patterns through back-projection error minimization.
 - ii. Light source pose estimation with Least Squares
 - iii. Contour detection.
 - iv. 3D reconstruction of the sequence of cast shadows.
- 7. Neural Networks
 - a. Set of lab assignments with software “NeuroSolutions” (with Prof. Luís Borges de Almeida):
 - i. Learning logic functions with ADALINE.
 - ii. Classification: Recognition of digits with MLP.
 - iii. Regression: Image Restoration with MLP.
 - iv. Recurrent Networks: Learning Linear Filters and Binary functions.
 - v. Statistical Properties of Neural Networks: Estimation of mean, variance, median, percentiles and Bayesian posterior probabilities.
 - vi. Unsupervised and Supervised Learning: Radial Basis Functions
 - vii. Clustering and Vector Quantization
 - viii. Hopfield Networks
 - ix. Principal Component Analysis
- 8. Automation and Industrial Processes
 - a. Lab assignment of “Cell Automation for Quality Control” (with Prof. Paulo Oliveira):
 - i. Assembly of a Cell composed of Conveyor Belts (3), Robot Manipulators (2), proximity and defect sensors (5), push actuators (2).
 - ii. Modeling and Analysis of the Cell using Discrete Event Systems
 - iii. Trajectory Programming for a Robotic Manipulator
 - iv. Cell programming in GRAFCET using a PLC’s
- 9. Algorithms and Data Structures
 - a. Lab assignment on Tree data structures (with Prof. Carlos Bispo)
 - i. Tree representations in file format
 - ii. In-fix and Post-Fix tree traversal
 - iii. Sweep depth and sweep breadth tree search.
 - iv. Ordering and Balancing tests.
- 10. Athens course at IST “Introduction to Robotics”, 2007.
 - a. Organization and production of contents
- 11. Best Summer Course on Intelligent Robotics Systems, module “Perception”, 1996.
 - a. Organization and production of contents (with Prof. José Santos-Victor)

7 Supervision Activities

7.1 Concluded PhD Theses

1. Luís Palma Nunes Mendes (2019-2024)

Title	Real-Time Optical Flow Velocimetry for Natural and Built Environments
Program	PhD Degree in Electrical and Computer Engineering, IST, U. Lisboa
Supervisor	Prof. Alexandre Bernardino (IST)
Co-supervisors	Prof. Rui Miguel Lage Ferreira (IST), Dra. Ana Margarida da Costa Ricardo (IST)
Classification	Pass with distinction
Funding	Fundação para a Ciência e Tecnologia
Current Job	Technical Project Manager at CleverThis and Freelance Software Developer

2. João Manuel Caeiro Avelino (2017-2023)

Title	Robots meet people: socially-aware engagement behaviours
Program	PhD Degree in Electrical and Computer Engineering, IST, U. Lisboa
Supervisor	Prof. Alexandre Bernardino (IST)
Co-supervisors	Prof. Rodrigo Ventura (IST), Prof. Leonel Garcia-Marques (Fac. Psicologia, U.Lisboa)
Classification	Pass with distinction
Funding	Fundação para a Ciência e Tecnologia
Current Job	Computer Systems and AI Scientist, uRoboptics, Lisbon, Portugal

3. Pedro Manuel Quintela Ribeiro (2017-2023)

Title	A novel magnetoresistive tactile soft sensor, inspired by biological cilia
Program	PhD Degree in Technological Physics Engineering, IST, U. Lisboa.
Supervisor	Prof. Susana Freitas (IST)
Co-supervisors	Prof. Lorenzo Jamone (IST and QMUL), Prof. Alexandre Bernardino (IST)
Classification	Pass with distinction
Funding	Fundação para a Ciência e Tecnologia
Current Job	Design Engineer, ASML, Veldhoven, Netherlands

4. Pedro Emanuel Antunes Ouro Vicente (2016-2021)

Title	Continuous Body Schema Adaptation in Humanoid Robots
Program	PhD Degree in Electrical and Computer Engineering, IST, U. Lisboa.
Supervisor	Prof. Alexandre Bernardino (IST)
Co-supervisors	Prof. Lorenzo Jamone (IST and QMUL), Prof. José Santos-Victor (IST)
Classification	Pass with distinction
Funding	Fundação para a Ciência e Tecnologia
Current Job	Computer Vision scientist at Sensei, Lisbon, Portugal.

5. Miguel Sérgio de Abreu Neto (2014-2021)

Title	Design, Development and Integration of a Flexible Array of Magnetoresistive Sensors, for Tactile Sensing.
Program	PhD Degree in Leaders for Technical Industries, IST, U. Lisboa.
Supervisor	Prof. Susana Freitas (IST)
Co-supervisors	Prof. Alexandre Bernardino (IST)
Classification	Pass
Funding	Fundação para a Ciência e Tecnologia, MIT Portugal
Current Job	NA

6. Min Hun Lee (2015-2021)

Title	Interactive Hybrid Intelligence Systems for Human-AI/Robot Collaboration: Improving the practices of physical stroke rehabilitation
Program	PhD Degree in Electrical and Computer Engineering, IST, U. Lisboa.
Supervisor	Prof. Alexandre Bernardino (IST)
Co-supervisors	Prof. Daniel Siewiorek (CMU) and Prof. Asim Smailagic (CMU)
Classification	Pass with Distinction
Funding	Fundação para a Ciência e Tecnologia and CMU Portugal
Current Job	Assistant Professor at SMU, Singapore.

7. Rui Miguel Horta Pimentel de Figueiredo (2014-2020)

Title	Space-Variant Vision Mechanisms for Resource-Constrained Humanoid Robot Applications
Program	PhD Degree in Electrical and Computer Engineering, IST, U. Lisboa.
Supervisor	Prof. Alexandre Bernardino (IST)
Co-supervisors	Prof. Hélder de Jesus Araújo (UC)
Classification	Pass with Distinction
Funding	Fundação para a Ciência e Tecnologia, RBCog Doctoral Program
Current Job	Post-doctoral Researcher at Aalborg University, Denmark.

8. João Guilherme Antunes Martins (2015-2020)

Title	Leveraging Context for Multi-Label Action Recognition and Detection in Video
Program	IST-CMU Dual Degree in Electrical and Computer Engineering
Supervisor IST	Prof. Alexandre Bernardino (IST)
Supervisors CMU	Prof. Daniel Siewiorek (CMU), Prof. Asim Smailagic (CMU)
Classification	Grade A (CMU)
Funding	Fundação para a Ciência e Tecnologia, CMU Portugal
Current Job	Machine Learning scientist at Apple, US.

9. Nuno Alexandre Antunes Martins Pessanha-Santos (2015-2020)

Title	Fixed-wing UAV tracking in outdoor scenarios for autonomous landing.
Program	PhD Degree in Electrical and Computer Engineering, IST, U. Lisboa.
Supervisor	Prof. Alexandre Bernardino (IST)
Co-supervisors	Prof. Vitor Lobo (UNL)
Classification	Pass with Distinction
Funding	-
Current Job	Assistant Professor at the Military Academy, Portugal.

10. Gonalo Charters Santos Cruz (2013-2020)

Title	Autonomous aerial imagery analysis in maritime surveillance scenarios
Program	PhD Degree in Electrical and Computer Engineering, IST, U. Lisboa.
Supervisor	Prof. Alexandre Bernardino (IST)
Co-supervisors	Prof. Ant3nio Pedro Aguiar (FEUP)
Classification	Pass with Distinction
Funding	-
Current Job	Researcher at the Portuguese Air Force Research Center

11. Giovanni Saponaro (2010-2019)

Title	Gestures and Object Affordances for Human-Robot Interaction
Program	PhD Degree in Electrical and Computer Engineering, IST, U. Lisboa.
Supervisor	Prof. Alexandre Bernardino (IST)
Co-supervisors	Prof. Giampiero Salvi (KTH and NTNU) and Dr. Lorenzo Jamone (IST and QMUL)
Classification	Pass with Distinction
Funding	Funda3o para a Ci4ncia e Tecnologia
Current Job	Computer Vision scientist at Eyeware, Lda, Lisbon, Portugal

12. Mehmet Hasan Mutlu (2014-2019)

Title	Perception, control, and planning for (self-)reconfigurable modular robots.
Program	IST-EPFL Joint Degree in Electrical and Computer Engineering
Supervisor IST	Prof. Alexandre Bernardino (IST)
Supervisor EPFL	Prof. Auke Ijspeert (EPFL)
Classification	Pass
Funding	Funda3o para a Ci4ncia e Tecnologia, RBCog Doctoral Program
Current Job	Senior Electronics Engineer at ANYbotics, Zurich, Switzerland

13. Athira Muraleedharan Nambiar (2014-2017)

Title	Towards automatic long term Person Re-identification System in video surveillance
Program	PhD Degree in Electrical and Computer Engineering, IST, U. Lisboa.
Supervisor	Prof. Alexandre Bernardino (IST)
Co-Supervisors	Prof. Jacinto Nascimento (IST), Prof. Jos4 Santos-Victor (IST)

Title	Towards automatic long term Person Re-identification System in video surveillance
Classification	Pass with Distinction and Honour
Funding	Fundação para a Ciência e Tecnologia
Current Job	Assistant Professor at SMIST, Chennai, India

14. Nuno Miguel Banheiro Moutinho (2011-2017)

Title	Humanoid Robot Head Continuous Calibration for Active Stereo Vision, using Non-Linear Filtering Techniques
Program	PhD Degree in Electrical and Computer Engineering, IST, U. Lisboa.
Supervisor	Prof. Alexandre Bernardino (IST)
Co-Supervisors	Prof. José Gaspar (IST)
Classification	Pass with Distinction
Funding	EC FP7 Project RoboSoM
Current Job	CTO at Sensei, Lda., Lisbon, Portugal.

15. Alexandre Tuleu (2011-2017)

Title	Hardware, Software and Control Considerations Towards Low-Cost Compliant Quadruped Robots
Program	IST-EPFL Joint Degree in Electrical and Computer Engineering
Supervisor IST	Prof. Alexandre Bernardino (IST)
Supervisor EPFL	Prof. Auke Ijspeert (EPFL)
Classification	Pass
Funding	Fundação para a Ciência e Tecnologia, RBCog Doctoral Program
Current Job	Freelance Developer and Consultant, Lausanne, Switzerland

16. Dario António Bacellar Figueira (2009-2016)

Title	Automatic Person ReIdentification for Video Surveillance Applications
Program	PhD Degree in Electrical and Computer Engineering, IST, U. Lisboa.
Supervisor	Prof. Alexandre Bernardino (IST)
Co-Supervisor	Prof. Jacinto Nascimento (IST)
Classification	Pass with Distinction
Funding	Fundação para a Ciência e Tecnologia
Current Job	Computer Vision Scientist at Ocado Technology, London, UK

17. Matteo Taiana (2008-2015)

Title	A methodology for training and evaluating Pedestrian Detectors: towards practical applications
Program	PhD Degree in Electrical and Computer Engineering, IST, U. Lisboa.
Supervisor	Prof. Alexandre Bernardino (IST)
Co-Supervisor	Prof. Jacinto Nascimento (IST)
Classification	Pass with Distinction
Funding	Fundação para a Ciência e Tecnologia
Current Job	Post-doc Researcher at IIT, Genoa, Italy

Title	A methodology for training and evaluating Pedestrian Detectors: towards practical applications
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18. Ricardo da Silveira Cabral (2009-2015)

Title	Unified Low-rank Models for Visual Learning
Program	PhD Degree in Electrical and Computer Engineering, IST, U. Lisboa and Carnegie Mellon University, Pittsburgh, U.S. (Dual Degree)
Supervisors (IST)	Prof. João Paulo Costeira, Prof. Alexandre Bernardino
Supervisor(CMU)	Prof. Fernando de La Torre
Classification	Pass
Funding	Fundação para a Ciência e Tecnologia, CMU-Portugal
Current Job	Senior Engineering Manager at Mixed Reality@Meta, Zurich, Switzerland

19. Hugo Miguel Gomes da Silva (2009-2014)

Title	A Probabilistic Approach to Stereo Visual Egomotion
Program	PhD Degree in Electrical and Computer Engineering, IST, U. Lisboa
Supervisor	Prof. Alexandre Bernardino (IST)
Co-Supervisor	Prof. Eduardo Alexandre Pereira da Silva (ISEP/IPP)
Classification	Pass with Merit
Funding	Fundação para a Ciência e Tecnologia
Current Job	Senior Researcher at INESC Technology and Science, Porto, Portugal

20. Urbain Prieur (2009-2013)

Title	High-level planning of dexterous in-hand manipulation using a robotic hand
Program	PhD Degree in Electrical and Computer Engineering, IST, U. Lisboa Université Pierre et Marie Curie, Paris, France (co-tutelle)
Supervisor (IST)	Prof. Alexandre Bernardino
Supervisor(UPMC)	Prof. Véronique Perdereaux
Classification	Pass
Funding	EC FP7 Project
Current Job	Robotics Project Leader at Exotec Solutions, Croix, France,

21. Jonas Ruesch (2008-2013)

Title	A Computational Approach on the Co-Development of Visual Sensorimotor Structures
Program	PhD Degree in Electrical and Computer Engineering, IST, U. Lisboa
Supervisor	Prof. Alexandre Bernardino (IST)
Co-Supervisor	Prof. José Santos-Victor (IST)
Classification	Pass with Merit
Funding	Fundação para a Ciência e Tecnologia
Current Job	Senior Consultant, bbv Software Services Corp. Zurich, Switzerland.

22. Plinio Moreno (2004-2008)

Title	Component Models for Visual Object Recognition using Gabor Filters
Program	PhD Degree in Electrical and Computer Engineering, IST, U. Lisboa
Supervisor	Prof. José Santos-Victor (IST)
Co-Supervisor	Prof. Alexandre Bernardino (IST)
Classification	Pass
Funding	Fundação para a Ciência e Tecnologia
Current Job	Senior Researcher, ISR-Lisboa.

7.2 Ongoing PhD Theses

1. Hugo Alexandre Rodrigues Simão (2021-)

Title (tentative)	Value-based design methodologies in human-robot interaction.
Program	PhD Degree in Computer Science, Fac. Ciências, U. Lisboa.
Supervisor	Prof. Tiago Guerreiro (FCUL)
Co-Supervisors	Prof. Alexandre Bernardino (IST), Prof. Jodi Forlizzi (CMU, US)
Funding	Fundação para a Ciência e Tecnologia

2. Ana Rita Mendes Cóias (2022-)

Title (tentative)	Solutions for Unsupervised Stroke Rehabilitation with Low-cost Computer and Robotic Based Systems
Program	PhD Degree in Electrical and Computer Engineering, IST, U. Lisboa
Supervisor	Prof. Alexandre Bernardino
Co-Supervisor	Prof. Asim SAMILAGIC (CMU)
Funding	Fundação para a Ciência e Tecnologia

3. José Pedro Ribeiro Gomes (2023-)

Title (tentative)	Deep Learning Based Models for Human Action Representations
Program	PhD Degree in Electrical and Computer Engineering, IST, U. Lisboa, Carnegie Mellon University, Pittsburgh, US (Dual Degree)
Supervisor (IST)	Prof. Alexandre Bernardino
Supervisor(CMU)	Prof. Fernando de La Torre
Funding	Fundação para a Ciência e Tecnologia, CMU Portugal

4. Maria Rita Ribeiro da Fonseca Verdelho (2023-)

Title (tentative)	Personalizing the therapy of melanoma patients using machine learning
Program	PhD Degree in Electrical and Computer Engineering, IST, U. Lisboa
Supervisor	Prof. Catarina Barata (IST)
Co-Supervisor	Prof. Alexandre Bernardino (IST)
Funding	Fundação para a Ciência e Tecnologia

5. Diogo Alexandre Oliveira Silva (2023-)

Title (tentative)	Towards Probabilistic Object Detection in Images and Videos.
Program	PhD Degree in Electrical and Computer Engineering, IST, U. Lisboa.
Supervisor	Prof. Alexandre Bernardino (IST)
Co-Supervisor	Dr. Gonalo Cruz (Air Force Research Center)
Funding	NA

6. Joo Miguel Barradas Luzio (2024-)

Title (tentative)	Semantic Attention for Robots
Program	PhD Degree in Electrical and Computer Engineering, IST, U. Lisboa.
Supervisor	Prof. Alexandre Bernardino (IST)
Co-Supervisor	Dr. Plinio Moreno (ISR-Lisboa)
Funding	FCT

7. Ayman Bnousaad (2024-)

Title (tentative)	Remote Sensing for Prediction of Harmful Algal Blooms
Program	PhD Degree in Electrical and Computer Engineering, IST, U. Lisboa.
Supervisor	Prof. Alexandre Bernardino (IST)
Co-Supervisor	Dr. El Khalil Cherif (ISR-Lisboa)
Funding	FCT

8. Joo Francisco Felizardo (2025-)

Title (tentative)	Modelling and Optimization of Cochlear Implants
Program	PhD Degree in Electrical and Computer Engineering, IST, U. Lisboa.
Supervisor	Prof. Alexandre Bernardino (IST)
Co-Supervisor	Prof. John Van Opstal (University of Nijmegen, Netherlands)
Funding	Internal Funding

9. Vinicius Menezes Souza (2025-)

Title (tentative)	Perception for Action in Self-Driving Vehicles
Program	PhD Degree in Electrical and Computer Engineering, IST, U. Lisboa.
Supervisor	Prof. Alexandre Bernardino (IST)
Co-Supervisor	Prof. Jos Gaspar (IST)
Funding	FCT

10. Tiago Gouveia (2025-)

Title (tentative)	Situational Awareness for Planning Missions
Program	PhD Degree in Electrical and Computer Engineering, IST, U. Lisboa.
Supervisor	Prof. Alexandre Bernardino (IST)
Co-Supervisor	Prof. nio Chambel (Academia Militar)
Funding	NA

7.3 Supervised and Co-Supervised MSc Theses

1. José Luís Lopes da Cunha. **Road Accident Severity Prediction in Portugal**. Supervised by Prof. José Silvestre da Serra Silva (Military Academy) and Prof. Alexandre Bernardino (IST). Military Academy 2025.
2. André Filipe Rento Teodósio. **The Role of Trajectory Prediction in the Perception of Approaching Agents**. Supervised by Prof. Plinio Moreno (IST) and Prof. Alexandre Bernardino (IST). IST 2025.
3. Inês Cardoso Paiva. **Dual Field of View Gaze Estimation with Knowledge Distillation for Mobile Robots**. Supervised by Prof. Alexandre Bernardino (IST) and Prof. Plinio Moreno (IST). IST 2025.
4. Sofia Alexandre Nobre dos Santos. **Search for and detect objects with foveal vision systems using Visual-Language Models**. Supervised by Prof. Alexandre Bernardino (IST) and Prof. Plinio Moreno (IST). IST 2025.
5. David Wang Ye. **AI-Driven Conversational User Interfaces for Incident Management Systems**. Supervised by Prof. Alexandre Bernardino (IST). IST 2025.
6. Filipe Martins Tendeiro. **FireTools: Integrated Tools for Enhanced Wildfire Detection and Segmentation**. Supervised by Prof. Alexandre Bernardino (IST) and Prof. Nuno Pessanha-Santos (Military Academy). IST 2025.
7. Miguel Alexandre Marques Gomes Brandão Bandeira. **Ground-Based Monocular Vision System for UAV Tracking: Performance Evaluation in Realistic Simulations**. Supervised by Prof. Alexandre Bernardino (IST) and Prof. Nuno Pessanha-Santos (Military Academy). IST 2025.
8. Tomás Canivari Crispim da Costa Luís. **Swimmer and Boat Detection and Tracking in Search and Rescue Operations in Maritime Environments using UAVs**. Supervised by Prof. Alexandre Bernardino (IST) and Prof. Bruno Damas (Naval School). IST 2025.
9. Francisco Reis Nogueira. **Comprehension of Multilingual Expressions Referring to Target Objects in Visual Inputs**. Supervised by Prof. Bruno Martins (IST) and Prof. Alexandre Bernardino (IST). IST 2025.
10. Luís Pedro Soares Marmoto Gaspar Lopes. **Generalized Referring Expression Segmentation on Aerial Photos**. Supervised by Prof. Bruno Martins (IST) and Prof. Alexandre Bernardino (IST). IST 2025.
11. Francisco Miguel Pessanha Pardal Ribeiro. **Artificial Pain Representation for Remote Collaborative Manipulation**. Supervised by Prof. Alexandre Bernardino (IST) and Prof. José Santos-Victor (IST). IST 2025.
12. João Paiva de Castelo-Branco. **Integrating Deep Reinforcement Learning and Bayesian Active Inference for Object Search in Mobile Robotics**. Supervised by Prof. Alexandre Bernardino (IST) and Prof. José Santos-Victor (IST). IST 2025.
13. Guilherme Gonçalves Gutierrez de Lima. **Garden Eel Detection Using Deep Learning Camouflaged Object Detection Techniques**. Supervised by Prof. Alexandre Bernardino (IST) and Prof. Helena Sofia Pinto (IST). IST 2025.
14. Tomás da Silva Faria Lampreia Gomes. **Camouflaged Rays Detection With Above Water Cameras in Oceanário de Lisboa**. Supervised by Prof. Helena Sofia Pinto (IST) and Prof. Alexandre Bernardino (IST). IST 2025.

15. Afonso Maria Soares Moita de Deus Araújo. ***SMART Hand Rehabilitation after Stroke.*** Supervised by Prof. Alexandre Bernardino (IST) and Prof. Athanasios Vourvopoulos (IST). IST 2024.
16. Afonso Serrão Carogo de Carvalho. ***Hyperspectral/multispectral image fusion with a neural network regularization.*** Supervised by Prof. Alexandre Bernardino (IST) and Prof. Thomas Oberlin (ISAE-SupAero, Toulouse, France). IST 2024.
17. André Viúla Pires. ***Thermography for Vineyard Water Status Estimation, A Comparative Analysis of Thermal Imaging and Other Predictors of Plant Water Across Diverse Vineyard Conditions.*** Supervised by Prof. Alexandre Bernardino (IST) and Dr. Gonçalo Filipe dos Santos Victorino (ISA-UL). IST 2024.
18. Bruno Miguel da Silva Cebola. ***Comparative Analysis of Approaches to Perform Human Activity Recognition in Public Transportation Hubs.*** Supervised by Prof. Alexandre Bernardino (IST) and Eng. João Tiago Pinto de Almeida Dias Fonseca (Thales Portugal). IST 2024.
19. César Mateus Caires Batista Rosa. ***Leveraging Mirror Geometry for Monocular 3D Reconstruction of Human Body Pose in Rehabilitation Therapy.*** Supervised by Prof. Alexandre Bernardino (IST) and Eng. Gonçalo Reis Rodrigues Chambel (Clynx Health). IST 2024.
20. Diogo Costa Ravasco. ***Computer Vision Based Approach to Mackerel Population Counting from Single Images.*** Supervised by Prof. Alexandre Bernardino (IST) and Prof. Helena Sofia Pinto (IST). IST 2024.
21. Diogo Filipe Baptista da Silva. ***Real-Time Control of Dexterous Robot Hands with Tactile and Force Sensors.*** Supervised by Prof. Alexandre Bernardino (IST) and Dr. Plinio Moreno Lopez (ISR-IST). IST 2024.
22. João Francisco Mendes da Costa Felizardo. ***Computational Auditory Process Simulator for Optimising the Parameters of Cochlear Implants.*** Supervised by Prof. Alexandre Bernardino (IST) and Prof. John Van Opstal (Donders Institute, Nijmegen, Netherlands). IST 2024.
23. Francisco Tavares Braz Mêda. ***Skin-Inspired Magnetoresistive-based Tactile Sensor for Force Characterization in Distributed Areas.*** Supervised by Prof. (IST) Susana Isabel Pinheiro Cardoso de Freitas (IST) and Prof. Alexandre Bernardino (IST). IST 2024.
24. Gonçalo Cachinho Matias da Silva Matos. ***Active Visual Object Search for Mobile Robots.*** Supervised by Prof. Alexandre Bernardino (IST) and Dr. Plinio Moreno Lopez (ISR-IST). IST 2024.
25. Hugo Daniel Sousa Lima. ***Benchmarking Real-Time Fire and Smoke Segmentation in RGB Images for Wildfire Monitoring.*** Supervised by Prof. Alexandre Bernardino (IST) and Major Gonçalo Charters dos Santos Cruz (Portuguese Air Force). IST 2024.
26. Inês Cristina Catanas Forte Reis Pinto. ***Detection and Tracking of Drones in Infrared Images.*** Supervised by Prof. Alexandre Bernardino (IST) and Prof. José Silvestre Serra da Silva (Military Academy). IST 2024.
27. João Pedro Figueiredo Gomes de Oliveira. ***Area Coverage optimization with UAV Swarms.*** Supervised by Prof. Alexandre Bernardino (IST) and Prof. José Silvestre Serra da Silva (Military Academy). Military Academy 2024.
28. João Rafael Lopes Pedrosa. ***Calibration of extrinsic and intrinsic parameters of a camera + projector.*** Supervised by Prof. Alexandre Bernardino (IST). IST 2024.

29. Lara Gemmrich. ***Estimating Sea Ice Thickness in the Arctic using ROV-based Underwater Hyperspectral Imaging***. Supervised by Prof. Alexandre Bernardino (IST) and Prof. Martin Ludvigsen (NTNU, Norway). IST 2024.
30. Maximilian J. Vieweg. ***Segmentation of Maritime Vehicles from Video Sequences***. Supervised by Prof. Alexandre Bernardino (IST) and Prof. Maria Margarida Campos da Silveira (IST). IST 2024.
31. Miguel Esteves Teixeira. ***How does the brain control the eye movements? An analysis-by-synthesis approach***. Supervised by Prof. Alexandre Bernardino (IST) and Prof. Pedro José Oliveira Sebastião (IST). IST 2024.
32. Nuno Miguel Caldeira Miguéns Diogo. ***Active Health Data Collection Using Social Robot***. Supervised by Dr. Plinio Moreno Lopez (ISR-IST) and Prof. Alexandre Bernardino (IST). IST 2024.
33. Ricardo Alexandre Ramos Rodrigues. ***Effectiveness of a Humorous Robot in Reducing Anxiety in Clinical Procedures***. Supervised by Dr. Plinio Moreno Lopez (ISR-IST) and Prof. Alexandre Bernardino (IST). IST 2024.
34. Vinicius Meneses Souza. ***Improving Autonomous Navigation for Car-like Vehicles with Dual GNSS Integration***. Supervised by Prof. José António da Cruz Pinto Gaspar (IST) and Prof. Alexandre Bernardino (IST). IST 2024.
35. Vasco Duarte Vieira Rodrigues. ***Sensor Fusion for Deep Object Detection In Ship Automation***. Supervised by Prof. Alexandre Bernardino (IST) and Eng. Diogo Costa Arreda (SeaAI). IST 2023.
36. João Miguel Barradas Luzio. ***Semantic-Based Active Perception for Humanoid Visual Tasks***. Supervised by Prof. Alexandre Bernardino (IST) and Prof. Plinio Moreno Lopez (IST). IST 2023.
37. Mateus da Silva Dias Libano Monteiro. ***Real-Time Vehicle Turning-Movement Counting using Object Detection and Tracking***. Supervised by Prof. Alexandre Bernardino (IST) and Prof. Sofia Pinto (IST). IST 2023.
38. Madalena Raposo de Lima. ***Estudo da Aplicabilidade dos Exercogs na Plataforma PEPE em Séniores que Frequentam Estruturas de Apoio à Pessoa Idosa (in portuguese)***. Supervised by Prof. Maria Vânia Silva Nunes (Univ. Católica), and Prof. Alexandre Bernardino (IST). Univ. Católica / ISPA 2023.
39. Daniel Nobre Chagas. ***Combining Active Learning with Neural Networks for Robot Self-Calibration***. Supervised by Prof. Alexandre Bernardino (IST) and Prof. Plinio Moreno Lopez (IST). IST 2023.
40. Nuno Daniel Matos Martins. ***Reconhecimento de Impressões Digitais em Cenários de Crime (in portuguese)***. Supervised by Prof. José Silvestre Serra Silva (Military Academy), and Prof. Alexandre Bernardino (IST). IST / Military Academy 2023.
41. Ricardo Jorge Dias Martins. ***Automatic detection of aircraft fuselage defects from photographs***. Supervised by Prof. Alexandre Bernardino (IST) and Prof. Antonio Miguel Perez Romero (Univ. Seville). IST 2023.
42. Francisco Mansilha Pena Galante. ***Active Robot Perception: Visual Object Search in Fisheye Vision***. Supervised by Prof. Alexandre Bernardino (IST) and Prof. Plinio Moreno Lopez (IST). IST 2023.

43. João Paulo de Abrantes Raio e Lucas Lavrador. ***Post-Stroke Gait Rehabilitation Assistive System***. Supervised by Prof. Alexandre Bernardino (IST) and Prof. Plinio Moreno Lopez (IST). IST 2023.
44. Ricardo Luís Mota de Quadros Tenreiro. ***Deep Learning Optical Flow for the Analysis of Fluids***. Supervised by Prof. Alexandre Bernardino (IST) and Prof. Rui Ferreira (IST). IST 2023.
45. Franziska Stephanie Auer. ***Detecting Swimmers in Open Water***. Supervised by Prof. Alexandre Bernardino (IST) and Eng. Lukasz Dzianach (Callboats). IST 2023.
46. Luís Miguel de Brito Ramos Costa. ***Validation of Deep Learning Optical Flow Algorithms for Applications in Fluid Mechanics***. Supervised by Prof. Alexandre Bernardino (IST) and Prof. Rui Ferreira (IST). IST 2023.
47. Tom Miguel dos Santos. ***Multispectral and Multiscale Feature-Level Fusion for Pedestrian Detection in Road Environments***. Supervised by Prof. Alexandre Bernardino (IST) and Eng. Diogo Arreda (SeaAI). IST 2023.
48. Alexandre Luís Martins Gomes de Andrade Fonseca. ***Sensor Fusion for Object Detection based on Test-Time Augmentation***. Supervised by Prof. Alexandre Bernardino (IST) and Eng. Diogo Arreda (SeaAI). IST 2023.
49. André Domingues Gomes Branco. ***Influence of Cognitive Workload Features in Motor Imagery Brain-Robot Interfacing Performance***. Supervised by Prof. Athanasios Vourvopoulos (IST) and Prof. Alexandre Bernardino (IST). IST 2023.
50. Bernardo Alves Rocha. ***Cross-view Generalization in Action Recognition: Feature Design for Transitioning from Exocentric to Egocentric Views***. Supervised by Prof. Alexandre Bernardino (IST) and Prof. Plinio Moreno Lopez (IST). IST 2023.
51. Alexandra Esperança Gonçalves. ***Non-Verbal Cues on Robot-Group Persuasion***. Supervised by Prof. Alexandre Bernardino (IST) and Prof. Plinio Moreno Lopez (IST). IST 2023.
52. Alexandre Yu Jin. ***3D Reconstruction using images obtained by an unmanned aerial vehicle***. Supervised by Prof. Alexandre Bernardino (IST) and Prof. José Silva (Military Academy, Portugal). IST 2022.
53. António Pedro Pacheco Figueiredo Carvalho. ***Learning Mobile Robot's Push Action Models for Box Transportation in Logistics Application***. Supervised by Prof. Alexandre Bernardino (IST) and Prof. Claudio Zito (TII, UAE). IST 2022.
54. Beatriz Tavares Silva. ***A Navigation and Perception System for an Autonomous Race Car***. Supervised by Prof. Alexandre Bernardino (IST) and Prof. Claudio Zito (TII, UAE), IST 2022.
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191. André Dionísio Falcão. ***Controlo Adaptativo do nível de Hipnose em Anestesia***. Supervised by Prof. Alexandre Bernardino (IST) and Prof. João Miranda Lemos (IST). IST 2009.
192. Daniela Filipa Gonçalves Pamplona. ***Gaussian Foveation***. Supervised by Prof. Carlos Alves (IST) and Prof. Alexandre Bernardino (IST). IST 2008.
193. Samuel António Tojal Ferreira. ***Predtrack - Predição de Trajectórias de Alvos Móveis : Aplicação ao Controlo de uma Cabeça Robótica***. Supervised by Prof. José Santos-Victor (IST) and Prof. Alexandre Bernardino (IST). IST 2008.
194. Luís Miguel Correia Guilherme. ***Controlo de um Gerador de Ondas Hidráulicas em Canal com Absorção Activa***. Supervised by Prof. Alexandre Bernardino (IST) and Dr. João Palma (LNEC). IST 2008 (pre-bologna).
195. Hugo Miguel Gomes da Silva. ***Sistema de Visão em Tempo Real para Aplicações Robóticas em Sistemas Autónomos***. Supervised by Prof. Eduardo Silva (ISEP) and Prof. Alexandre Bernardino (IST). IST 2008. (pre-bologna).
196. Matteo Taiana. ***3D model-based tracking with one omnidirectional camera and particle filters***. Supervised by Prof. Matteo Matteucci (Politecnico di Milano) and Prof. Alexandre Bernardino (IST). Politecnico di Milano 2008.
197. Cláudia Alexandra Magalhães Soares. ***Reconhecimento de vocábulos com modelos antropomórficos de síntese de voz: aplicação à robótica humanóide***. Supervised by Prof. José Santos-Victor (IST) and Prof. Alexandre Bernardino (IST). IST 2007.

198. Ricardo Manuel Candeias da Silva Santos. ***Evaluation of Visual Servoing Techniques for Robotic Applications***. Supervised by Prof. Alexandre Bernardino (IST). IST 2007.
199. Paulo Jorge Pereira Carreiras. ***Predgrab - Predição de Trajetórias de Alvos Móveis: Aplicação ao Controlo de um Braço Robot***. Supervised by Prof. José Santos-Victor (IST) and Prof. Alexandre Bernardino (IST). IST 2007.
200. Jorge Grande Calvo. ***Aprendizagem supervisionada de mapas visuo-motores: Métodos lineares e recursivos***. Supervised by Prof. Alexandre Bernardino (IST) and Prof. Javier Traver (UJI), Univ. Jaume I, Catellon de la Plana, Espanha, 2006.
201. Renata Patrícia Ramos Ferreira and Joel Pedro Nascimento Lopes. ***CITYMATCH: Monitorização da Dinâmica Urbana Combinando Imagem Aérea e Laser***. Supervised by Prof. João Paulo Costeira (IST) and Alexandre Bernardino (IST), IST 2006 (Pre-bologna Licenciatura Degree)
202. João Paulo Gonçalves Ferreira. ***CITY LASER: Criação de Modelos 3D de Edifícios em Mapas Digitais de Superfície para Planeamento Urbano***. Supervised by Prof. Alexandre Bernardino (IST) and João Paulo Costeira (IST), IST 2006 (Pre-bologna Licenciatura Degree)
203. João Tiago Rodrigues Marques and Marco Gonçalo de Sousa Neves. ***Vision Based Control of an Autonomous Blimp (VIDEOBLIMP)***. Supervised by Prof. José Santos-Victor (IST) and Prof. Alexandre Bernardino (IST), IST 2005 (Pre-bologna Licenciatura Degree)
204. Júlio Jesus León Pérez. ***Técnicas de segmentação de cor para seguimento de alvos - aplicação à aprendizagem de tarefas em Robots Humanoides***. Supervised by Prof. Alexandre Bernardino (IST) and Prof. José Santos-Victor (IST). IST 2005 (Pre-bologna Licenciatura Degree).
205. Rudolfo Santos and Arménio Pereira. ***Interpretação Automática de Mapas para processamento de imagem aérea***. Supervised by Prof. João Paulo Costeira (IST) and Alexandre Bernardino (IST), IST 2004 (Pre-bologna Licenciatura Degree)
206. Filipe Manuel da Silva Metelo and Luís Ricardo Garcia Campos. ***Vision Based Control of an Autonomous Blimp (VIDEOBLIMP)***. Supervised by Prof. José Santos-Victor (IST) and Alexandre Bernardino (IST), IST 2003 (Pre-bologna Licenciatura Degree)
207. João Alves. ***Captura de Movimento Antropomórfico***. Supervised by Prof. José Santos-Victor (IST) and Alexandre Bernardino (IST), IST 2003 (Pre-bologna Licenciatura Degree).
208. Ricardo Marranita. ***VideoSkeleton - Reconhecimento Antropomórfico do Corpo Humano***. Supervised by Prof. José Santos-Victor (IST) and Alexandre Bernardino (IST), IST 2003 (Pre-bologna Licenciatura Degree)
209. Rodrigo Matias. ***Sistema de Visão para Monitorização de Tráfego Automóvel***. Supervised by Prof. José Santos-Victor (IST) and Alexandre Bernardino (IST), IST 2003 (Pre-bologna Licenciatura Degree)
210. António Matos, Bertolino Ferreira, and Humberto Martins. ***Veículo Celular Teleoperado com Controlo Baseado em Visão***. Supervised by Prof. José Santos-Victor (IST) and Alexandre Bernardino (IST), IST 1999 (Pre-bologna Licenciatura Degree)
211. José Vitorino, Nília Oliveira, and Nuno Gomes. ***Patinhas - Robot Móvel com Pernas***. IST 1998. Supervised by Prof. José Santos-Victor (IST) and Alexandre Bernardino (IST), IST 1999 (Pre-bologna Licenciatura Degree)

212. Ofélia Malheiros, Rui Gonçalves, and João Ribeiro. ***Seguimento de alvos móveis utilizando a cabeça robótica MEDUSA***. Supervised by Prof. José Santos-Victor (IST) and Alexandre Bernardino (IST), IST 1998 (Pre-bologna Licenciatura Degree)
213. Alexandre Miguel Lopes Barroso and Sjoerd van der Zwaan. ***Desenvolvimento de um sensor avançado em Robótica: "Olho Composto"***. Supervised by Prof. José Santos-Victor (IST) and Alexandre Bernardino (IST), IST 1998 (Pre-bologna Licenciatura Degree)
214. Paulo Silva. ***Ambiente Virtual de Visão Estereoscópica com Controlo Activo***. Supervised by Prof. José Santos-Victor (IST) and Alexandre Bernardino (IST), IST 1998 (Pre-bologna Licenciatura Degree)

7.4 PostDoc Researchers

1. Ângelo Cardoso, Postdoctoral Fellowship, Apr. 2014 – Feb. 2015 (BIOMORPH Project)
2. Lorenzo Jamone, Postdoctoral Fellowship, Feb. 2013 – Jan. 2015 (POETICON++ Project), Apr. 2016 – Mar. 2017 (LARSyS Funding).
3. Ricardo Ribeiro, Postdoctoral Fellowship, Mar. 2014 – Feb. 2015 (SEAGULL Project), Apr. 2015 – Dec. 2018 (AHA Project)
4. Nino Cauli, Postdoctoral Fellowship, Feb. 2016 – Jul. 2016 (AHA Project)
5. Milad Niknejad, Postdoctoral Fellowship, Jun. 2019 – Feb. 2022 (VOAMAS Project)
6. Tanmay Verlekar, Postdoctoral Fellowship, Dec. 2019 – Nov. 2020 (FIREFRONT Project)
7. Reza Javanmard Alitappeh, Postdoctoral Fellowship, Sep. 2020 – Nov. 2022 (ORIENT Project)
8. El Khalil Cherif, Apr. 2021 – Mar. 2022 (FIREFRONT Project), Apr. 2022 – Mar. 2023 (ISR Funding)
9. Rui Figueiredo, Postdoctoral Fellowship, Jun. 2021 – Aug. 2021 (HAVATAR Project)

7.5 Research Assistants

1. Julio Gomes, Research Initiation Grant, Apr. 2006 – Sept. 2006 (Robot-Cub Project)
2. Ana Faria, Research Initiation Grant, Oct. 2007 - Mar. 2008 (URUS Project)
3. Dario Figueira, Research Grant, Jan. 2009 – Jun. 2009 (URUS Project)
4. Matteo Taiana, Research Initiation Grant, Oct. 2008 – Jun. 2009 (BIOLOOK Project)
5. José Santos, Research Grant, Feb. 2009 – Aug. 2009 (BIOLOOK Project)
6. Daniela Pamplona, Research Initiation Grant, Mar. 2009 – Set. 2009 (BIOLOOK Project)
7. Christian Wressnegger, Research Grant, Jul. 2009 – Jun. 2010 (ISR Funding)
8. Nuno Moutinho, Research Initiation Grant, Feb. 2010 – Aug. 2010 (BIOLOOK Project)
9. Ashish Jain, Research Initiation Grant, Feb. 2010 – Aug. 2010 (BIOLOOK Project)
10. Pedro Osório, Research Initiation Grant, Nov. 2010 – Apr. 2011 (HANDLE Project)
11. Filipe Veiga, Research Initiation Grant, Nov. 2010 – Apr. 2011 (HANDLE Project)
12. Martim Brandão, Research Grant, Nov. 2010 – Apr. 2011 (ROBOSOM Project)
13. David Antunes, Research Grant, Oct. 2011 – Mar. 2012 (DICORE2S Project)

14. Jitesh Pramodray, Research Grant, Set. 2011 –Feb. 2012 (DICORE2S Project)
15. João Pimentel, Research Grant, Set. 2011 – Mar. 2013 (DICORE2S Project)
16. Bruno Nascimento, Research Grant, Set. 2011 – Feb. 2012 (DICORE2S Project)
17. Rui Figueiredo, Research Initiation Grant, Set. 2011 – Feb. 2012 (FIRST-MM Project)
18. Ricardo Santos, Research Initiation Grant, Set. 2011 – Aug. 2013 (FIRST-MM Project)
19. Pedro Vicente, Research Initiation Grant, Set. 2011 – Aug. 2013 (FIRST-MM Project)
20. João Gomes, Research Grant, Jan 2012 – Dec. 2012 (MAIS+S Project)
21. André Grilo, Research Grant, Feb. 2012 – Jan. 2013 (DICORE2S Project)
22. Ana Correia, Research Grant, Nov. 2012 – Aug. 2013 (DICORE2S Project)
23. Jaime Bernardo, Research Grant, Oct. 2012 – Jul. 2013 (DICORE2S Project)
24. Fábio Reis, Research Grant, Feb. 2012 – Jan. 2013 (DICORE2S Project)
25. João Costa, Research Grant, Set. 2012 – Mar. 2013 (Poeticon++ Project)
26. Afonso Gonçalves, Research Grant, May 2013 – Apr. 2014 (Poeticon++ Project)
27. Matteo Taiana, Research Grant, May 2013 – Apr. 2014 (Poeticon++ Project), May 2014 – Apr. 2015 (SEAGULL Project).
28. Rui Figueiredo, Research Grant, Aug. 2013 – Jul. 2014 (Poeticon++ Project)
29. Filipe Veiga, Research Grant, Aug. 2013 – Jul. 2014 (Poeticon++ Project)
30. Dario Figueira, Research Grant, Aug. 2013 – Dez. 2015 (Poeticon++ Project)
31. Ana Curralo, Research Grant, Jun. 2014 – Jan. 2015 (BIOMORPH Project)
32. David Ribeiro, Research Grant, Oct. 2014 - Dec. 2014 (ISR Funding INCENTIVO)
33. Nuno Moutinho, Research Grant, Feb. 2014 – Jul. 2014 (Poeticon++ Project)
34. Athira Nambiar, Research Grant, Feb. 2014 – Jul. 2014 (Poeticon++ Project)
35. Alexandre Antunes, Research Grant, Jul. 2014 – Nov. 2015 (Poeticon++ Project)
36. Andreia Moço, Research Grant, Jul. 2014 – Dec. 2014 (Poeticon++ Project)
37. Julio Alves, Research Initiation Grant, Apr. 2014 – Sep. 2014 (Poeticon++ Project)
38. Pedro Vicente, Research Grant, Feb. 2014 – Sep. 2015 (Poeticon++ Project)
39. Ricardo Santos, Research Grant, Feb. 2014 – Dec. 2015 (Poeticon++ Project)
40. Carlos Cardoso, Research Grant, Dec. 2014 – Dec 2015 (Poeticon++ Project)
41. Christian Vismara, Research Grant, Oct. 2014 – Oct. 2016 (AHA Project)
42. Hugo Simão, Research Grant, Sep. 2016 - May 2019 (LARSyS Funding)
43. João Domingos, Research Grant, Sep. 2016 - Nov. 2016 (LARSyS Funding)
44. João Lages, Research Initiation Grant, Jun. 2017 – Nov. 2017 (LARSyS Funding)
45. Giovanni Saponaro, Research Grant, Set. 2017 – Feb. 2018 (LARSyS Funding)
46. Tiago Paulino, Research Grant, Nov. 2017 – Apr. 2018 (LARSyS Funding)
47. Ana Luísa Fernandes, May 2018 – Jul. 2018 (AHA Project)

48. Heitor Cardoso, Research Grant, Jan 2018 – Dec 2018 (AHA Project), Jan 2019 – Jun 2019 (ISR Funding)
49. João Avelino, Research Grant, Jan 2016 – Sep. 2016 (AHA Project)
50. João Salvado, Research Grant, Out 2016 – Mar 2017 (AHA Project)
51. Nino Cauli, Research Grant, Mar 2015 – Aug 2015 (AHA Project)
52. Rui Garcia Figueiredo, Research Grant, Mar 2018 – Dec 2018 (AHA Project)
53. José Coelho, Research Initiation Grant, Oct 2018 – Dec 2018 (AHA Project)
54. Manuel Reis, Research Initiation Grant, Oct 2018 – Dec 2018 (AHA Project)
55. João Alves, Research Initiation Grant, Oct 2018 – Dec 2018 (AHA Project)
56. Rodrigo Zenha, Research Grant, Sep. 2018 – Dec 2018 (LARSyS Funding)
57. Pedro Abreu, Research Initiation Grant, May 2017 – Sep. 2018 (SPARSIS Project)
58. Isabel Silva, Research Initiation Grant, Sep. 2017 – Feb. 2018 (SPARSIS Project)
59. Duarte Dornellas, Research Grant, Sep. 2018 – Feb. 2019 (ELEVAR Project)
60. Jorge Gonçalves, Research Grant, Sep. 2018 – Feb. 2019 (ELEVAR Project)
61. Mariana Martins, Research Grant, Jun. 2018 – Sep. 2019 (ORIENT Project)
62. Carlos Aleluia, Research Grant, Jun. 2018 – Sep. 2019 (ORIENT Project)
63. Tiago Mendes, Research Grant, Apr. 2022 – Sep. 2022 (VOAMAI Project)
64. Tiago Garcia, Research Research Grant, Apr. 2022 – Jan. 2023 (VOAMAI Project)
65. Ana Rita Córias, Research Grant, Jul. 2021 – Dec. 2021 (ISR Funding)
66. Min Hun Lee, Research Grant, Jul. 2021 – Dec. 2021 (ISR Funding)
67. Bernardo Dias, Research Grant, Dec. 2019 – Mar. 2022 (ORIENT Project)
68. Rui Cardoso, Research Grant, Dec. 2019 – Feb. 2020 (ORIENT Project)
69. Alexandre Dias, Research Grant, Sep. 2022 – Feb. 2023 (HAVATAR Project)
70. Ayman Bnousaad, Research Grant, Apr. 2023 – Jun. 2023 (ASTRIIS Project)
71. Mariana Alves, Research Grant, Apr. 2023 – Jun. 2023 (ASTRIIS Project)
72. Henrique Granado, Research Grant, Apr. 2022 – Dec. 2022 (ORIENT Project)
73. José Gomes, Research Grant, Jun. 2022 – Nov. 2022 (ORIENT Project)
74. Tiago Fonseca, Research Grant, Mar. 2022 – Aug. 2022 (HAVATAR Project)
75. Francisco Faria, Research Grant, Mar. 2022 – May. 2023 (SHITF-HRI Project)
76. Henrique Fernandes, Research Grant, Sep. 2022 – May 2023 (HAVATAR Project)
77. Bruno Miguel da Silva Cebola, Research Grant, Jul. 2023 – May 2024 (Thales Funding)
78. Chaimaa Oulad Dahman, Research Grant, Jul. 2024 – Dez 2024 (HAVATAR Project)
79. Gonçalo Nunes, Research Grant, Jul. 2024 – Dez 2024 (HAVATAR Project)
80. Vinicius Meneses Souza, Research Grant, Apr. 2025 – Jan 2025 (CRAI Project)
81. Hugo Matias, Research Grant, Apr. 2025 – Sep. 2025 (Own Research Funds)

82. João Francisco Felizardo, Research Grant, Mar 2025 – Feb 2026 (CRAI Project)
83. Miguel Estevens Teixeira, Research Grant, Mar 2025 – Dez 2025 (CRAI Project)
84. João Paiva de Castelo Branco, Research Grant, March 2026 - (Own Research Funds)

7.6 Visiting Students

1. Lorenz Gerstmayr, Universität Tübingen, Germany, BSc student, 2004.
2. Anshul São, IIT Karagphur, India, Summer Internship, 2006.
3. Jonas Ruesch, University of Zurich, Switzerland, RobotCub Project Internship, 2007.
4. Manuel Marín, Universidad de Granada, Spain, PhD student, 2006.
5. Jorge Grande Calvo, Universidad de Valencia, Spain, BSc, 2006.
6. Verica Kronic, University of Belgrade, Serbia, IAESTE Internship, 2007-2008.
7. Bartosz Tworek, University of Technology and Science, Krakow, Poland, IAESTE Internship, 2007.
8. Jurgen Leitner, University of Vienna, Austria, Summer Internship, 2007.
9. José Pena Pastor, Politecnico de Madrid, Spain, Summer Intership, 2007.
10. Mohit Khurana, IIT Guwahati, India, Summer Intership, 2008.
11. Ivana Cingovska, Univ. St. Cyril and Methodius, Macedonia, IAESTE Internship, 2008.
12. Carla González, Univ. Carlos III de Madrid, Spain, Ph.D student 2009.
13. Afshin Deghan, Univ. Teeran, Iran, IAESTE Internship, 2009.
14. Ashish Jain, Manipal Institute of Technology, India, IAESTE Internship, 2010.
15. Kristijan Petkov, Univ. St. Cyril and Methodius, Macedonia, IAESTE Intership, 2010.
16. Adam Owczarek, Univ. Lodz, Poland, IAESTE Internship, 2011.
17. Kira Neubehler, KIT, Germany, IAESTE Internship, 2012.
18. Huu Quang Nguyen, ULB, Brussels, Belgium, IAESTE Internship, 2014.
19. Alberto Sabater Bailón, University of Zaragoza, PhD student, 2019.
20. Maarten Jongeneel, TU/e, BSc Student, 2019.
21. Stefan van der Palen, TU/e, BSc Student, 2021.
22. Sara Marqués Villarroja, Universidad Carlos III de Madrid, PhD Student 2021.
23. Seyedsaber Mohammadi, Italian Institute of Technology, PhD student, 2022.

8 Publications

In this section I list the scientific publications I have (co-)authored and have been validated by the scientific community:

- Papers in peer-reviewed international scientific journals, indexed common journal indexing databases (Web of Science, Scopus, Crossref, Google Scholar).

- Papers in peer-reviewed international conference and symposia with formal program committees.
- Chapter in edited books, typically with invitation and revision of the editors, to present material relevant to the topic of the book.
- Other papers in peer-reviewed international meeting, where I included both short papers and abstracts that were presented in peer-reviewed conferences and symposia, as well as full papers in workshops and other scientific events with *ad hoc* program committees.

For all published articles, I include the digital object identifier (DOI). If DOI is not available, and alternative link to other related web resources will be provided. If some data is not available, will be denoted as NA.

8.1 Papers in Peer-Reviewed International Scientific Journals

1. M. Osório, A. Bernardino, A. Wichert (2026) “Shape matters: Few-shot object classification from high-information contour features”, *Neural Computation*, accepted.
2. M. R. Lima, C. Pombo, C. Gameiro, B. Lopes, H. Cardoso, A. Bernardino, M. V. S. Nunes (2026) “Development of personalized exercogs for older adults care: A feasibility and single-arm study”, *International Journal of Clinical and Health Psychology*, 26(2), <https://doi.org/10.1016/j.ijchp.2026.100674>
3. V. Souza, J. Gaspar, J. F. P. Fernandes, A. Bernardino (2026) “Localization for car-like vehicles using loosely coupled dual-GNSS integration”, *Discover Vehicles* 2(1). <https://doi.org/10.1007/s44465-026-00004-5>
4. X. Liu, F. A. Longchamp, L. Zunino, L. Gevers, L. R. Schneider, S.I. Bothner, A. Guignard, A. Crespi, G. Bellegarda, A. Bernardino, E. A. Naumann, A. J. Ijspeert (2026) “Energy efficiency and neural control of continuous versus intermittent swimming in a fishlike robot”, *Science Robotics* 11(110). <https://doi.org/10.1126/scirobotics.adw7868>
5. J. Luzio, A. Bernardino, P. Moreno (2026) “SemBA-FAST: Semantic-based Bayesian attention applied to foveal active visual search tasks”, *Neurocomputing*. <https://doi.org/10.1016/j.neucom.2026.132860>
6. R. Zenha, B. Denoun, A. Cavallaro, A. Bernardino, L. Jamone (2025) “Let’s DENSE: a novel protocol for efficiently collecting dense and diverse data for tactile slip detection in robotic grasping”, *npj Robot*, 3, 36. <https://doi.org/10.1038/s44182-025-00055-y>
7. A. R. Córias, M. Hun Lee, A. Bernardino, A. Smailagic, M. Mateus, D. Fernandes, S. Trapola (2025) “Learning Frame-Level Classifiers for Video-Based Real-Time Assessment of Stroke Rehabilitation Exercises From Weakly Annotated Datasets” in *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 33, pp. 3334-3345. <https://doi.org/10.1109/TNSRE.2025.3602548>
8. A. Pires, A. Bernardino, G. Victorino, J. Miguel Costa, C. M. Lopes, J. Santos-Victor, (2025) “Scalable thermal imaging and processing framework for water status monitoring in vineyards”, *Computers and Electronics in Agriculture*, Volume 239, Part A, <https://doi.org/10.1016/j.compag.2025.110931>
9. X. Liu, M. D. Loring, L. Zunino, K. E. Fouke, F. A. Longchamp, A. Bernardino, A. J. Ijspeert, E. A. Naumann (2025) “Artificial Embodied Circuits Uncover Neural Architectures of Vertebrate Visuomotor Behaviors”, *Science Robotics*, 10, <https://doi.org/10.1126/scirobotics.adv4408>

10. F. Mêda, F. Näf, T. Fernandes, A. Bernardino, L. Jamone, G. Tavares, S. Cardoso (2025) “Skin-Inspired Magnetoresistive-based Tactile Sensor for Force Characterization in Distributed Areas”, *Sensors*, 25(12):3724. <https://doi.org/10.3390/s25123724>
11. H. Harkat, H. Ahmed, J. M.P. Nascimento, A. Bernardino, (2025) “Early fire detection using wavelet based features”, *Measurement*, 242A, 115881. <https://doi.org/10.1016/j.measurement.2024.115881>
12. J. Avelino, P. Moreno, F. Correia, L. Garcia-Marques, R. Ventura, A. Bernardino (2025), “Evaluating Greeting Models for Human-Robot Engagement During First Encounters”, *International Journal of Social Robotics*, 17:297-314. <https://doi.org/10.1007/s12369-025-01213-w>
13. S. Ferreira, J. Marmeleira, J. Del Pozo Cruz , N. Leite, A. Bernardino, A. Moradell, A. Raimundo (2024), “Effects of an exercise program with augmented reality on functional fitness and physical activity of community-dwelling older adults”, *Frontiers in Sports and Active Living*, 6. <https://doi.org/10.3389/fspor.2024.1447866>
14. A. Prakash, S. Thejaswin, A. Nambiar, A. Bernardino (2024), “Exploring Fusion Techniques and Explainable AI on Adapt-FuseNet: Context-adaptive Fusion of Face and Gait for Person Identification”, *IEEE Transactions on Biometrics, Behavior, and Identity Science*, 6(4):515-527. <https://doi.org/10.1109/TBIOM.2024.3405081>
15. H. Simão, D. Gonçalves, A. C. Pires, L. Abreu, A. Bernardino, J. Forlizzi, T. Guerreiro (2024) “I want to send a message to my friend: Exploring the Shift of Agency to Older Adults in HRI”, *International Journal of Social Robotics*, 16:1721-1734. <https://doi.org/10.1007/s12369-024-01128-y>
16. A. J. Van Opstal, R. J. Alitappeh, A. John and A. Bernardino (2024), “Realistic 3D Human Saccades Generated by a 6-DOF Biomimetic Robotic Eye Under Optimal Control”, *Frontiers in Robotics and AI, Sec Robot Vision and Artificial Perception*, vol. 11, <https://doi.org/10.3389/frobt.2024.1393637>
17. L. Mendes, A. Ricardo, A. Bernardino, and R. Ferreira (2024) “A Hybrid PIV/Optical Flow Method for Incompressible Turbulent Flows”, *Water* 16, no. 7: 1021. <https://doi.org/10.3390/w16071021>
18. R. J. Alitappeh, A. John, B. Dias, A. J. van Opstal and A. Bernardino (2024) “Emergence of human oculomotor behavior in a cable-driven biomimetic robotic eye using optimal control,” in *IEEE Transactions on Cognitive and Developmental Systems*, 6(4):1546-1560. <https://doi.org/10.1109/TCDS.2024.3376072>
19. S. Ferreira, A. Raimundo, J. del Pozo-Cruz, A. Bernardino, N. Leite, H. M. Yoshida, & J. Marmeleira (2024), Effects of Multimodal Exercise With Augmented Reality on Cognition in Community-Dwelling Older Adults. *Journal of the American Medical Directors Association*, 25(6). <https://doi.org/10.1016/j.jamda.2024.01.023>
20. N. Greggio, A. Bernardino (2024), “Unsupervised incremental estimation of Gaussian mixture models with 1D split moves”, *Pattern Recognition*, 150, 110306. <https://doi.org/10.1016/j.patcog.2024.110306>
21. N. Martins, J.S. Silva, A. Bernardino (2024), “Fingerprint Recognition in Forensic Scenarios”. *Sensors*, 24, 664. <https://doi.org/10.3390/s24020664>
22. H. Badda, E.K. Cherif, H. Boulaassal, M. Wahbi, O. Yazidi Alaoui, M. Maatouk, A. Bernardino, F. Coren, O. El Kharki (2023), “Improving the Accuracy of Random Forest

- Classifier for Identifying Burned Areas in the Tangier-Tetouan-Al Hoceima Region Using Google Earth Engine”, *Remote Sensing*, 15(17):4226. <https://doi.org/10.3390/rs15174226>
23. T. Marto, A. Bernardino, G. Cruz (2023), “Fire and Smoke Segmentation Using Active Learning Methods”, *Remote Sensing*, 15(17):4136. <https://doi.org/10.3390/rs15174136>
 24. R. P. de Figueiredo, A. Bernardino (2023), “An overview of space-variant and active vision mechanisms for resource-constrained human inspired robotic vision”, *Autonomous Robots*, pp 1-17. <https://doi.org/10.1007/s10514-023-10107-7>
 25. N. Pessanha-Santos, V. Lobo, A. Bernardino (2023), “Fixed-Wing Unmanned Aerial Vehicle 3D-Model-Based Tracking for Autonomous Landing”, *Drones*, 7(4):243, pp 1-25. <https://doi.org/10.3390/drones7040243>
 26. J. Correia, A. Bernardino and R. Ribeiro (2023), “Learning Performance Models of Distributed Computer Vision Methods for Decision Making in Detection and Tracking Algorithms in UAVs”, *IEEE Internet of Things Journal*, 10(14), pp 12486-12495. <https://doi.org/10.1109/JIOT.2023.3247589>
 27. T. Garcia, R. Ribeiro, A. Bernardino (2023), “Wildfire aerial thermal image segmentation using unsupervised methods: a multilayer level set approach”, *International Journal of Wildland Fire*, 32(3), pp 435-447. <https://doi.org/10.1071/WF22136>
 28. M. Hun Lee, D. Siewiorek, A. Smailagic, A. Bernardino, S. Bermúdez i Badia (2023), “Design, development, and evaluation of an interactive personalized social robot to monitor and coach post-stroke rehabilitation exercises”, *User Modeling and User-Adapted Interaction*, 33, pp 545–569. <https://doi.org/10.1007/s11257-022-09348-5>
 29. D. Dias, J. Silva, A. Bernardino (2023), “The Prediction of Road-Accident Risk through Data Mining: A Case Study from Setubal, Portugal”, *Informatics*, 10(1), 17, pp 1-15. <https://doi.org/10.3390/informatics10010017>
 30. H. Harkat, J. M. P. Nascimento, A. Bernardino, H. F. T. Ahmed (2023), “Fire images classification based on a handcraft approach”, *Expert Systems with Applications*, 212, pp 1-13. <https://doi.org/10.1016/j.eswa.2022.118594>
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8.2 Papers In Peer-Reviewed International Conference Proceedings

1. J. Luzio, A. Bernardino and P. Moreno (2026) “Cost-Efficient Multi-Scale Fovea for Semantic-Based Visual Search Attention”, International Joint Conference on Neural Networks (IJCNN’26), accepted.
2. J. Castelo-Branco, A. Bernardino and J. Santos-Victor (2026) “Integrating Deep RL and Bayesian Inference for Object Search in Mobile Robotics”, *IEEE International Conference*

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3. A. Teodósio, A. Bernardino and P. Moreno (2026) “The Role of Trajectory Prediction in the Perception of Approaching Agents”, *IEEE International Conference on Autonomous Robot Systems and Competitions (ICARSC'26)*, accepted
4. I. C. Paiva, A. Bernardino and P. Moreno (2026) “Dual Field of View Gaze Estimation with Knowledge Distillation for Mobile Robots”, *IEEE International Conference on Autonomous Robot Systems and Competitions (ICARSC'26)*, accepted
5. M. Campos, N. Pessanha Santos and A. Bernardino (2026) “Vision-Based Unmanned Aerial Vehicle Pose Correction Using a Digital Model”, *IEEE International Conference on Autonomous Robot Systems and Competitions (ICARSC'26)*, accepted
6. D. Silva, A. Bernardino, G. Cruz (2026) “Conformal Aerial Geo-Localization Through Visual Place Recognition” *International Conference on Unmanned Aircraft Systems (ICUAS)*, accepted
7. H. Simão, D. Gonçalves, N. M. Khanuja, V. Nisi, A. Bernardino, T. Guerreiro, and J. Forlizzi (2026) “Values Across Contexts: Understanding How Older Adults Enact What Matters Through Technology”. In *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI '26)* 257, 1–16. <https://doi.org/10.1145/3772318.3791780>
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8.5 Papers in Peer-Reviewed National Journals or Conference Proceedings

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2. P. R. Martins, J. S. S., A. Bernardino (2021), “Face Detection and Alignment Using On-the-Wild Multispectral Images”, *Portuguese Conference on Pattern Recognition (RecPad 2021)*, 2 pages. <https://recpad2021.uevora.pt/wp-content/uploads/2021/12/proceedings-recpad2021.pdf>

3. L. Pereira, B. Ferreira and A. Bernardino (2020), “Real-Time 3D Tracking of Simple Objects with an RGB Camera”, *Portuguese Conference on Pattern Recognition (RecPad 2020)*, 2 pages. https://recpad2020.uevora.pt/wp-content/uploads/2020/10/RECPAD_2020_paper_36.pdf
4. L. Chambino, J. Silva and A. Bernardino (2020), “Multispectral Images Applied to Face Recognition”, *Portuguese Conference on Pattern Recognition (RecPad 2020)*, 2 pages. https://recpad2020.uevora.pt/wp-content/uploads/2020/10/RECPAD_2020_paper_3.pdf
5. A. R. Córias and A. Bernardino (2020), “Assessment of Motor Compensation Patterns in Stroke Rehabilitation Exercises”. *Portuguese Conference on Pattern Recognition (RecPad 2020)*, 2 pages. https://recpad2020.uevora.pt/wp-content/uploads/2020/10/RECPAD_2020_paper_32.pdf
6. A. Filipe, A. Bernardino and P. Moreno (2020), “Learning to Grasp Objects in Virtual Environments through Imitation”, *Portuguese Conference on Pattern Recognition (RecPad 2020)*, 2 pages. https://recpad2020.uevora.pt/wp-content/uploads/2020/10/RECPAD_2020_paper_26.pdf
7. G. Cunha, A. Bernardino, P. Vicente, R. Ribeiro and P. Moreno (2020), “Active robot Learning for Efficient Body-Schema Online Adaptation”. *Portuguese Conference on Pattern Recognition (RecPad 2020)*, 2 pages. https://recpad2020.uevora.pt/wp-content/uploads/2020/10/RECPAD_2020_paper_23.pdf
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11. C. Pires, A. Bernardino and B. Damas (2020), “Ship Segmentation in Aerial Images for Maritime Surveillance”. *Portuguese Conference on Pattern Recognition (RecPad 2020)*. https://recpad2020.uevora.pt/wp-content/uploads/2020/10/RECPAD_2020_paper_25.pdf
12. M. J. Henriques, T. Pinto, J. Andrade, D. Dornellas, J. Gonçalves, R. Ribeiro, A. Bernardino, J. Santos-Victor, A. Lopes Baptista (2019), “Project ELEVAR – Study of Vertical Structures with Robotized Aircrafts. A Contribution to Visual Inspections of Engineering Works”, *Congresso de Ensaios e Experimentação em Engenharia Civil (TESTE 2019)*, abstract. <https://doi.org/10.5281/zenodo.3355354>
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33. A. Bernardino, C. Silva, J. Santos-Victor, C. Pinto-Ferreira (1994), “Uma Abordagem Biológica em Visão Activa”, *Actas do 1º Encontro Português de Controlo Automático* (in Portuguese).

8.6 Theses

1. Binocular Head Control with Foveal Vision: Methods and Applications, *Ph.D. Thesis*, Instituto Superior Técnico, Universidade Técnica de Lisboa, 2004.
2. Seguimento Binocular de Alvos Móveis Baseado em Imagens Log-Polar (In Portuguese), *M.Sc. Thesis*, Instituto Superior Técnico, Universidade Técnica de Lisboa, 1996.

9 Impact Factors

9.1 Citations

My overall publication impact was analyzed by using the citation reports of Clarivate’s Web of Science, Elsevier’s Scopus, and Google Scholar, according to the profiles and reports summarized in Table 1 and Figures 1, 2 and 3. My 10 most cited publications and corresponding citations in the last few years are shown in Figure 4.

Table 1 – Profiles and citation metrics in Web of Science, Scopus, and Google Scholar.

	Web of Science	Scopus	Google Scholar
Profile	G-1316-2010	7003407125	auFmh9MAAAAJ
Report (27/07/2023)	https://drive.google.com/file/d/1hlkSJCjBu8D3v9_0Wbu6doJAYw7LKtcr/view?usp=sharing	https://drive.google.com/file/d/1htk2NB3StvbmR1v1UvC81yPWpuaxZbX4/view?usp=sharing	https://drive.google.com/file/d/1hxnGDIEB62qJCfri5TclwdJzn6aqNF8/view?usp=sharing

	Web of Science	Scopus	Google Scholar
Number of articles with citation data	269	296	401
Number of citations	3133	4686	7875
Number of citations excluding self-citations	2637	NA	NA
h-index	24	30	42

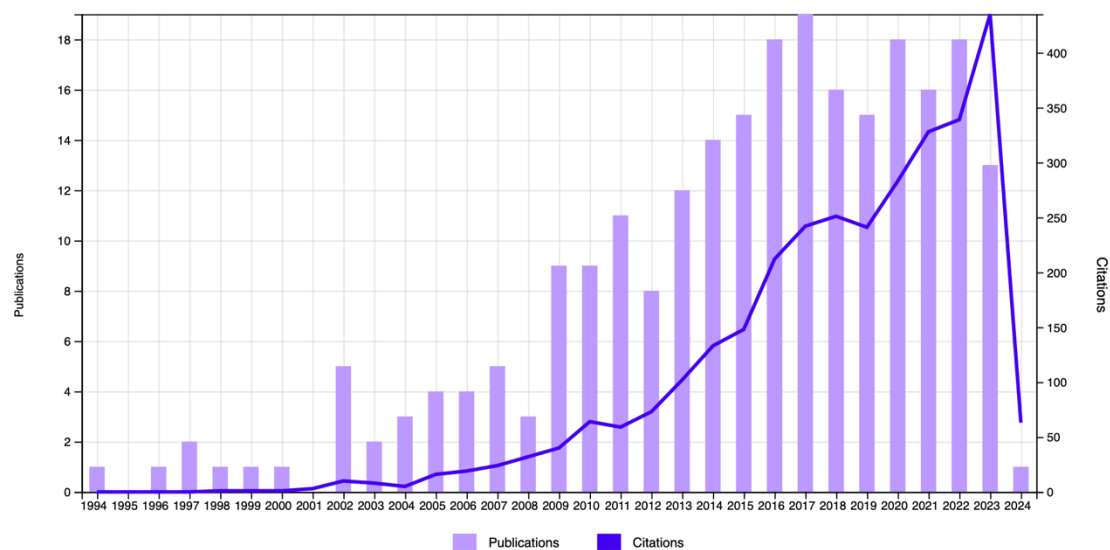


Figure 1 – Web of Science: Citations and publications over time

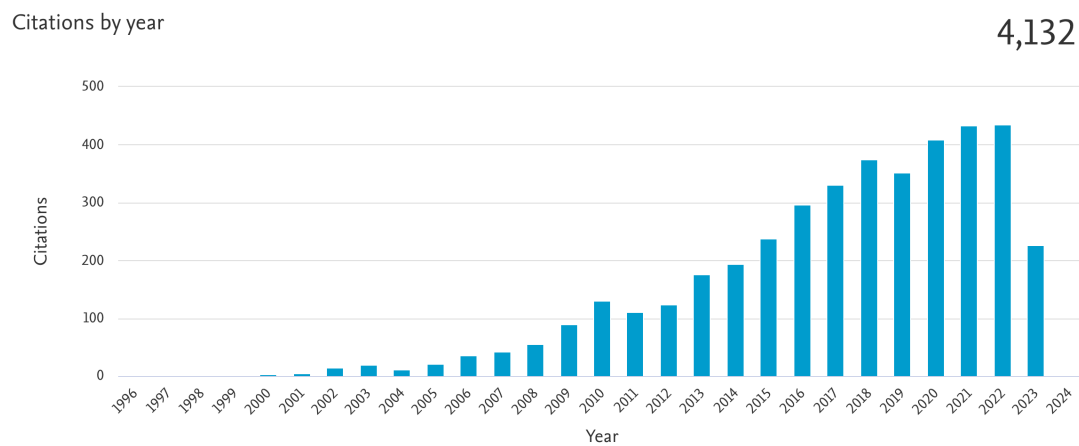


Figure 2 – Scopus : Citations per year

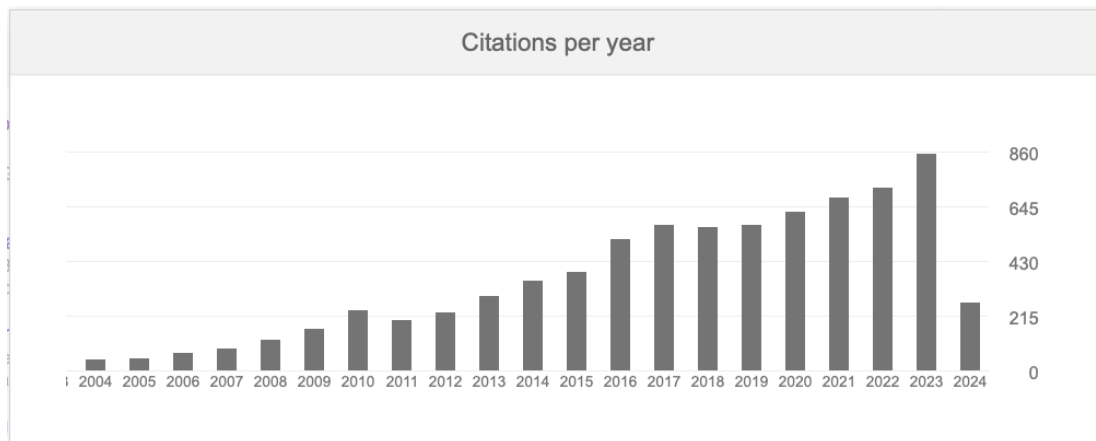


Figure 3: Google Scholar: Citations per year.

245 Publications				Sort by: Citations: highest first ▾				< 1 of 5 >				Citations							
												< Previous year					Next year >	Average per year	Total
												2020	2021	2022	2023	2024			
Total												283	328	339	435	63	116.04	3,133	
1 The iCub humanoid robot: An open-systems platform for research in cognitive development Metta, G. , Natale, L. (—); Montesano, L. Oct-nov 2010 NEURAL NETWORKS 23 (8-9), pp.1125-1134												31	37	22	22	5	24.6	369	
2 Learning object affordances: From sensory-motor coordination to imitation Montesano, L. , Lopes, M. (—); Santos-Victor, J. Feb 2008 IEEE TRANSACTIONS ON ROBOTICS 24 (1), pp.15-26												15	7	10	9	1	13.24	225	
3 Design of the Robot-Cub (iCub) head Belra, R. , López, M. (—); Saltarén, R. IEEE International Conference on Robotics and Automation (ICRA) 2006 2006 IEEE INTERNATIONAL CONFERENCE ON ROBOTICS AND AUTOMATION (ICRA), VOLS 1-10, pp.94-+												7	5	5	8	0	6.26	119	
4 Matrix Completion for Weakly-Supervised Multi-Label Image Classification Cabral, R. , De la Torre, F. (—); Bernardino, A. Jan 2015 IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE 37 (1), pp.121-135												16	12	11	6	3	11.5	115	
5 Unifying Nuclear Norm and Bilinear Factorization Approaches for Low-rank Matrix Decomposition Cabral, R. , De la Torre, F. (—); Bernardino, A. IEEE International Conference on Computer Vision (ICCV) 2013 2013 IEEE INTERNATIONAL CONFERENCE ON COMPUTER VISION (ICCV), pp.2488-2495												16	17	12	11	2	9.58	115	
6 A review of log-polar Imaging for visual perception in robotics Traver, V.J. and Bernardino, A. Apr 30 2010 ROBOTICS AND AUTONOMOUS SYSTEMS 58 (4), pp.378-398												9	9	4	5	0	7.13	107	
7 Mosaic-based navigation for autonomous, underwater vehicles Gracias, N.R. , van der Zwaan, S. (—); Santos-Victor, J. Oct 2003 IEEE JOURNAL OF OCEANIC ENGINEERING 28 (4), pp.609-624												6	2	3	2	0	4.27	94	
8 Detection and classification of highway lanes using vehicle motion trajectories Melo, J. , Nafati, A. (—); Santos-Victor, J. Jun 2006 IEEE TRANSACTIONS ON INTELLIGENT TRANSPORTATION SYSTEMS 7 (2), pp.188-200												6	6	1	2	2	4.84	92	
9 Affordances in Psychology, Neuroscience, and Robotics: A Survey Jamone, L. , Ugur, E. (—); Santos-Victor, J. Mar 2018 IEEE TRANSACTIONS ON COGNITIVE AND DEVELOPMENTAL SYSTEMS 10 (1), pp.4-25												16	21	13	14	3	12	84	
10 Multimodal saliency-based bottom-up attention a framework for the humanoid robot iCub Ruesch, J. , Lopes, M. (—); Pfeifer, R. IEEE International Conference on Robotics and Automation 2008 2008 IEEE INTERNATIONAL CONFERENCE ON ROBOTICS AND AUTOMATION, VOLS 1-9, pp.962-+												10	3	2	3	1	4.06	69	

Figure 4 -Web of Science : 10 most cited publications.

9.2 Journal Impact Factors

Scientific journals are indexed in services that account for several impact indicators that are typically computed from the number of citations received by the journal’s published articles. Two of the most renowned journals ranking services are Clarivate Analytics² Web of Science Journal Citation Reports (JCR) and the SCImago Lab’s Journal & Country Rank (SJR). These services use different types of metrics to access the relative importance of a journal within its field. Although there is a large debate about the use of these indexes to assess scientific work, they carry some value to broadly consider the quality of the reviewing process in the different journals.

The journals where my papers have been published are listed in Table 2. For each journal we include the following metrics, when available, obtained from the Journal Citation Reports (JCR) and the SCImago Journal Ranks (SJR):

- The 2022 Journal Impact Factor (JIF³) obtained from the JCR.
- The 5-Year Journal Impact Factor⁴ obtained from JCR.
- The 2022 Journal Citation Index (JCI⁵), obtained from JCR.
- The categories where the journal is indexed that are most related to my research area in JCR and SJR.
- The rank of the journal in each category according to JCR.
- The quartile of the journal in each category according to JCR (computed with the JIF or, if not available, the JCI) and SJR (computed with the SJR2 indicator⁶).
- Number of papers I have published in those journals.

Whenever the JCR impact factor is not available, the journal ranking quartiles reported are obtained by JCI, and displayed *in italic font*.

Journal	2022 JIF	5- year JIF	2022 JCI	Category @ JCR (SJR)	Rank@ JCR categ.	Quart @ JCR (SJR)	#
Autonomous Robots	3.5	4.6	0.72	Robotics (Artif. Intel.)	15/30	Q2 (Q1)	2
Drones	4.8	5.5	0.92	Remote Sensing (Artif. Intel.)	14/33	Q2 (Q2)	1
IEEE Internet of Things Journal	10.6	11.1	2.61	EE Eng. (Signal Proc.)	13/275	Q1 (Q1)	1

²Formerly known as The Institute for Scientific Information (ISI).

³The ratio of the number of current year citations to the source items published in that journal during the previous two years (<https://clarivate.com/webofsciencegroup/essays/impact-factor/>)

⁴The ratio of the number of citations in the previous 5 years to the number of articles published in the same period (<https://clarivate.com/webofsciencegroup/essays/impact-factor/>)

⁵The Journal Citation Indicator provides a field-normalized measure of citation impact where a value of 1.0 means that, across the journal, published papers received a number of citations equal to the average citation count in that category (https://clarivate.com/wp-content/uploads/dlm_uploads/2021/05/Journal-Citation-Indicator-discussion-paper-2.pdf)

⁶Vicente P. Guerrero-Bote, Félix Moya-Anegón (2012), “A further step forward in measuring journals’ scientific prestige: The SJR2 indicator”, *Journal of Informetrics*, 6(4):674-688.

Journal	2022 JIF	5- year JIF	2022 JCI	Category @ JCR (SJR)	Rank@ JCR categ.	Quart @ JCR (SJR)	#
International Journal of Wildland Fire	3.1	3.8	1.19	Forestry (Forestry)	13/69	Q1 (Q1)	1
User Modeling and User-Adapted Interaction	3.6	5.7	0.58	Cybernetics (CS Applic.)	10/24	Q2 (Q1)	1
Informatics	3.1	2.7	0.53	Interdiscipl. App. (Comp. Networks)	97/164	Q3 (Q2)	1
Expert Systems with Applications	8.5	8.3	1.70	EE Eng. (Artif. Intel.)	23/275	Q1 (Q1)	1
Remote Sensing	5.0	5.6	1.02	Remote Sensing (Earth Sciences)	12/33	Q2 (Q1)	3
Frontiers in Robotics and AI	3.4	3.9	0.67	Robotics (Artif. Intel.)	23/42	Q3 (Q2)	4
International Journal of Environmental Research and Public Health	4.6	4.8	0.93	Public Health (Public Health)	71/201	Q2 (Q2)	1
Sensors	3.9	4.1	0.89	EE Eng. (EE Eng.)	100/275	Q2 (Q2)	7
Journal of Neuroengineering and Rehabilitation	5.1	5.8	1.35	Biomedical Eng. (Health Informatics)	26/96	Q2 (Q1)	1
International Journal of Social Robotics	4.7	4.9	1.05	Robotics (EE Eng.)	12/30	Q2 (Q1)	3
IEEE Access	3.9	4.1	0.89	EE Eng. (Engineering)	100/275	Q2 (Q1)	3
Experiments in Fluids	2.4	2.7	0.64	Mechanics (Comp. Mech.)	70/133	Q3 (Q1)	1
Robotics and Autonomous Systems	4.3	4.6	0.84	Robotics (Ctr & Syst. Eng.)	13/30	Q2 (Q1)	9
PLOS Computational Biology	4.3	5.4	1.10	Comp. Biol. (Mod. & Sim.)	10/55	Q1 (Q1)	1
Journal of Intelligent & Robotic Systems	3.3	3.4	0.70	Robotics (EE Eng.)	17/30	Q3 (Q1)	5
Marine Technology Society Journal	0.8	0.9	0.21	Ocean Eng. (Ocean Eng.)	14/16	Q4 (Q3)	1
Proceedings of the ACM on Human-Computer Interaction	NA	NA	NA	NA (HCI)	NA	NA (Q2)	1

Journal	2022 JIF	5- year JIF	2022 JCI	Category @ JCR (SJR)	Rank@ JCR categ.	Quart @ JCR (SJR)	#
SoftwareX	3.4	3.3	0.88	Software Eng.	38/108	Q2	1
Journal of Field Robotics	8.3	7.5	1.52	Robotics (Ctr & Syst. Eng.)	5/30	Q1 (Q1)	1
IEEE Robotics and Automation Letters	5.2	5.9	1.23	Robotics (Ctr & Syst. Eng.)	10/30	Q2 (Q1)	4
IEEE Transactions on Cognitive and Developmental Systems	5.0	4.8	1.13	Robotics (Artif. Intel.)	11/30	Q2 (Q1)	3
International Journal of Robotics Research	9.2	9.6	1.80	Robotics (EE Eng.)	4/30	Q1 (Q1)	1
IEEE Transactions on Circuits and Systems for Video Technology	8.4	7.1	1.89	EE Eng. (EE Eng.)	25/275	Q1 (Q1)	1
ACM Computing Surveys	16.6	18	4.16	Comp. Science (Comp. Science)	3/111	Q1 (Q1)	1
IEEE Transactions on Geoscience and Remote Sensing	8.2	8.8	1.9	EE Eng. (EE Eng.)	27/275	Q1 (Q1)	1
Paladyn Journal of Behavioural Robotics	NA	NA	NA	NA (Artif. Intel.)	NA	NA (Q3)	1
Advanced Robotics	2	2.6	0.46	Robotics (Ctr & Syst. Eng.)	26/30	Q4 (Q2)	1
International Journal of Advanced Robotic Systems	2.3	2.3	0.41	Robotics (Artif. Intel.)	24/30	Q4 (Q3)	2
IEEE Transactions on Magnetics	2.1	2.0	0.46	EE Eng. (EE Eng.)	179/275	Q3 (Q2)	1
Mechanism and Machine Theory	5.2	5.2	1.37	Mechanical Eng. (CS Applic.)	19/135	Q1 (Q1)	1
Pattern Recognition	8.0	8.4	1.66	Artif. Intel. (Artif. Intel.)	25/145	Q1 (Q1)	1
IEEE Transactions on Systems, Man and Cybernetics : Systems	8.7	9.5	2.18	Aut. & Ctr. Systems (EE Eng.)	6/65	Q1 (Q1)	1

Journal	2022 JIF	5- year JIF	2022 JCI	Category @ JCR (SJR)	Rank@ JCR categ.	Quart @ JCR (SJR)	#
IEEE Robotics & Automation Magazine	5.7	6.4	1.04	Autom. & Ctr. Syst. (Ctr. & Syst. Eng.)	15/65	Q1 (Q1)	1
Methods of Information in Medicine	1.7	2.5	0.4	Information Systems (Health Informatics)	128/158	Q4 (Q3)	1
Computer Vision and Image Understanding	4.5	5.2	0.98	Artif. Intel. (CV & PR)	62/145	Q2 (Q1)	1
International Journal of Pattern Recognition and Artificial Intelligence	1.5	1.4	0.23	Artif. Intel. (Artif. Intel.)	126/145	Q4 (Q3)	1
Pattern Recognition Letters	5.1	4.8	0.87	Artif. Intel. (Artif. Intel.)	52/145	Q2 (Q1)	2
Neurocomputing	6.0	6.0	1.01	Artif. Intel. (Artif. Intel.)	41/145	Q2 (Q1)	2
IEEE Transactions on Pattern Analysis and Machine Intelligence	23.6	26.7	6.76	Artif. Intel. (Artif. Intel.)	2/145	Q1 (Q1)	1
International Journal of Machine Intelligence and Sensory Signal Processing	NA	NA	NA	NA (NA)	NA	NA (NA)	1
Adaptive Behavior	1.6	1.5	0.54	Artif. Intel. (Artif. Intel.)	124/145	Q4 (Q3)	1
IEEE Transactions on Autonomous Mental Development (2017)	2.9	2.16	NA	Artif. Intel. (NA)	30/132	Q1 (NA)	1
IEEE Transactions on Systems, Man, and Cybernetics, Part B: Cybernetics (2014)	6.2	6.2	NA	Autom. & Ctr. Syst. (Ctr. & Syst. Eng.)	2/58	Q1 (Q1)	1
Machine Vision and Applications	3.3	2.6	0.54	EE Eng. (CV & PR)	114/275	Q2 (Q2)	1
Neural Networks	7.8	8.4	1.58	Artif. Intel. (Artif. Intel.)	28/145	Q1 (Q1)	1

Journal	2022 JIF	5- year JIF	2022 JCI	Category @ JCR (SJR)	Rank@ JCR categ.	Quart @ JCR (SJR)	#
IEEE Transactions on Robotics and Automation (2004)	2.1	NA	NA	Autom. & Ctr. Syst. (Ctr. & Syst. Eng.)	2/46	Q1 (Q1)	1
IEEE Transactions on Robotics	7.8	10.2	1.79	Robotics (Ctr. & Syst. Eng.)	7/30	Q1 (Q1)	1
IEEE Transactions on Image Processing	10.6	13.3	2.01	Artif. Intel. (Software)	13/145	Q1 (Q1)	1
IEEE Transactions on Intelligent Transportation Systems	8.5	9.0	2.18	EE Eng. (CS Applic.)	23/275	Q1 (Q1)	1
IEEE Journal of Oceanic Engineering	4.1	4.2	1.17	EE Eng. (EE Eng.)	97/275	Q2 (Q1)	1

10 Project Grants

On the description of the funding, for each project I show both the global budget and the budget for the participating institution, in the format (global budget / local budget).

10.1 Project Grants as Global Coordinator

1. HAVATAR

Title: Healthcare Added-Value Applications of Tele-Autonomous Robots

Dates: 2021-03-01 to 2024-02-28 (successfully concluded)

Reference: <https://doi.org/10.54499/PTDC/EEI-ROB/1155/2020>

Funding: FCT (248K€ / 130K€)

Institutions: *Instituto de Sistemas e Robótica, IST-ID*

Instituto de Sistemas e Robótica – Universidade de Coimbra

Hospital da Luz GLSMED Learning Health, S.A.

Role: Global Coordinator (Principal Investigator)

Abstract: The HAVATAR project developed new software skills for telepresence robots that increase their level of autonomy, the teleoperation comfort of the user, and its social presence at the remote site. Two academic partners with long-standing international experience in robotics projects (IST-ID Lisboa and ISR-UC Coimbra) will collaborate for the development of the tele-autonomy skills and their integration in the healthcare added-value applications, that will be tested in realistic scenarios provided by Hospital da Luz Learning Health (HL.LH). One application will be hosted at the HL.LH Simulation Center, where practitioner students, health professionals or other actors will remotely access a telepresence robot to dynamically

observe simulated clinical practices and interact with the local peers (e.g. simulated surgical acts or simulated Intensive Care Unit in a pandemic scenario). In this scenario, we will also study the issue of having multiple remote users sharing the same local space and interacting with each other via their proxy robots. The other application will deploy one telepresence robot in the home of a patient and let the relatives or health professionals interact and program the robot to acquire added value variables and patient reported data. In this application, we will study how extended periods of autonomy can be useful for healthcare professionals.

2. FIREFRONT

Title: Real-Time Forest Fire Mapping and Spread Forecast Using Unmanned Aerial Vehicles

Dates: 2019-03-01 to 2023-02-28 (successfully concluded)

Funding: FCT with reference PCIF/SSI/0096/2017 (377K€ / 102K€)

Institutions: *Instituto de Sistemas e Robótica, IST-ID*

Aeroclube de Torres Vedras - Centro de Ciência do Ar e do Espaço

Associação para o Desenvolvimento da Aerodinâmica Industrial

Força Aérea Portuguesa

Instituto de Telecomunicações

UAVision, Engenharia de Sistemas, Ltd

Role: Global Coordinator (Principal Investigator)

Abstract: This project develops solution to support firefighting actions in forest fires through the realtime detection and tracking of fire fronts and reburns. This is achieved by processing the information acquired from manned and unmanned aerial vehicles equipped with specialized sensing and communication systems, that overfly the affected areas. This information can be made available to the coordination and fighting units on a graphical interface the localization of the fire events in georeferenced coordinates. Forecasts of the fire front evolution, images of the fire area, wind speed and direction, and other meteorological data of interest can also be supplied. This information is of great value for decision making on firefighting actions.

3. VOAM AIS

Title: Computer Vision for the Operation of Unmanned Aerial Vehicles in Maritime and Wildfire Scenarios

Dates: 2019-02-01 to 2023-01-31 (successfully concluded)

Funding: FCT with reference PTDC/EEI-AUT/31172/2017 (222K€ / 154K€)

Institutions: *Instituto de Sistemas e Robótica, IST-ID*

Força Aérea Portuguesa

Marinha Portuguesa

Role: Global Coordinator (Principal Investigator)

Abstract: This project developed novel methods for target detection and tracking in airborne and seaborne image sequences. The availability of low-cost visual sensors (visible and IR) and the recent developments on convolutional neural-networks and

correlation filters are cost-effective and energy-efficient solutions for detection and tracking of targets capable of running onboard on air or sea vehicles. The results of the project have a direct impact on strategic national activities such as ocean and forest monitoring. To demonstrate the applicability of the methods, three case scenarios are considered: (i) detection of forest fires from airborne images, (ii) detection and tracking of sea vessels from airborne and seaborne images, and (iii) detection, tracking and pose estimation of aircrafts from seaborne images for teleguidance of unmanned aerial vehicles (UAV's). These demonstrations will inform the responsible institutions for the protection of the environment and coastal operations of a novel technology to support their activities.

4. AHA

Title: Augmented Human Assistance

Dates: 2014-08-01 to 2018-12-31 (successfully concluded)

Funding: FCT with reference CMUP-ERI/HCI/0046/2013 (590K€/307K€)

Institutions: *Instituto de Sistemas e Robótica, IST-ID*

Faculdade de Motricidade Humana, Universidade de Lisboa

Madeira Interactive Technologies Institute M-ITI

NOVA.ID.FCT

PLUX, Engenharia de Biosensores, Lda.

YDreams - Informática, SA

Carnegie Mellon University (US)

Role: Global Coordinator (Principal Investigator)

Abstract: This project is a novel, integrative and cross-disciplinary approach of 4 portuguese universities, CMU and 2 portuguese industry partners that combines innovation and fundamental research in the areas of human computer interaction, robotics, serious games and physiological computing. AHA has developed a new generation of ICT based solutions that have the potential to transform healthcare by optimizing resource allocation, reducing costs, improving diagnoses and enabling novel therapies, thus increasing quality of life. The project proposes the development and deployment of a novel Robotic Assistance Platform designed to support healthy lifestyle, sustain active aging, and support those with motor deficits.

5. LIMOMAN

Title: Developmental Learning of Internal Models for Robotic Manipulation based on Motor Primitives and Multisensory Integration

Dates: 2014-05-01 to 2016-04-30 (successfully concluded)

Funding: EU-FP7-PEOPLE-2013-IEF, grant agreement 628315 (147K€)

Institutions: *Instituto de Sistemas e Robótica, Instituto Superior Técnico*

Role: Global Coordinator

Abstract: LIMOMAN (developmental Learning of Internal Models for robotic Manipulation based on motor primitives and multisensory integration) addresses the key problem of improving the ability of current robots in dexterous manipulation,

inspired by three main aspects of the human motor control system: internal models, learning and multisensory integration. Internal models that encode the robot sensorimotor capabilities and the dynamics of its interaction with the environment are exploited to generate predictions that improve both robot perception and motion control. These models are either acquired from scratch and/or progressively refined and adapted through learning and optimization techniques, using sensorimotor data collected by the robot during goal-directed actions. The combination of different sensory modalities (vision, proprioception, touch) and the use of probabilistic (Bayesian) techniques permits to achieve robot behaviors that are effective, robust and safe.

6. BIOMORPH

Title: Bio-inspired self-organizing robot computational morphologies

Dates: 2014-02-01 to 2015-07-31 (successfully concluded)

Funding: FCT with reference EXPL/EEI-AUT/2175/2013 (49K€)

Institutions: *Instituto de Sistemas e Robótica, IST-ID*

Role: Coordinator (Principal Investigator)

Abstract: This project progressed towards the development of highly efficient, scalable and adaptable control structures for autonomous robotic agents, using biological principles for the creation of sparse sensorimotor couplings and prediction mechanisms. The developed methods were implemented and tested in concrete robotic test beds to prove the proposed concepts in realistic scenarios and compared them with more classical control approaches.

7. BIOLOOK

Title: Biomimetic Oculomotor Control for Humanoid Robots

Dates: 2007-10-01 to 2010-11-30 (successfully concluded)

Funding: FCT with reference FCT - PTDC/EEA-ACR/71032/2006 (90K€)

Institutions: *Instituto de Sistemas e Robótica, IST-ID*

University of Uppsala (SW)

Role: Global Coordinator (Principal Investigator)

Abstract: In this project we addressed the following scientific topics in oculomotor control: (i) how to perform human-like eye-head movements and postures in the execution of tasks and interaction with humans; (ii) how oculomotor processes are learned and developed through childhood, and how can be exploited to create adaptive autonomous systems; and (iii) how robot behaviour can be modulated to implicitly communicate emotions, intentions and goals to human users. The studied methodologies were implemented and tested both in realistic dynamical simulations and in prototype humanoid robotic platforms available in the consortium.

8. FOVEAR

Title: Foveal Vision for Robotics Applications

Dates: 2006-04-01 to 2008-03-31 (successfully concluded)

Funding: FCT (Luso-Spain Integrated Action) with reference E-47/06 (4K€)

Institutions: *Instituto de Sistemas e Robótica, Instituto Superior Técnico*

Role: Principal Investigator

Abstract: The FOVEAR Project developed computer vision technology for robotics applications with foveal vision sensors. These sensors yield a significant reduction of the amount of information to process at any time, while still allowing wide fields of view and high acuity in the center of the visual field, like in the human retina. This allows the real-time operation of robotic systems in generic and unconstrained environments.

10.2 Project Grants As Coordinator of Participating Institution

1. ORIENT

Title: Goal-directed eye-head coordination in dynamic multisensory environment.

Dates: 2019-07-01 to 2022-12-31 (successfully concluded)

Reference: <https://doi.org/10.3030/693400>

Funding: H2020-EU.1.1, ERC-215-AdG, n. 693400 (2523K€/314K€)

Institutions: *Stichting Radboud Universiteit, Nijmegen* (NL)

Instituto de Sistemas e Robótica, Instituto Superior Técnico

Instituto de Sistemas e Robótica, IST-ID

Role: Participating Institution Coordinator

Abstract: Rapid object identification is crucial for survival of all organisms but poses daunting challenges if many stimuli compete for attention, and multiple sensory and motor systems are involved in the processing, programming and generating of an eye-head gaze-orienting response to a selected goal. How do normal and sensory-impaired brains decide which signals to integrate (“goal”), or suppress (“distracter”) ? Audiovisual (AV) integration only helps for spatially and temporally aligned stimuli. However, sensory inputs differ markedly in their reliability, reference frames, and processing delays, yielding considerable spatial-temporal uncertainty to the brain. Vision and audition utilize coordinates that misalign whenever eyes and head move. Meanwhile, their sensory acuities vary across space and time in essentially different ways. As a result, assessing AV alignment poses major computational problems, which so far have only been studied for the simplest stimulus-response conditions.

2. ELEVAR

Title: Study of Vertical Structures with Robotized Aircrafts

Dates: 2017-04-01 to 2019-03-31 (successfully concluded)

Funding: PT2020 - LISBOA-01-0247-FEDER-17924 (101K€)

Institutions: *Tekever ASDS Lda*

Instituto de Sistemas e Robótica, Instituto Superior Técnico

Laboratório Nacional de Engenharia Civil

Role: Participating Institution Coordinator

Abstract: The inspections and assessments of the structural integrity of large infrastructures (eg. dams, bridges and high voltage cables) are inadequate because they

are not performed in a systematic manner and they greatly depend on the personal criteria of the inspection engineers. Additionally, certain areas of these structures or difficult to reach by conventional means and the locations where data can be acquired are limited which in turn can compromise the quality of the gathered information and make these inspections costly. Some of the already existing solutions based on multicopters are completely dependent on a certified operator and/or GPS, lacking the ability to perform an autonomous survey without the support of an infrastructure that supplies the platform with global positioning signals. The solution proposed by ELEVAR consists of utilizing an autonomous unmanned aircraft integrated with sensors and vision-based navigation algorithms to solve the above-mentioned problems.

3. iPEPE

Title: Cognitive Rehabilitation with Portable Exergame Platforms for the Elderly

Dates: 2020-06-01 to 2023-05-31 (successfully concluded)

Funding: P2020 with reference LISBOA-06-4234-FSE-000079 (30K€ / 74K€)

Institutions: *Instituto das Irmãs Hospitaleiras do Sagrado Coração de Jesus*

Instituto de Sistemas e Robótica, Instituto Superior Técnico

Role: Participating Institution Coordinator

Abstract: This project developed and tested augmented reality exergames for the cognitive stimulation of the elder people living in nursing homes or care institutions. These games put together physical and cognitive tasks in entertaining and motivating activities and have been shown to have benefits at the physical, cognitive, and social dimensions. In this project a special attention is given to patient with dementia, to assess if the proposed approach can be applied to this fraction of the population.

4. SEAGULL

Title: Intelligent Systems for the Maritime Situational Awareness using Unmanned Aerial Vehicles.

Dates: 2013-07-01 to 2015-06-30 (successfully concluded)

Funding: QREN - FCOMP-01-0202-FEDER-034063 (870K€/97K€)

Institutions: *Critical Software, SA*

Instituto de Sistemas e Robótica, Instituto Superior Técnico

Faculdade de Engenharia da Universidade do Porto

Força Aérea Portuguesa

Marinha Portuguesa

Role: Participating Institution Coordinator

Abstract: Portugal is a maritime nation having the largest exclusive economic zone (EEZ) of Europe. Upon acceptance of the proposal submitted to the United Nations (UN), the Portuguese EEZ will be extended to an area larger than 2 million square kilometres. In addition, Portugal is known as a specialist in maritime matters not only by historical reasons but also due to its contemporary activities on the sea. It is crucial for Portugal to maintain the safety of human life at sea, maintain the safety of its maritime area, and protect the sea and coastline natural environments. The SEAGULL project consists on the use of unmanned air vehicles (UAV's) as a not

only viable but also cheaper solution to help with these difficult tasks on such a large area. The objective is to develop an intelligent system that couples the UAV with sensors like visual, infra-red and hyperspectral cameras along with GPS receivers and other sensors. Its purpose is to fly autonomously over the sea while scanning it to automatically detect, identify, report and follow targets like boats, shipwrecks, lifeboats, boats with suspicious behaviour, and also hydrocarbyls and other chemical pollutants.

5. DICO(RE)2S

Title: Discount Coupon Recommendation and Redemption System

Dates: 2011-07-01 to 2016-06-30 (successfully concluded)

Funding: EU-FP7-SME-262451 (1428K€/216K€)

Institutions: *Fundacio Barcelona Media* (ES)

Nova Ventus Consulting SL (ES)

Nextage SRL (IT)

Aitek SPA (IT)

Observit – Tecnologias de Visão por Computador, Lda

Instituto de Sistemas e Robótica, Instituto Superior Técnico

Role: Participating Institution Coordinator

Abstract: DICO(RE)2S is an SME European Consortium devoted to develop and deploy a coupon-based discount campaign platform that will exploit state-of-the-art information and communication technologies and processes for providing consumers and retailers/manufactures with a personalized and efficient environment for maximizing customer satisfaction and business profitability. The distributed network of coupon redemption points will implement pattern recognition systems for reading and validating coupons as well as security encryption and transmission systems that will guarantee the integrity of the overall coupon redemption strategy as well as the reliability of the data.

6. HDA

Title: High-Definition Analytics.

Dates: 2010-05-01 to 2013-10-30 (successfully concluded)

Funding: QREN - FCOMP-01-0202-FEDER-013750 (536K€/203K€)

Institutions: *Observit – Tecnologias de Visão por Computador, Lda*

Instituto de Sistemas e Robótica, Instituto Superior Técnico

Role: Participating Institution Coordinator and Global Scientific Coordinator

Abstract: HD-Analytics is a QREN research project aimed at the development of new algorithms and methods to address important video analytics challenges, in the context of new technology available in the IP camera market: high-resolution multi-megapixel cameras and high-resolution omni-directional cameras. Thus, new algorithms and methods are required to deal with the increased amount of data.

7. HANDLE

Title: Developmental pathway towards autonomy and dexterity in robot in-hand manipulation

Dates: 2009-02-01 to 2013-01-31 (successfully concluded)

Funding: EU-FP7-ICT-231640 (8190K€/749K€)

Institutions: *Universite Pierre et Marie Curie, Paris 6* (FR)

Universitaet Hamburg (DE)

Universidad Carlos III de Madrid (ES)

Commissariat a L'Energie Atomique et Aux Energies Alteratives (FR)

Instituto de Sistemas e Robótica, Instituto Superior Técnico

Faculdade de Ciências e Tecnologia da Universidade de Coimbra

Orebro University (SW)

Kings College London (UK)

The Shadow Robot Company Limited (UK)

Role: Participating Institution Coordinator and Global Technical Manager

Abstract: The HANDLE project aims at understanding how humans perform the manipulation of objects to replicate grasping and skilled in-hand movements with an anthropomorphic artificial hand, and thereby move robot grippers from current best practice towards more autonomous, natural, and effective articulated hands. The project implies not only focusing on technological developments but also working with fundamental multidisciplinary research aspects to endow the robotic hand system with advanced perception capabilities, high level feedback control and elements of intelligence that allow recognition of objects and context, reasoning about actions and a high degree of recovery from failure during the execution of dexterous tasks.

8. VEMUCARV

Title: Spatial Validation of Complex Urban Grids in Virtual Imersive Environments

Dates: 2009-02-01 to 2013-01-31 (successfully concluded)

Funding: FCT - POCTI/AUR/48123/2002 (25K€)

Institutions: *Institute of Mechanical Engineering, Instituto Superior Técnico*

Instituto de Sistemas e Robótica, Instituto Superior Técnico

Câmara Municipal de Lisboa

Role: Participating Institution Coordinator

Abstract: The main goals of this project are related to the semi-automatic acquisition and maintenance of 3D virtual reality models of urban areas. It is intended to use registered aerial images and low altitude laser range scans to acquire 3D data of city structure. This data will be processed in order to segment relevant structures for urban planning (buildings, roads, green areas, etc). Range information provides a very rich description of 3D structure but lack photometric information. Aerial photos provide this information, allowing to pre-segment regions based on color and texture. The main scientific innovation of this project is the combined use of 2D (aerial images) and 3D (range scans) to simplify and improve the building extraction process. Most

current approaches use one or the other types of data exclusively. The final result provides a computer model which stands for a mix geometry-image database that can interface to GIS software available (at CML), as well as the generation of real-time walkthrough with thematic information. The results of this project are to be integrated on Lisbon City Hall public computational facilities.

10.3 Project Grants As Team Member

1. RePAIR

Title: Reconstructing the Past: Artificial Intelligence and Robotics Meet Cultural Heritage

Dates: 2021-06-01 to 2024-11-30 (ongoing)

Funding: H2020 FETOPEN - Grant agreement 964854 (3522K€/575K€)

Institutions: *Universita Ca'Foscari, Venezia* (IT)

Instituto de Sistemas e Robótica, IST-ID

Ben-Gurion University of the Negev (IR)

Fondazione Istituto Italiano di Tecnologia (IT)

Rheinische Friedrich-Wilhelms-Universität Bonn (DE)

Ministero Della Coltura (IT)

Role: Team Member

Abstract: The physical reconstruction of shattered artworks is one of the most labour-intensive steps in archaeological research. Dug out from excavation sites are countless ancient artefacts, such as vases, amphoras and frescoes, that are damaged. The EU-funded RePAIR project will facilitate the reconstruction process to bring ancient artworks back to life. Specifically, it will develop an intelligent robotic system that can autonomously process, match and physically assemble large fractured artefacts in a fraction of the time required by humans. This new system will be tested on iconic case studies from the UNESCO World Heritage Site of Pompeii. It will restore two world-renowned frescoes, which are in thousands of broken pieces and currently in storerooms.

2. ShiftHRI

Title: Exploring the Transfer of Agency to Older Adults in HRI

Dates: 2022-03-01 to 2023-05-28 (successfully concluded)

Funding: FCT, CMU/TIC/0026/2021 (63K€/26K€)

Institutions: *LASIGE*

Instituto de Sistemas e Robótica, IST-ID

Carnegie Mellon University (US)

Role: Team Member

Abstract: Our vision is to advance research on human-robot interaction to facilitate a new type of relationship between elders and assistive robots. This relationship will be built on a deep understanding of human values and what is critical to older adults. Our previous work shows that values are manifested through interactions

with products, and that this becomes more critical for elders as they experience physical, cognitive, and emotional decline. In addition, what elders value can change dynamically, even day to day, as elders experience decline. In ShiftHRI, we will start by exploring important values and co-designing HRI scenarios that help people to fulfil them. This paradigm shift has the potential to significantly impact the research done in the area. Contrary to the state of the art, where most robots have strict predefined user interfaces and routines, shifting agency to older adults brings several challenges to the control of those robots and in the service they offer.

3. IntelligentCare

Title: Intelligent Multimorbidity Management System

Dates: 2020-04-01 to 2023-06-30 (successfully concluded)

Funding: LISBOA-01-0247- FEDER-045948 (3522K€/497K€)

Institutions: *GLSMED Learning Health S.A.*

Instituto de Sistemas e Robótica, Instituto Superior Técnico

Instituto de Sistemas e Robótica, IST-ID

Hospital da Luz S.A.

Instituto De Engenharia De Sistemas E Computadores, Investigação e Desenvolvimento Em Lisboa

Priberam Informática S.A.

Carnegie Mellon University (US)

Role: Team Member

Abstract: This project focus on managing the issue of a growing population with multimorbidity (co-occurring diseases), a worldwide problem to the sustainability of the healthcare system. IntelligentCare aims to develop a patient-centric solution to manage multimorbidity conditions by using analytical methods to explore data from the electronic health records and measures reported remotely by the patients. Under the value-based healthcare framework, the project will develop methods for early signalization and characterization of patients, identification of personalized clinical pathways based on evidence, and based on patient value vision, together with monitoring and prediction of hospital interactions. To achieve these goals research will be conducted in advanced analytical algorithms, particularly deep learning methods, to fully explore data. This is an innovative approach to evaluate the multimorbidity condition by introducing the concept of value to the patient in the process of characterization and prediction of patient interactions with the hospital.

4. ASTRIIS

Title: Atlantic Sustainability Through Remote and In-situ Integrated Solutions

Dates: 2020-03-01 to 2023-06-30 (successfully concluded)

Funding: POCI-01-0247-FEDER-046092 (891K€/273K€)

Institutions: *Tekever*

+Atlantic CoLAB

CEiiA

Hidromod, modelação em engenharia

Institute for Systems and Robotics, IST

ISQ

LSTS, Underwater Systems and Technology Laboratory, FEUP

MARETEC, Marine Environment and Technology Center, IST

CERENA, Center of Natural Resources and Environment, IST

Ocean Infinity

Ocean Scan, Marine Systems & Technology Lda

Spin.Works, S.A.

WavEC Offshore Renewables

Universidade do Minho

Universidade do Algarve

Role: Team Member

Abstract: This project is a Portuguese effort to support the sustainable growth of the blue economy by developing technical and scientific knowledge for the design, demonstration and implementation of integrated and customizable information products and services. The project focuses on sectors of the blue economy with high potential for value creation, such as marine renewable energy, aquaculture, coastal tourism, or recreational boating, and is oriented to the needs of users, which include investors, regulators, policymakers, consumers, and scientific researchers. The ISR team participated in the development of remote sensing solutions for environmental monitoring of the Atlantic national coastal waters.

5. RBCog-Lab

Title: Robotics, Brain and Cognition Lab

Dates: 2017-06-01 to 2021-11-30 (successfully concluded)

Funding: FCT - Portuguese Roadmap of Research Infrastructures 01/SAICT/2016 SAICT Proj. 22084 (300K€)

Institutions: *Instituto de Sistemas e Robótica, Instituto Superior Técnico*

Instituto de Sistemas e Robótica, IST-ID

Role: Team Member

Abstract: The Robotics, Brain and Cognition Lab (RBCogLab) is part of the Portuguese Roadmap of Research Infrastructures. RBCog-Lab supports an integrated multidisciplinary research endeavour (neuroscience, developmental psychology, cognitive science, psychophysics, machine learning and other engineering subjects) not only promoting novel robotic methodologies but also contributing to a better understanding of the human brain. To be effective and generate a positive impact in the society, this multidisciplinary research needs to be supported by advanced equipment and specialized technical expertise, that might not be available in small research and academic centres with low budgets. The RBCogLab research infrastructure aims at

filling this gap by providing the scientific community with a centralized open laboratory resource for robotics experiments and human measurements, offering both state of the art devices and specialized know-how.

6. ALHTOUR

Title: Assisted Living Technologies for the Health Tourism Sector

Dates: 2016-01-01 to 2018-12-31 (successfully concluded)

Funding: H2020-TWINN-2015 Grant Agreement 692311 (1175K€/40K€)

Institutions: *Universidade de Lisboa*

Faculdade de Motricidade Humana

Faculdade de Medicina da Universidade de Lisboa

Faculdade de Ciências da Universidade de Lisboa

Instituto de Ciências Sociais

Instituto Superior Técnico

Katholieke Universiteit Leuven (BE)

Universiteit Maastricht (NL)

Università degli studi di Macerata (IT)

Role: Coordinator of activities at the Institute for Systems and Robotics, Instituto Superior Técnico.

Abstract: The ALHTOUR project aims to strengthen and stimulate scientific excellence, and innovation capacity in technologies for independent living, applied to the health tourism at UL (Portugal), by preparing for the set-up of a “Health Tourism Living Lab”, identified as a key driver for territorial development.

7. SPARSIS

Title: Sparse Modeling and Estimation of Motion Fields

Dates: 2016-07-01 to 2019-12-31 (successfully concluded)

Funding: FCT - PTDC/EEI-PRO/0426/2014 (199K€)

Institutions: *Instituto de Sistemas e Robótica, Associação do Instituto Superior Técnico para a Investigação e o Desenvolvimento*

Instituto De Engenharia De Sistemas E Computadores, Investigação e Desenvolvimento Em Lisboa

Instituto de Telecomunicações

Role: Workpackage Coordinator

Abstract: In this project we will explore the use of sparse techniques to improve the estimation of multiple motion fields as well as space varying switching matrix (stochastic matrix) that describes the switching process associated to the movement of each target. As scientific outcomes of the project, we expect to obtain reliable estimates for the model parameters (many hundreds), to remove the over smoothed character of the field estimates. We also expect to speed up the estimation procedure bringing it closer to real time applications and extend these techniques to

multicamera settings. We also expect to develop an adaptive algorithm for the on-line estimation of multiple motion fields able to update in a recursive way the number of fields and the field estimates whenever new information arrives.

9. INSIDE

Title: Intelligent Networked Robot Systems for Symbiotic Interaction with Children with Impaired Development

Dates: 2014-08-01 to 2018-07-31 (successfully concluded)

Funding: FCT with reference CMUPERI/HCI/0051/2013 (590K€/964K€)

Institutions: *INESC-ID*

ISR-Lisboa, LARSyS, IST-ID

Hospital Garcia de Horta EPE

IDMIND – Engenharia de Sistemas, LDA

PLUX, Engenharia de Biosensores, LDA

VoiceInteraction – Tecnologias de Processamento da Fala, S.A.

Carnegie Mellon University (US)

Role: Team Member

Abstract: The INSIDE initiative explores symbiotic interactions between humans and robots in joint cooperative activities, and addresses the following key research problems: (i) How can robots plan their course of action to coordinate with and accommodate for the actions of their human teammates? (ii) How can task, context and environment information collected from a set of networked sensors (such as cameras and biometric sensors) be exploited to create more natural and engaging interactions between humans and robots involved in a joint cooperative activity in a physical environment? INSIDE developed new hardware and software solutions that support the aforementioned research towards the deployment of a real-world interaction scenario where children with impaired development interact with a networked robot system in a joint cooperative task with therapeutical purposes.

8. Poeticon++

Title: Robots Need Language: A Computational Mechanism for Generalisation and Generation of New Behaviours in Robots

Dates: 2012-01-01 to 2015-12-31 (successfully concluded)

Funding: EU-FP7-ICT-288382 (3880K€/470K€)

Institutions: *Fondazione Istituto Italiano do Tecnologia (IT)*

Instituto de Sistemas e Robótica, Instituto Superior Técnico

Cognitive Systems Research Institute (GR)

Athina (GR)

University of Plymouth (UK)

The University of Maryland Foundation, Inc (US)

Role: Deputy Scientific Coordinator at Participating Institution

Abstract: Handling novel situations beyond learned schemas or set behaviours is still a quest in engineering cognitive and embodied systems. Powerful generalisation mechanisms are necessary for any agent to operate effectively in real-world environments. POETICON++ suggests that natural language can be used as a learning tool for generalisation of learned behaviours & perceptual experiences and generation of new behaviours & experiences (creativity). The main objective of POETICON++ is the development of a computational mechanism for such generalisation of motor programs and visual experiences for robots. To this end, it will integrate natural language and visual action/object recognition tools with motor skills and learning abilities.

9. AuReRo

Title: Human-robot interaction with field robots using augmented reality and interactive mapping

Dates: 2011-04-01 to 2014-09-30 (successfully concluded)

Funding: FCT - PTDC/EIA-CCO/113257/2009 (130K€)

Institutions: *Instituto de Sistemas e Robótica, Instituto Superior Técnico*

Madeira Interactive Technologies Institute M-ITI

Role: Co-Principal Investigator

Abstract: To achieve effective human-robot interaction (HRI), we propose presenting a pair of stereo camera feeds from a robot to an operator through an Augmented Reality (AR) head-mounted display (HMD). This will provide an immersive experience to the controller and has the capacity to display rich contextual information. To further enhance this display, we will superimpose mapping data generated by SLAM-6D (Simultaneous Localization And Mapping) algorithms over the video input using Augmented Reality (AR) techniques.

10. FirstMM

Title: Flexible Skill Acquisition and Intuitive Robot Tasking for Mobile Manipulation in the Real World

Dates: 2010-02-01 to 2013-07-31 (successfully concluded)

Funding: EU-FP7-ICT-248258 (2997K€/351K€)

Institutions: *Albert-Ludwigs-Universität Freiburg, (DE)*

Instituto de Sistemas e Robótica, Instituto Superior Técnico

Ecole Polytechnique Federale de Lausanne (CH)

Kuka Laboratories GmbH (DE)

Fraunhofer Institute (DE)

Technische Universität Berlin (DE)

Idryma Technologias Kai Erevnas (DE)

Role: Deputy Scientific Coordinator at Participating Institution

Abstract: Many industrial processes highly depend on the reliability and robustness of robotic manipulators. Research on mobile robots has led to systems that demonstrated the capability of safe and accurate navigation. The goal of the FIRST-MM

project is to integrate these two areas in the context of a real-world application scenario. The project will build upon and extend recent results in robot programming, navigation, manipulation, perception, learning by instruction, and statistical relational learning to develop advanced technology for autonomous mobile manipulation robots that can flexibly be instructed even by non-expert users to perform challenging manipulation and transportation tasks in real-world environments.

11. RoboSoM

Title: A Robotic Sense of Movement

Dates: 2009-12-01 to 2013-05-31 (successfully concluded)

Funding: EU-FP7-ICT-248366 (1659K€/391K€)

Institutions: *Scuola Superiore di Studi S'Anna*, (IT)

Instituto de Sistemas e Robótica, Instituto Superior Técnico

Centre National de la Recherche Scientifique (FR)

Waseda University (JP)

College de France (FR)

Role: Deputy Scientific Coordinator at Participating Institution

Abstract: The objective of RoboSoM is to investigate and to apply new approaches to the design and development of humanoid robots with advanced perception and action capabilities, showing robust, adaptive, predictive, and effective behaviour in the real world. The expected robot behaviour is the capability to follow a visual target by coordinating eye, head, and leg movements, with head stabilization, walking smoothly and effectively in an unstructured environment, with a robust reactive behaviour, improved by predictions.

12. MMCACC

Title: Modern Monte Carlo Algorithms for Computational Control

Dates: 2007-09-01 to 2010-12-31 (successfully concluded)

Funding: FCT - PTDC/EEA-ACR/70174/2006 (80K€)

Institutions: *Instituto de Sistemas e Robótica, Instituto Superior Técnico*

University of British Columbia (CA)

Role: Team Member

Abstract: The objective is to develop modern Monte Carlo methods to solve discrete time stochastic optimal control problems in both the fully and partially observed cases for nonlinear and/or non-Gaussian models. It aims to combine modern computational tools developed actively in statistics (Markov chain Monte Carlo, Sequential Monte Carlo aka Particle filters) to ideas developed in automatic control, operation research (gradient estimation) and reinforcement learning (value function and policy parameterization). The focus is in developing generic algorithms that can be applied in different domains and provide a unified framework for this type of problems.

13. URUS

Title: Ubiquitous Networking Robotics in Urban Settings

Dates: 2006-12-01 to 2009-11-30 (successfully concluded)

Funding: EU-FP6-IST-045062 (2600K€/80K€)

Institutions: *Universitat Politecnica de Catalunya* (ES)

Instituto de Sistemas e Robótica, Instituto Superior Técnico

Agencia d'Ecologia Urbana de Barcelona (ES)

Asociacion de la Investigacion Y Cooperacion Industrial de Andalucia "F. de Paula Rojas" (ES)

Centre National de la Recherche Scientifique (FR)

ETH Zurich (CH)

Robotech SRL (IT)

Scuola Superiore di Studi S'Anna (IT)

Telefonica Investigacion Y Desarrollo AS Unipersonal (ES)

The University of Surrey (UK)

Universidad de Zaragoza (ES)

Role: Scientific Coordinator of Computer Vision at Participating Institution

Abstract: European ancient cities are becoming difficult places to live due to noise, pollution, lack of quality facilities and security. Moreover, the average age of people living in large European cities is growing and in a short period of time there will be an important community of elderly people. City Halls are becoming conscious of this problem and are studying solutions, for example by reducing the areas of free circulation of cars. Free car areas imply a revolution in the planning of urban settings, for example, by imposing new means for transportation of goods to the stores, security issues, new ways of human assistance, etc. In this project we want to analyse and test the idea of incorporating a network of robots (which include humans, robots, intelligent sensors, intelligent devices and communications) in order to improve quality of life in such urban areas.

14. CONTACT

Title: Learning and Development of Contextual Action

Dates: 2005-09-01 to 2009-02-28 (successfully concluded)

Funding: EU-FP6-NEST-5010 (NA/380K€)

Institutions: *Universita degli studi di Genova*, (IT)

Instituto de Sistemas e Robótica, Instituto Superior Técnico

Universita degli studi di Ferrara (IT)

Uppsala Universitet (SW)

University of Stockholm (SW)

Role: Workpackage Scientific Coordinator at Participating Institution

Abstract: By spontaneously generating, observing and imitating manipulative gestures, infants quickly learn how to interact with their environment. At the same time, communicative abilities develop leading to a more and more complex use of

speech. Are these motor abilities developing independently? Or are there fundamentally similar mechanisms in the development of perception and production, for both speech and manipulation? We believe that the later is the case. This project represents an ambitious attempt to investigate this parallel development of manipulatory and speech related gestures from a multi-disciplinary perspective. Experts in Robotics, Neuroscience, and Child-Development will tightly collaborate to build an artifact (an embodied system) which will develop its motor capabilities both in manipulation and speech.

15. Robot-Cub

Title: Robotic Open-architecture Technology for Cognition, Understanding and Behaviour

Dates: 2004-09-01 to 2010-01-31 (successfully concluded)

Funding: EU-FP6-IST-2004-004370 (8500K€/594K€)

Institutions: *Universita degli studi di Genova* (IT)

Universita degli studi di Ferrara (IT)

Instituto de Sistemas e Robótica, Instituto Superior Técnico

Uppsala Universitet (SW)

Ecole Polytechnique Federale de Lausanne (CH)

University of Zurich (CH)

Fondazione European Brain Research Institute Rita Levi (IT)

Scuola Superiore di Studi S'Anna (IT)

Fondazione Istituto Italiano do Tecnologia (IT)

Telerobot S.R.L. (IT)

The University of Hertfordshire Higher Education Corporation (UK)

The University of Salford (UK)

The University of Sheffield (UK)

Role: Workpackage Scientific Coordinator at Participating Institution

Abstract: The main goals of RobotCub are (i) to create an open robotic platform for embodied research that can be taken up and used by the research community at large to further their approach to the development of humanoid-based cognitive systems and (ii) to advance our understanding of several key issues in cognition by exploiting this platform in the investigation of cognitive capabilities. The scientific objective of RobotCub is, therefore, to jointly design the mindware and the hardware of a humanoid platform to be used to investigate human cognition and human-machine interaction. We call this platform CUB or Cognitive Universal Body. It is worth remarking that the results of RobotCub will be fully open and consequently licensed following a General Public (GP) license to the scientific community.

16. INTELTRAF

Title: Automatic Monitoring of Traffic and Detection of Accidents in Highways

Dates: 2002-12-01 to 2005-01-31 (successfully concluded)

Funding: ADI, POSI Med. 1.3 (22K€)

Institutions: *Observit, Tecnologias de Visão por Computador Lda*
Instituto de Sistemas e Robótica, Instituto Superior Técnico,
Aitek SRL (IT)
Brisa, Auto-Estradas de Portugal.

Role: Deputy Coordinator at Participating Institution

Abstract: This project developed an Automated Traffic Surveillance system with Computer Vision techniques, including real-time algorithms for video sequence analysis of traffic scenes and the use of innovative extended field of view camera systems (panoramic), for price reduction and performance gain with respect to currently existing systems. We address the applications of measuring traffic flow and detecting abnormal events on highways and critical urban areas. Traffic flow monitoring records statistical data on traffic distribution along time (number of vehicles, average velocity, average wait time on queues, accidents, transgressions to traffic rules, etc.).

17. CAVIAR

Title: Context Aware Vision using Image-based Active Recognition

Dates: 2002-10-01 to 2005-09-30 (successfully concluded)

Funding: EU-FP5-IST-2001-37540 (1168K€/189K€)

Institutions: *The University of Edimburgh, (SC)*
Instituto de Sistemas e Robótica, Instituto Superior Técnico
Centre National de la Recherche Scientifique (FR)
Institut National de Recherche en Informatique et en Automatique (FR)
Universite Joseph Fourier Grenoble 1 (FR)

Role: Workpackage Coordinator at Participating Institution

Abstract: The main objective is to develop the theory of context-aware visual recognition systems. We will implement the theory in a complete closed-loop vision system, and apply it to two applications (city street surveillance and customer behaviour analysis). To achieve these objectives, we will develop new feature grouping, attention and appearance-based recognition processes. This will also require development of new techniques for acquiring, representing, and using visual context and situation knowledge.

18. MIRROR

Title: Mirror Neurons for Recognition

Dates: 2001-09-01 to 2004-08-31 (successfully concluded)

Funding: EU-FP5-IST-2000-28159 (720K€/160K€)

Institutions: *Universita degli studi di Genova, Italy (IT)*
Universita degli studi di Ferrara (IT)
Instituto de Sistemas e Robótica, Instituto Superior Técnico

Uppsala Universitet (SW)

Role: Workpackage Coordinator at Participating Institution

Abstract: The goals of MIRROR are (i) to realize an artificial system that learns to communicate with humans by means of body gestures and (ii) to study the mechanisms used by the brain to learn and represent gestures. The biological base is the existence in primates pre-motor cortex of a motor resonant system, called mirror neurons, activated both during execution of goal directed actions and during observation of similar actions performed by others. This unified representation may subserve the learning of goal directed actions during development and the recognition of motor acts, when visually perceived.

19. RESCUE

Title: Cooperative Navigation for Rescue Robots

Dates: 2001-09-01 to 2004-08-31 (successfully concluded)

Funding: FCT SRI/32546/99-00 (115K€)

Institutions: *Instituto de Sistemas e Robótica, Instituto Superior Técnico*

Role: Team Member

Abstract: This project fosters research on multi-agent robotic systems for search and rescue operations as its long-term goal. Currently, the project is focused on obtaining new results on outdoors perception and navigation, both for individual and cooperative robots.

20. NARVAL

Title: Navigation of Autonomous Robots Via Active Environmental Perception

Dates: 1996-03-01 to 2001-12-31 (successfully concluded)

Funding: EU-FP5-Esprit LTR Project -30185

Institutions: *Instituto de Sistemas e Robótica, Instituto Superior Técnico*

Université de Nice Sophia Antipolis, I3S-CNRS (FR)

Università degli studi di Genova (IT)

Thomson Sintra ASM (FR)

Role: Responsible for Cross-platform Computational Architecture

Abstract: The goal of this project is to develop non-intruding and reliable navigation systems giving the robot the ability to select natural landmarks, and to navigate with respect to them, extending in this way the autonomy range of autonomous robots operating in unknown and unstructured environments. Reliability is achieved by continuously controlling the uncertainty associated with knowledge of the environment and of the robot's position and orientation. In the context of the project, non-intrusiveness means that the robot must be able to operate without special purpose landmarks being added to its environment. Non-intrusive operation in unstructured environments precludes navigation with respect to a set of handcrafted landmarks and requires the robot's ability to infer its position from learned natural landmarks of the environment using its perception system.

21. SIVA

Title: Integrated System for Active Surveillance

Dates: 1998-01-01 to 2001-04-30 (successfully concluded)

Funding: FCT - PRAXIS 2/2.1/TPAR/2074/95

Institutions: *Instituto de Sistemas e Robótica, Instituto Superior Técnico*

Instituto de Sistemas e Robótica, Universidade de Coimbra

Role: Team Member

Abstract: With this project we aim to develop an autonomous system able to perform surveillance tasks in a structured environment. The system will be made up of several mobile platforms (agents). Each of them will be equipped with an active vision system, an odometry system and communication links enabling the platforms to communicate among themselves. The full system (including all the platforms) will have a high degree of autonomy. Each of the platforms will also be autonomous in terms of the navigation and surveillance tasks. The system will be designed to operate off-hours in structured environments such as supermarkets, military installations, public buildings, power plants, etc. The implication of its use during off-hours is that the primary condition for the detection of an intruder will be the occurrence of motion. Each one of the agents will perform the surveillance tasks based on the active vision system.

22. INFANTE

Title: Development of Vehicles and Advanced Systems for the Execution of Underwater Inspection Tasks

Dates: 1997-01-01 to 2001-12-31 (successfully concluded)

Funding: FCT - PRAXIS XXI 3/3.1/TPAR/2042/95

Institutions: *Instituto de Sistemas e Robótica, Instituto Superior Técnico*

RINAVE

Instituto Hidrográfico

Cintal – Universidade do Algarve

Role: Team Member

Abstract: The objectives of this project are the design and the construction of an autonomous underwater vehicle (AUV), and the development and integration of advanced systems for navigation, guidance and control, acoustic communications, and mission management. These systems will be tested in the lab. Once they are installed in the vehicle, they will be also tested in pool and sea trials.

11 Recognition And Visibility

11.1 Prizes

1. 2023 IAPR Best Biometrics Student Paper Award at the IEEE International Joint Conference in Biometrics, with paper “Adapt-FuseNet: Context-aware Multimodal Adaptive Fusion of Face and Gait Features Using Attention Techniques for Human Identification”.

2. Supervisor of the Recipient of the 2nd Place of the Best Iberian PhD Thesis in Robotics 2022 (Pedro Vicente, IST), granted by the Spanish Society for Research and Development in Robotics and the Portuguese Robotics Society.
3. Supervisor of the Recipient of the Best National PhD Thesis in Robotics Award of 2022 (Pedro Vicente, IST), granted by Portuguese Society of Robotics.
4. Supervisor of the Recipient of the Honorable Mention for the Best National PhD Thesis Award of 2019/2020 (Rui Figueiredo, IST), granted by the Portuguese Association of Pattern Recognition.
5. Best Poster Award at the Portuguese Conference of Pattern Recognition 2021, “Active robot Learning for Efficient Body-Schema Online Adaptation”, granted by the Portuguese Association of Pattern Recognition.
6. Finalist for Best Paper Award on Agri-Robotics, IROS 2020, with the paper “Fruit Quality Control by Surface Analysis Using a Bio-Inspired Soft Tactile Sensor” sponsored by YANMAR.
7. Highly Commended Paper Award IEEE ICARSC 2020, with the paper “2D visual servoing meets rapidly-exploring random trees for collision avoidance” granted by the Industrial Robot Journal.
8. Honorable Mention Demo Award “Custom-made exergames for older people: New inputs for multidimensional physical training” at the 5th Experiment@ International Conference 2019.
9. Finalist (Top 5 among 85 participants) of the Smart Ageing Prize, AAL Programme, AAL Europe – ICT for Ageing Well, 2018
10. Supervisor of the IBM Prize recipient of 2014 (Ricardo Cabral, IST), granted by IBM Portugal.
11. Best Conference Paper Award at IEEE ROBIO 2013, with the paper “Precision Grasp Synergies for Dexterous Robotic Hands”, granted by the IEEE Robotics and Automation Society.
12. Supervisor of the Best MSc Thesis Award of 2012 (Filipe Veiga, IST), granted by the Portuguese Society of Robotics.
13. Best paper on Control and Instrumentation at Robotica 2008, with the paper “Visual Self-Calibration of Pan-Tilt Kinematic Structures”, granted by the Portuguese Association of Automatic Control.
14. Finalist of the National Physics, Portuguese Society of Physics, 1988.

11.2 Invited Seminars

11.2.1 International

1. “HCI and HRI Technologies in the rehabilitation and physical/cognitive stimulation of older adults”, XXXVI Congreso Internacional de Geriatria y Gerontologia, 22 e 23 de Maio, 2026, Lugo, Espanha.
2. “Semantic Active Foveal Vision”, *Workshop on Active Methods in Autonomous Navigation*, ICRA 2023, London, June 2nd, 2023
3. “Active and Foveal Vision for Biologically Inspired Robots”, *Donders Institute for Brain, Cognition, and Behavior*, Nijmegen, Netherlands, September 29th, 2022.

4. "Active Perception with Foveal Sensors: a bio-inspired strategy for resource constrained robots", *MED22 Workshop: Active methods in autonomous navigation*, June 28th, 2022.
5. "Humanoid Robot Learning", *Queen Mary University of London*, March 12th, 2021.
6. "iCub and Friends, 10 years later", *Italian Institute of Technology*, February 8th, 2021.
7. "ISR/Vislab Robotics Research and the AHA Project", *EPFL, Lausanne*, June 11th, 2019.
8. "Towards Long-Term RE-ID", *SPARSIS Project Workshop*, IST Lisboa, Portugal 22-March 23rd, 2018.
9. "Robot Vision", *Second Winter School on Humanoid Robot Programming*, Santa Margherita Ligure, Italy, 9th February 2018.
10. "Learning by Self-Exploration", 4th Lucia PhD School on Artificial Intelligence and Robotics, IST, Lisboa, Sep. 7th 2017.
11. "Dealing with Uncertainty in Robot Grasping", *Conference on Electronics, Telecommunications and Computers*, 6th – 7th December 2016, Lisbon, Portugal.
12. "Co-development of visuo-motor structures", *Dagstuhl Seminars*, February 2014.
13. "Self-Organizing Retinas from Natural Motor Principles", *EPFL, Lausanne*, December 2013.
14. "Principles for the Co-Development of Sensorimotor Internal Models". *The World Inside the Brain: Internal Predictive Models in Humans and Robots*, Birmingham, UK, May 2013.
15. "Real-Time Vision: From Biology to Computer Algorithms", *Vibot Master*, 2-3 May 2011, Girona, Spain.
16. "Object Detection and Tracking on the iCub", *Workshop iCub and Friends*, IROS 2010.
17. "Biologically Motivated Strategies for Visual Attention", *Robot Learning Summer School*, 24 July 2009, Lisbon, Portugal
18. Chico: A baby robot for studies on cognition and development. *Department of Psychology, University of Uppsala*, Sweden, 5th June 2008.
19. Biologically Motivated Strategies for Real-Time Vision: Foveal Sensors and Visual Attention. *Vibot Master*. Girona, 28th April 2008.
20. Region Based Motion Estimation and Tracking with Log-Polar Images, *Universidade Jaume I, Castellon*, Espanha, 4th April 2008.
21. Robotic Vision: Applications to Humanoid Robots, *X Simposio de Señales, Imagenes y Visión Artificial*, Universidad del Valle, Cali, Colombia, September 2005.

11.2.2 National

1. "Tecnologias de Interação Homem-Máquina e Interação Humano-Robot na reabilitação física e cognitiva de adultos seniores", 4th AGE.COMM Congress - Challeges of Ageing Creativity, Connectivity and Inclusion, 13, 14, 15 Nov. 2025, Politécnico Castelo Branco
2. "Inteligência Artificial Aplicada à Saúde e Ambiente", *Conferência 75 Anos: Ousar o Futuro*, Beltrão Coelho, Instituto Português de Reumatologia, Hot Club de Portugal e a LPN - Liga para a Proteção da Natureza, Lisboa, Portugal, Dec. 15th, 2023.
3. "Veículos Autónomos", *Técnico+ Formação Avançada*, May 5th, 2023.

4. “Aplicações de visão artificial no contexto da defesa nacional”, *4th Encounter of Research and Development in Military Sciences*, Sintra, Portugal, November 16th, 2022.
5. “Robots and AI for Cognitive Interaction and Care”, Alexandre Bernardino and João Sanches, *Encontro Ciência*, Jun. 9th, 2019, Lisboa, Portugal.
6. “Tecnologias Digitais e Robóticas para Estimulação Física e Cognitiva”, *Congresso Nacional de Esclerose Múltipla*, Mar. 3rd, 2019.
7. “Robots and Games for Elderly Exercise and Rehabilitation”, *SEEEP/SESE Summer School 2019* – Lisbon 22-24 July.
8. “Interactive Robots: from Research to Application”, *Seminars on Innovation and Sustainable Development (SIDS)*, Dec. 6th, 2018, IST, Lisboa, Portugal
9. “New Interactive Technologies for Promoting Exercise in Elderly Care”, *+TALKS, Associação CURA+*, September 15th 2018, Faculdade de Farmácia da Univ. do Porto.
10. “AHA – Augmented Human Assistance”, *Encontro Ciência*, Jul. 3rd, 2018, Lisboa, Portugal.
11. “Social Robots and Serious Games for Promoting Exercise in Elderly Care”, *9th Symposium on Bioengineering*, 15th April 2018, Porto, Portugal.
12. “Realidade Aumentada e Robots Sociais em Envelhecimento Activo: Possíveis Intervenções Terapêuticas em Idosos Saudáveis e doentes de Parkinson”, *Congresso CNS*, April 7, 2018, Torres Vedras, Portugal.
13. “CogROS – Cognitive Robots and Systems for Assited Living and Working”, Alexandre Bernardino and Sergi Bermúdez I Badia. *Encontro Ciência*, Jul. 3rd, 2017.
14. “Social Robots: from research to application”, *4th Annual Conference Health@Lisbon, Rede Saúde, Universidade de Lisboa, Semana da Inovação*, May 4, 2017.
15. “The AHA project in the promotion of exercise for the elderly”, *Ciclo de Conferências rede SAÚDE 2017, Sessão Envelhecimento Activo*, FMH, April 10, 2017.
16. “Robots Sociais para a Promoção do Exercício”. *Escola Secundária Virgílio Ferreira*, Lisboa, February 15, 2016.
17. “Robots in Health and Assisted Living”, Alexandre Bernardino and João Sequeira, *Encontro Ciência*, July 2016, Lisboa, Portugal.
18. “Robotics Assisted Exercise for Users with Special Needs”. *Innovation in Health Robotics, Lisbon eHealth Summer Week*. June 29th, 2016.
19. “Robots for Healthy and Assisted Living”, *Encontro Ciência 2016*, Lisboa, Julho 2016.
20. “Robótica Assistiva para Envelhecimento Activo”, *Conferência Envelhecimento no sec XXI*, FMH, Lisboa, Outubro 2015.
21. “Reconhecimento de Objectos em Imagens com Dados de Treino Incompletos”, *XXI Congresso da Sociedade Portuguesa de Estatística*, Aveiro, Dezembro 2013.
22. “The DICORE2S project”, *Portuguese FP7 promotion office, Portuguese Industrial Association (AIP)*, Lisbon, 6th of July 2012.
23. Visão Robótica no VisLab-ISR : da Biologia à Engenharia, *1^{as} Jornadas de Engenharia Industrial*, Instituto Politécnico de Castelo Branco, Outubro 2005.

11.3 Scientific and Professional Societies

1. Institute of Electrical and Electronics Engineers, Senior Member, number 80360903, Robotics and Automation Society and Life Sciences Community.
2. ELLIS Society, Member, associated with the Lisbon Unit (LUMILIS).

12 External Engagement

12.1 Technology Transfer, Patents and Utility Models

1. Author of the National Patent “Mobile Platform for Physical Stimulation and Rehabilitation”. Alexandre Bernardino, José Santos-Victor, Plinio Moreno López, Ricardo Ribeiro, Ricardo Nunes, Hugo Simão, Ana Luísa Fernandes, Heitor Cardoso, Rui Garcia Figueiredo, João Avelino. PT117350, EP4382180, US2024342578, WO2023287314, 10/07/2021. INPI.
2. Author of “PEPE” Drawings (European Registration). Alexandre Bernardino, José Santos-Victor, Plínio Moreno Lopez, Ricardo Ribeiro, Ricardo Nunes, Hugo Simão, Heitor Cardoso, Rui Garcia Figueiredo, João Avelino, Ana Luísa Fernandes. Registration 015017934-0001, 2023, European Intellectual Property Office. This drawing is one of the results of the design of the PEPE platform (Portable Exergame Platform) in the scope of the AHA project.
3. Promoter of Contract the between IST and company IDMind - Engenharia de Sistemas, lda, for the Exclusive Use of Intellectual Property Rights of the Patent “Mobile Platform for Physical Stimulation and Rehabilitation”, July 2023.
4. Author of “Dicore2S” Software (National Registration). Alexandre Bernardino, João Pimentel, Jaime Bernardo, José Santos-Victor. Registration number 5466/2013, IGAC. This software was the result of the research performed at IST in the project DICORE2S and implemented an electronic coupon redemption system using a database, a smartphone app, and pattern recognition techniques. The software was acquired in its entirety by the company Observit Lda, with an income for IST of 50k€.
5. Founder of the company Observit Lda, in 2001, a spin-off dedicated to applications of computer vision to the video surveillance business.

12.2 Services to Industry

1. Co-responsible for a FCT Doctoral Studentship with Industry (Albatroz Engenharia, S.A.), by Vinicius Souza, 2023.
2. Co-responsible for a FCT Doctoral Studentship with Industry (Hospital da Luz S.A.), by Rita Verdelho, 2023.
3. Responsible for a MSc Thesis with formal agreement between IST and the company PRO-DRONE, by Inês Coelho, 2022.
4. Responsible for a MSc Thesis with formal agreement between IST and the company Mobis Parts Europe N.V., by João Espadinha, 2021.
5. Supervisor of a MSc Thesis with formal agreement between IST and the company Thales, by Bruno Cebola, 2023.
6. Responsible for a consulting and services contract (13K€) with ADELO “Associação de Desenvolvimento Local da Bairrada e Mondego” in the context of the utilization of aug-

mented reality platforms and exergames for physical and cognitive stimulation of older persons, 2019.

7. Responsible for a collaboration protocol between ISR and CRPCGC (Centro de Reabilitação de Paralisia Cerebral Calouste Gulbenkian – Santa Casa da Misericórdia) for research in the use of virtual and augmented reality games for the rehabilitation of patients with neurological diseases, in particular cerebral palsy, 2019.
8. Responsible for a collaboration protocol between ISR and Grupo José Mello for research in “Daily Living Functionality effectiveness and the perceived Quality of Life in Institutionalized elderly adults with PEPE’s exergames approach, 2018.
9. Responsible for a fund-raising donation (1.3K€) from GLINTT – Healthcare Solutions, S.A. for research in robotics for health, 2018.
10. Responsible for a collaboration protocol between LARSyS and CNS (Campus Neurológico Senior) for activities in research and education on the use of technology rehabilitation and improvement of life quality in patients with neurological diseases, 2017.
11. Evaluator of projects for the national innovation agency (ADI/ANI) in the programs IDEIA (2007), QREN (2008, 2009, 2010, 2011), P2020 (2015, 2016, 2017, 2018, 2021).

12.3 Organization of International Conferences and Workshops

1. IAV 2004 – 5th Symposium on Intelligent Autonomous Vehicles, Instituto Superior Técnico Lisbon, Portugal, July 5-7, 2004. **Local Organizing Committee.**
2. 4th International Symposium on “Imitation in Animals and Artifacts” at the AISB'07 Convention, Newcastle upon Tyne, UK, April 2-5, 2007. **Symposium Co-Chair.**
3. Workshop “Towards social humanoid robots: what makes interaction human-like?” at the IROS’13 Conference, Tokyo, Japan, November 3-7, 2013. **Organizing Committee.**
4. ICARSC 2017 – IEEE International Conference on Autonomous Robot Systems and Competitions, April 26-30, Coimbra, Portugal. **Program Chair.**
5. ICDL 2017 – The Seventh Joint IEEE International Conference on Development and Learning and on Epigenetic Robotics, Instituto Superior Técnico, Lisbon, Portugal, September 18-21, 2017. **Publicity Co-Chair.**
6. Workshop EgoVIP2021, “Egocentric vision for interactive perception, learning, and control”, at IROS 2021, Prague, Czech Republic (online), September 27 – October 1, 2021. **Organizing Committee.**
7. ICDL 2022 – IEEE International Conference on Development and Learning, London, UK, September 12-15, 2022. **Program Co-Chair.**
8. Workshop FOREST@ICPR2022 – Image Analysis for Forest Environmental Monitoring, at the 26th International Conference on Pattern Recognition, August 21st, 2022, Montréal, Quebec, Canada. **General Chair.**
9. 1st Workshop on Maritime Computer Vision (MaCVi), at the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), January 3rd, 2023, Waikoloa, Hawaii. **Organizing Committee.**
10. Workshop CVUIA@ICIP2026 - Computer Vision for Underwater Image Analysis, at the IEEE International Conference on Image Processing, 13th Sept. 2026, Tampere, Finland. **General Co-Chair.**

11. 16th International Conference on Information Science and Technology, October 1-6 2026, Coimbra, Portugal. **Program Co-Chair.**

12.4 Member of Program, Scientific or Technical Committees

1. ICDL 2005 – The 4th IEEE International Conference on Development and Learning, July 19-21, Osaka, Japan.
2. 4th International Symposium on Imitation in Animals and Artifacts at the AISB'07 Convention, Newcastle upon Tyne, UK, April 2007.
3. ICIAR 2006, International Conference on Image Analysis and Recognition, Póvoa do Varzim, Portugal, September 2006.
4. ICIAR 2007, International Conference on Image Analysis and Recognition, Montreal, Canada, August 2007.
5. VISAPP 2007, 2nd International Conference on Computer Vision Theory and Applications, March, 2007, Barcelona, Spain
6. VISAPP 2008, 3rd International Conference on Computer Vision Theory and Applications, January, 2008, Funchal, Portugal
7. EPIROB2006 - Sixth International Conference on Epigenetic Robotics:. Modeling Cognitive Development in Robotic Systems. Paris, France, September 2006.
8. EPIROB2005 - Fifth International Conference on Epigenetic Robotics:. Modeling Cognitive Development in Robotic Systems. Nara, Japan, July 2005.
9. HAREM 2005 - Workshop on Human Activity Recognition and Modeling , September 9th, 2005, Oxford, UK
10. ICVS 2009 - 7th International Conference on Computer Vision Systems, October 2009, Liège, Belgium.
11. ICIAR 2009 - International Conference on Image Analysis and Recognition, July 2009, Halifax, Canada.
12. ETFA 2009 - 14th IEEE International Conference on Emerging Technologies and Factory Automation, Track 7: Intelligent Robots and Systems. September 2009, Mallorca, Spain.
13. ICIAR 2010- International Conference on Image Analysis and Recognition, July 2010, Póvoa do Varzim, Portugal.
14. ETFA 2010 - 15th IEEE International Conference on Emerging Technologies and Factory Automation, Track 7: Intelligent Robots and Systems. September 2010, Bilbao, Spain.
15. IBPRIA 2011 - Iberian Conference on Pattern Recognition and Image Analysis, Las Palmas, Spain, June 2011.
16. ETFA 2011 - 16th IEEE International Conference on Emerging Technologies and Factory Automation, WiP Industry Practice Workshop. Toulouse, France, September 2011.
17. VISART Workshop @ ECCV 2012 - "Where Computer Vision Meets Art" Workshop, Florence, Italy, October 2012.
18. ICRA 2012 – IEEE International Conference on Robotics and Automation (Associate Editor), St. Paul, Minnesota, US, May 2012.
19. IROS 2012 - IEEE/RSJ International Conference on Intelligent Robot Systems (Associate Editor), Vilamoura, Portugal, October 2012.

20. ICRA 2013 - IEEE International Conference on Robotics and Automation (Associate Editor), Karlsruhe, Germany, May 2013.
21. ICDL-EPIROB 2013 - International Conference on Development and Learning and Epigenetic Robotics, Osaka, Japan, August 2013.
22. NEUROTECHNIX 2013 - International Congress on Neurotechnology, Electronics and Informatics, Vilamoura, Portugal, September 2013.
23. IROS 2013 - IEEE/RSJ International Conference on Intelligent Robot Systems (Associate Editor), Tokyo, Japan, November 2013.
24. ICRA 2014 – IEEE International Conference on Robotics and Automation (Associate Editor), Hong-Kong, China, June 2014.
25. IROS 2014 - IEEE/RSJ International Conference on Intelligent Robot Systems (Associate Editor), Chicago, Illinois, US, September 2014.
26. ICDL-EPIROB 2014 - International Conference on Development and Learning and Epigenetic Robotics (Associate Editor), Genoa, Italy, October 2014.
27. VS-RE-ID @ ECCV 2014 - Workshop on Visual Surveillance and Re-identification., Zurich, Switzerland, September 2014.
28. VISART II Workshop @ ECCV 2014 - "Where Computer Vision Meets Art" Workshop, Zurich, Switzerland, September 2014.
29. RECPAD 2015, Portuguese Conference on Pattern Recognition, October 20th, 2015, Faro, Portugal.
30. ROBOT 2015, Second Iberian Robotics Conference, November 19-21, 2015, Lisbon, Portugal.
31. IROS 2015 - IEEE/RSJ International Conference on Intelligent Robot Systems (Associate Editor), September, 2015, Hamburg, Germany.
32. ACM-SAC 2015 - ACM Symposium on Applied Computing, Salamanca, Spain, 2015.
33. RECPAD 2016, Portuguese Conference on Pattern Recognition, October 28th, 2016, Aveiro, Portugal.
34. IROS 2016 - IEEE/RSJ International Conference on Intelligent Robots and Systems, October 9-14, 2016, Daejeon, Korea.
35. RECPAD 2017, Portuguese Conference on Pattern Recognition, October 27th, 2017, Portuguese Military Academy, Campus Amadora, Portugal.
36. ROBOT 2017, Third Iberian Robotics Conference, November 22-24, 2017, Seville, Spain.
37. IJCAI 2017 – International Joint Conference on Artificial Intelligence, August 19-25, 2017, Melbourne, Australia.
38. IROS 2017 - IEEE/RSJ International Conference on Intelligent Robots and Systems, September 24-28, 2017, Vancouver, Canada.
39. ICARSC 2018 - IEEE International Conference on Autonomous Robot Systems and Competitions, April 25-27, 2018, Torres Vedras, Portugal.
40. ICARSC 2019 - IEEE International Conference on Autonomous Robot Systems and Competitions, April 24-26, 2019, Gondomar, Porto, Portugal.

41. ICRA 2021 - IEEE International Conference on Robotics and Automation (Associate Editor), May 30th - June 5th, 2021, Xi'An, China.
42. R2T2 Symposium: Robotics Research for Tomorrow's Technology, March 10-11, 2021, Abu Dhabi, UAE.
43. ICRA 2023 - IEEE International Conference on Robotics and Automation (Associate Editor), May 29th - June 2nd, 2023, London, UK.

12.5 Editorial Boards

1. Associate Editor of the "IEEE Robotics and Automation Letters" journal (RA-L), since 2024.
2. Associate Editor of the "Frontiers in Robotics and AI", section of Humanoid Robotics, since 2014.
3. Guest Associate Editor of "Frontiers in Robotics and AI", section of Robot Vision and Artificial Perception, since 2014.
4. Editor of the Special Issue "Image Analysis for Forest Environmental Monitoring" in the "Remote Sensing" journal, 2023.
5. Topic Editor of "Biologically Inspired Vision Mechanisms for Resource-Constrained Robotics Applications" of Frontiers in Robotics and AI – Robot Vision and Artificial Perception, 2021-2022.
6. Topic Editor of "Robots that Learn and Reason: Towards Learning Logic Rules from Noisy Data" of Frontiers in Robotics and AI – Computational Intelligence in Robotics, 2018-2019.

12.6 Evaluation of Academic Project Proposals

1. Evaluator of 2 ERC Starting Grant research proposals, 2024.
2. Evaluator of 5 ELLIS PhD Proposals, 2025.

12.7 Reviewing Activity

12.7.1 Journals

3. IEEE Robotics and Automation Magazine, *IEEE* (**IF: 5.7**), 1998(2), 2016.
4. IEEE Trans. on Systems, Man and Cybernetics, *IEEE* (**IF: 8.7**), 2001, 2007, 2008, 2010.
5. IEEE Trans. on Pattern Analysis and Machine Intelligence, *IEEE* (**IF: 23.6**), 2001, 2010.
6. IEEE Trans. on Image Processing, *IEEE* (**IF: 10.6**), 2006, 2017.
7. IEEE Trans. on Robotics, *IEEE* (**IF: 7.8**), 2006(2), 2007, 2008(4), 2009(5), 2010(4), 2011(3), 2012, 2013, 2016, 2022.
8. Autonomous Robots, *Springer* (**IF: 3.5**), 2006.
9. International Journal of Humanoid Robotics, *World Scientific* (**IF: 1.5**), 2006, 2008, 2009, 2010(3), 2011, 2019(2).
10. Pattern Analysis and Applications, *Springer* (**IF: 3.9**), 2007.
11. Signal Processing, *Elsevier* (**IF: 4.4**), 2007.

12. Advanced Robotics, *Taylor & Francis* (**IF: 2.0**), 2007.
13. IEEE Trans, on Circuits and Systems for Video Technology, *IEEE* (**IF: 8.4**), 2007, 2008(3), 2009(2), 2010.
14. IEEE Trans. on Intelligent Transportation Systems, *IEEE* (**IF: 8.5**), 2007.
15. Robotics and Autonomous Systems, *Elsevier* (**IF: 4.3**), 2009, 2011, 2013, 2016, 2018, 2019.
16. IEEE Trans. on Neural Systems and Rehabilitation Engineering, *IEEE* (**IF: 4.9**), 2009.
17. EURASIP Journal on Advances in Signal Processing, *Springer* (**IF: 1.9**), 2010.
18. Journal of Real-Time Image Processing, *Springer* (**IF: 3.0**), 2010, 2011.
19. International Journal of Robotics Research, *SAGE* (**IF: 9.2**), 2010, 2011, 2013.
20. Adaptive Behavior, *SAGE* (**IF: 1.6**), 2010.
21. Pattern Recognition, *Elsevier* (**IF: 8.0**), 2011, 2016.
22. Axioms, *MDPI* (**IF: 2.0**), 2013.
23. IET Biometrics, *John Wiley & Sons* (**IF: 2.0**), 2013.
24. IEEE Trans. on Autonomous Mental Development, *IEEE* (**IF: 2.9**), 2014.
25. Neurocomputing, *Elsevier* (**IF: 6.0**), 2014.
26. Applied Bionics and Biomechanics, *IOS Press* (**IF: 2.2**), 2015.
27. Computer Vision and Image Understanding, *Academic Press* (**IF: 4.5**), 2015.
28. Frontiers in Systems Neuroscience, *Frontiers Media* (**IF: 3.0**), 2015.
29. International Journal on Computer Vision, *Springer* (**IF: 19.5**), 2015, 2020, 2021.
30. Computer Journal, *Oxford University Press* (**IF: 1.4**), 2015.
31. Frontiers in Robotics and AI, *Frontiers Media* (**IF: 3.4**), 2016, 2017(2), 2020.
32. IEEE Robotics and Automation Letters, *IEEE* (**IF: 5.2**), 2016, 2017(2), 2021.
33. IET Computer Vision, *John Wiley & Sons* (**IF: 1.7**), 2016.
34. Electronics Letters, *John Wiley & Sons* (**IF: 1.1**), 2018.
35. Image and Vision Computing, *Elsevier* (**IF: 4.7**), 2018, 2019.
36. International Journal of Pattern Recognition and Artificial Intelligence, *World Scientific* (**IF: 1.5**), 2020.
37. IEEE Trans. on Medical Robotics and Bionics, *IEEE* (**IF: 3.7**), 2020.
38. Journal of Intelligent & Robotic Systems, *Springer*, (**IF: 3.3**), 2021.

12.7.2 Conferences

1. SIRS – Symposium on Intelligent Robot Systems, 1998, 1999, 2000.
2. IROS – International Conference on Intelligent Robots and Systems, 1999, 2005(2), 2007(4), 2008(2), 2009(4), 2010(4), 2011(6), 2012, 2014, 2016, 2017(3), 2018(3), 2021.
3. ECCV - European Conference on Computer Vision, 2000(4), 2002(4), 2006(2).
4. ICPR – International Conference on Pattern Recognition, 2002(3), 2020.

5. BMVC - British Machine Vision Conference, 2004(2), 2005(3).
6. EPIROB – International Workshop on Epigenetic Robotics, 2004(2), 2005, 2006(4), 2007(3).
7. IAV -Symposium on Intelligent Autonomous Vehicles, 2004(7).
8. CVPR - IEEE International Conference on Computer Vision and Pattern Recognition, 2005(4), 2015.
9. HAREM - Workshop on Human Activity Recognition and Modelling, 2005(4).
10. ICDL – International Conference on Development and Learning, 2005(3), 2009, 2010,
11. LARS - IEEE Latin American Robotics Symposium, 2005.
12. VS-PETS - Joint IEEE International Workshop on Visual Surveillance and Performance Evaluation of Tracking and Surveillance, 2005(2).
13. CONTROLO - Portuguese Conference on Automatic Control, 2006(3).
14. ICRA - IEEE International Conference on Robotics and Automation, 2006(4), 2008, 2009, 2010(2), 2011(2), 2012(2), 2013(2), 2014(2), 2015(2), 2016(4), 2017(2), 2018, 2020, 2021.
15. ITSC – IEEE International Conference on Intelligent Transportation Systems, 2006(4), 2007(2), 2008(3), 2009(3), 2010(2), 2012.
16. ICIAR – International Conference on Image Analysis and Recognition, 2006(4), 2007(4), 2008(4), 2009(3), 2010(5).
17. IBPRIA – Iberian Conference on Pattern Recognition and Image Analysis, 2007(5), 2010(7).
18. AISB - International Symposium on Imitation in Animals and Artifacts, 2007.
19. VISAPP - International Conference on Computer Vision Theory and Applications, 2007, 2008.
20. COGSYS – International Conference on Cognitive Systems, 2008.
21. ICVS - International Conference on Computer Vision Systems, 2008(5), 2009(5), 2011(3), 2013(6), 2015(3), 2017(3).
22. ROBIO-IEEE International Conference on Robotics and Biomimetics, 2008.
23. RSS- Robotics: Science and Systems Conference, 2008(2), 2015(2), 2016(2).
24. IFAC WC’– IFAC World Congress, 2009.
25. ETFA - IEEE International Conference on Emerging Technologies and Factory Automation, 2009(3), 2010(3), 2011(6).
26. IECON - Annual Conference of the IEEE Industrial Electronics Society, 2009(2).
27. SAFEPROCESS - IFAC Symposium on Fault Detection, Supervision and Safety of Technical Processes, 2009.
28. BIOROB – IEEE International Conference on Biomedical Robots and Biomechatronics, 2010.
29. RO-MAN – IEEE International Symposium on Robot and Human Interactive Communication, 2010, 2012, 2017.

30. HUMANOIDS – IEEE International Conference on Humanoid Robots, 2010, 2011, 2013(2), 2014, 2015(2), 2016.
31. ROBOTICA – Conference on Mobile Robots and Competitions, 2010.
32. ECCV-RE-ID – Workshop on Re-identification, 2012(2), 2014(2).
33. ECCV-VISART – “Where the Computer Vision Meets Art” Workshop, 2012(4), 2014(3), 2016(3).
34. CAEPIA - Conference of the Spanish Association for Artificial Intelligence, 2013(3).
35. CETC – Conference on Electronics, Telecommunications and Computers, 2013(3).
36. ICDL-EPIROB – Joint IEEE International Conference on Development and Learning and on Epigenetic Robotics, 2013(4), 2014(2), 2016, 2017(2), 2021, 2022.
37. NEUROTECHNIX – International Congress on Neurotechnology, Electronics and Informatics, 2013.
38. ROBOT – Iberian Robotics Conference, 2013, 2017.
39. ECC – European Control Conference, 2014, 2018, 2019.
40. HRI – ACM-IEEE International Conference on Human-Robot Interaction, 2014.
41. IV – IEEE Intelligent Vehicles Symposium, 2014.
42. SAC- ACM Symposium on Applied Computing, 2015(2), 2016(5), 2017(1).
43. ISCW - ACM International Symposium on Wearable Computers, 2016.
44. ICANN - International Conference on Artificial Neural Networks, 2016.
45. INTERSPEECH - Conference of the International Speech Communication Association, 2017(3).
46. MED – Mediterranean Conference on Control and Automation, 2017, 2018.
47. ICAR – International Conference on Advanced Robotics, 2019.
48. ICARSC - IEEE International Conference on Autonomous Robot Systems and Competitions, 2019(4), 2020(4).
49. EXPAT – Experiment International Conference, 2019.
50. IFAC World Congress, 2020.
51. SSSR - IEEE International Symposium on Safety, Security, and Rescue Robotics, 2020.

12.8 Communications to the Public

1. Article in online magazine (in portuguese), “A magia da eletrotecnia”, 23-04-2024, https://ionline.sapo.pt/artigo/812700/a-magia-da-eletrotecnia?seccao=Opiniao_i
2. Interview for TV show “Sociedade Civil: Tecnologia no AR” at national TV channel RTP2, September 2023.
3. Interview for news agency LUSA regarding the FIREFRONT Project with the news showing up in several national newspapers: Observador, Expresso, Diário de Notícias, April 16th, 2023.

4. “Boas Práticas de Igualdade de Género no Instituto Superior Técnico”, Marta Pile, Alexandre Bernardino, Maria Beatriz Silva, Filipa David; in *Igualdade de Género nas Organizações, Do retorno do investimento às boas práticas*, Editora D’Ideias, ISBN 978-989-53817-6-0, 2022.
5. “Celebrar as Mulheres na Engenharia”, Beatriz Silva, Alexandre Bernardino; i-online 23-11-2021, https://ionline.sapo.pt/artigo/753765/-celebrar-as-mulheres-na-engenharia?seccao=Opiniao_i
6. “O futuro da Engenharia é Feminino”, Beatriz Silva, Alexandre Bernardino; in *IDEIAS sobre Ciência, Tecnologia e Ensino Superior pela Comunidade do Instituto Superior Técnico*. IST Press 2021, ISBN 978-989-8481-87-0
7. Demonstration of FIREFRONT Project at European Researchers Night in Lisboa, Museum of Natural History and Science, under the theme “Science for Climate”, September 24th, 2021.
8. Presentation of the FIREFRONT Project at the 75th Anniversary of ACTV (Aeroclube de Torres Vedras) to the Portuguese President and the Secretary of State for Civil Protection, October 23rd, 2021.
9. Interview about the FIREFRONT Project published on the national newspaper “Público”, July 4th, 2019.
10. Participation on the interview “Robots Made in Portugal” for national magazine “DN Insider”, Nov. 25th 2018.
11. Research Project Profile: AHA – Augmented Human Assistance, ISVR Newsletter. International Society for Virtual Rehabilitation, December 2016.
12. Project Showcase: AHA – Augmented Human Assistance, Robótica, 98, 2015.
13. Presentation of the AHA Project and the ICT 2015 Exhibition, Lisbon, October 20th-22nd, 2015.
14. Interview about the AHA project for TV show “Exame Informática”, transmitted by TV channel SIC Notícias, June 13th, 2015.
15. Interview about the AHA project published in the national newspaper “I-online”, May 4th, 2015.
16. Interview about robotic manipulation for TV show “Falar Global” transmitted by TV channel SIC Notícias, April 10, 2010.
17. Organizes and Participates in the lab demonstration module “Robotic Vision” for frequent high school visits to the department, with more than 500 visitors/year (before 2020).
18. Demonstrations of the iCub Humanoid Robot during the students’ events JEEC 2008 and JEEC 2009.
19. Demonstrations of the iCub Humanoid Robot at science fairs Ciência Viva 2008, Innovation Days 2009.

13 Academic and Research Service

13.1 Management Positions

1. Adjunct Coordinator of the Scientific Area of Systems, Decision and Control (ACSDC), of the Dept. Elec. and Computer Engineering at IST, since January 2025.

2. Member of the Executive Board of the Institute for Systems and Robotics, since January 2023.
3. Coordinator at IST of the Dual Degree Master Programme between IST and KTH, on Control, Robotics and Artificial Intelligence, since 2020.
4. Joint coordinator of the Gender Balance group of the Instituto Superior Técnico, from January 2020 to January 2026.
5. Coordinator of the Thematic Line “Interaction” of the Associated Laboratory of Robotics and Engineering Systems (LARSyS), since 2013.
6. Member of the Scientific Board of ISR, since 2013.
7. Chair of the IEEE Robotics and Automation Society Portuguese Chapter, 2014-2019.
8. Secretary of the IEEE Robotics and Automation Society Portuguese Chapter, 2012-2013.
9. Adjunct Coordinator of the Computer and Robot Vision (VisLab) at ISR, since 2004.
10. Member of the Executive Board of the Department of Electrical and Computer Engineering at IST, April 2010 – Dec. 2016.
11. Responsible by the Library of the Department of Electrical and Computer Engineering at IST, October 2008 – April 2010.
12. Member of the Plenary of Pedagogical Council of IST, 2003-2004.
13. Member of the Senate of the Technical University of Lisbon, 2003-2004.

13.2 Tutoring

- 21 students of Aerospace Engineering in 2020/2021.
- 14 students of Electrical and Computer Engineering in 2019/2020.
- 20 students of Electrical and Computer Engineering in 2015/2016.
- 26 students of Electrical and Computer Engineering in 2007/2008.
- 25 students of Electrical and Computer Engineering in 2005/2006.

13.3 Juries of Academic Degrees Examinations

13.3.1 PhD Thesis Committees

1. Maria João Santos Lopes de Sousa, “Cooperative Aerial Robotics and Artificial Intelligence for Wildfire Detection and Monitoring Systems”, *Instituto Superior Técnico, Universidade de Lisboa*, Portugal, 2023.
2. Ali Hammoud, “Planning robotic in-hand manipulation to reproduce skilled dexterous manipulation tasks”, *Sorbonne University*, Paris, France, 2023.
3. Sara Costa Freitas, “Hyperspectral Imaging for Remote Marine Litter Detection and Classification using Learning based Approaches”, *Universidade do Porto*, Portugal, 2023.
4. Nuno Ricardo Ferreira Duarte, “Non-verbal Communication between Humans and Robots: Imitation, Mutual Understanding and Inferring Object Properties”, *Instituto Superior Técnico, Universidade de Lisboa, and École Polytechnique Fédérale de Lausanne*, 2023.
5. Simone Mentasti, “Embedded AI and Sensor Fusion for Autonomous Vehicles”, *Politecnico di Milano*, Italy, 2022.

6. Sergio Luis Herrera González, “Software Engineering Methods for Distributed and Model-Driven Development”, *Politecnico di Milano*, Italy, 2022.
7. Rubén Delgado Escaño, “Automatic Extraction of Biometric Descriptors Based on Gait”, *Universidad de Málaga*, Spain, 2022.
8. Rocio Nahime Torres, “Analysis of Geografic Data for Environmental Modeling”, *Politecnico di Milano*, Italy, 2022.
9. Nicola A. Piga, “Hybrid Architectures for Object Pose and Velocity Tracking at the Intersection of Kalman Filtering and Machine Learning”, *University of Genova*, Italy, 2022.
10. Gouthaman Kv, “Towards Robust Visual Question Answering”, *Indian Institute of Technology Madras*, Chennai, 2022.
11. Falar Jabar Rahim, “Assessment and Optimization of Omnidirectional Images Viewport Rendering”, *Instituto Superior Técnico, Universidade de Lisboa*, Portugal, 2022.
12. Alessandro Brusaferri, “Machine Learning for Optimization of Energy Intensive Industrial Processes”, *Politecnico do Milano*, Italy, 2022.
13. Maria Beatriz Quintino Ferreira, “Classification of Visual Data with Unreliable Annotations: a Real Case in the Fashion Domain”, *Instituto Superior Técnico, Universidade de Lisboa*, Portugal, 2021.
14. Marco Ciccone, “On Iterative and Conditional Computation for Visual Representation Learning”, *Politecnico di Milano*, Italy, 2021.
15. Alessandro Gabrielli, “Studying user acceptance of autonomous driving in the wild”, *Politecnico Milano*, Italy, 2021.
16. Afonso Rodrigues Gonçalves, “Fitness Applications for Healthy Older Adults Using Large Projection Displays: Methodology, Design, Assessment and Field Validation”, *Universidade da Madeira*, Funchal, Portugal, 2021.
17. David José da Silva Aresta Belo, “Learning Biosignals with Deep Learning”, *Universidade Nova de Lisboa*, 2021.
18. Chandrama Sarker, “Automated Detection of Flooded Areas using Machine Learning Methods”, *The Queensland University of Technology*, Australia, 2021.
19. Sheikh Shanawaz Mostafa, “Obstructive Sleep Apnea Detection using Fourth Level Devices”, *Instituto Superior Técnico, Universidade de Lisboa*, Portugal, 2020.
20. Sandy Carmo Relva Rodrigues Abreu, “Improving Machine Learning Prediction and Forecasting Models Used in Photovoltaic Monitoring systems”, *Instituto Superior Técnico, Universidade de Lisboa*, Portugal, 2020.
21. Clebeson Canuto dos Santos, “Sistema de Reconhecimento de Gestos e Ações em Tempo Real Baseado em Visão Computacional”, *Universidade Federal do Espírito Santo*, Vitoria, Brasil, 2020.
22. Tiago Guiomar Ribeiro, “Creating the Illusion of Life in Autonomous Social Robots”, *Instituto Superior Técnico, Universidade de Lisboa*, Portugal, 2019.
23. Tanmay Tulsidas Verlekar, “Gait Analysis in Unconstrained Environments”, *Instituto Superior Técnico, Universidade de Lisboa*, Portugal, 2019.
24. Seyed Hamidreza Mohades Kasaei, “Interactive Open-Ended Learning for 3D Object Recognition”, *Universidade de Aveiro*, Portugal, 2019.

25. Alireza Sepas-Moghaddam, “Light Field Based Biometric Recognition and Presentation Attack Detection”, *Instituto Superior Técnico, Universidade de Lisboa*, Portugal, 2019.
26. Joshin Parakkulangarayil Krishnan, “Complex-Values Image Restoration for Interferometric Phase Denoising and Phase Retrieval”, *Instituto Superior Técnico, Universidade de Lisboa*, Portugal, 2019.
27. Mohammad M. Aref, “Vision-Based Macro-Micro Mobile Manipulation”, *Tampere University*, Finland, 2018.
28. Lina Zhuang, “Inverse Problems in Hyperspectral Imaging: Denoising, Inpainting, Unmixing, and Compressive Sensing”, *Instituto Superior Técnico, Universidade de Lisboa*, Portugal, 2018.
29. Jaeseok Kim, “The development of mobile manipulation tasks for domestic environment”, *Scuola Universitaria Superiore Pisa*, Italy, 2018.
30. José Jerónimo Moreira Rodrigues, “3D pose estimation for bin-picking: A data-driven approach using multi-light images”, *Carnegie Mellon University*, Pittsburgh, US, 2018.
31. Seyed Jafar Hosseini Shamoushaki, “Deformable Surface Reconstruction Through Optimization-Based Techniques”, *Universidade de Coimbra*, Portugal, 2017.
32. Samaneh Khoshrou, “Learning in Evolving Video Streams”, *Universidade do Porto*, Portugal, 2017.
33. Ricardo Filipe Alves Martins, “Development of techniques for haptic exploration and recognition of objects”, *Universidade de Coimbra*, Portugal, 2017.
34. Miguel Ângelo Pires Farrajota, “Human pose and action recognition using neural networks”, *Universidade do Algarve*, Faro, Portugal, 2017.
35. Carlos Jorge Andrade Mariz Santiago, “Segmentation of the Left Ventricle in Cardiac Magnetic Resonance”, *Instituto Superior Técnico, Universidade de Lisboa*, Portugal, 2017.
36. Ana Catarina Fidalgo Barata, “Automatic Detection of Melanomas using Dermoscopy Images”, *Instituto Superior Técnico, Universidade de Lisboa*, Portugal, 2017.
37. Jorge Manuel Soares de Almeida, “Active Tracking of Dynamic Multivariate Agents using Vectorial Range Data”, *Universidade de Aveiro*, Portugal, 2016.
38. Michel Olivier Dominici, “Control Architectures for Robotic Assistance in Beating Heart Surgery”, *Universidade de Coimbra*, Portugal, 2016.
39. Silvio Brás Filipe, “Biologically Motivated Keypoint Detection for RGB-D Data”, *Universidade da Beira Interior*, Covilhã, Portugal, 2016.
40. Alina Liliana Trifan, “Time-constrained colored object detection for intelligent robots”, *Universidade de Aveiro*, Portugal, 2016.
41. Asheer Bachoo, “Unsupervised Maritime Target Detection”, *University of Cape Town*, South Africa, 2016.
42. Luka Lukic, “Visuomotor Coordination in Reach-to-Grasp Tasks: From Humans to Humanoids and Vice Versa”, *Instituto Superior Técnico, Universidade de Lisboa*, Portugal, 2015.
43. António Miguel Marques Rodrigues Teixeira Lourenço, “Keypoint Detection, Matching, and Tracking in Images with Non-linear Distortion: Applications in Medical Endoscopy and Panoramic Vision”, *Universidade de Coimbra*, Portugal, 2015.

44. André Miguel Pinheiro Dias, "Multi-Robot 3D Target Estimation Under Uncertainty", *Instituto Superior Técnico, Universidade de Lisboa*, Portugal, 2015.
45. Bruno Duarte Damas, "Learning and Sensorimotor Coordination of Anthropomorphic Robotic Systems", *Instituto Superior Técnico, Universidade de Lisboa*, Portugal, 2014.
46. Carla Elena González Uzcátegui, "A memetic approach to the inverse kinematics problem for robotic applications", *Universidad Carlos III de Madrid*, Espanha, 2014.
47. Robert Krug, "Optimization-Based Robot Grasp Synthesis and Motion Control", *Orebro University*, Sweden, 2014.
48. Jaime Afonso do Nascimento Carvalho Martins, "Face and Object Recognition by 3D Cortical Representations", *Universidade do Algarve*, Faro, Portugal, 2014.
49. Diego Resende Faria, "Probabilistic Learning of Human Manipulation of Objects Towards Autonomous Robotic Grasping", *Universidade de Coimbra*, Portugal, 2014.
50. Carlos Serra Toro, "Pedestrian Classification from Digital Images: Evaluation of Unconventional Learners and a New Descriptor", *Universitat Jaume I*, Castellon de la Plana, Espanha, 2014.
51. Ricardo Daniel Rita Beira, "Dexterous Mechanical Systems for Intuitive Telemanipulation in Minimally Invasive Surgery", *École Polytechnique Fédérale de Lausanne*, Switzerland, 2013.
52. Sébastien Gay, "Visual Control of Legged Robots for Locomotion in Complex Environments using Central Pattern Generators", *École Polytechnique Fédérale de Lausanne*, Switzerland, 2013.
53. Francisco Marcelo Gomes Quinderé, "Comunicação Humano-Robô Através de Linguagem Falada", *Universidade de Aveiro*, Portugal, 2013.
54. Rui Paúl Oliveira, "A Stereo Vision System Based on Soft Computing Techniques for Human Robot Interaction", *Universidad de Granada*, Spain, 2013.
55. Ângelo Miguel Aparício Cardoso, "Hierarchical Models for Visual Recognition", *Instituto Superior Técnico, Universidade de Lisboa*, Portugal, 2013.
56. Stefano Boccardo, "Tissue Engineering Platforms for Epidermis Like Growth and In-Vivo Vascularization", *Università degli Studi di Genova*, Italy, 2013.
57. Alice Tomassini, "Time Perception in the Haptic System", *Università degli Studi di Genova*, Italy, 2013.
58. Andrea del Prete, "Control of Contact Forces using Whole-Body Force and Tactile Sensors: Theory and Implementation on the iCub Humanoid Robot", *Università degli Studi di Genova*, Italy, 2013.
59. Fouzhan Housseini, "Low-Power Lightweight Vision Supercomputer: Algorithms and Architectures", *Università degli Studi di Genova*, Italy, 2013.
60. Francesco de Bertoldi, "Inter-trial Variability: A New Method to Evaluate the Reliability of Single-Subject fMRI", *Università degli Studi di Genova*, Italy, 2013.
61. Francesco Rea, "From Perception to Cognition: A Quest for Effective Active Vision in Human-Robot Interaction", *Università degli Studi di Genova*, Italy, 2013.
62. Gioele Gavazzi, "On the Relationships Between Time Perception of Visual Motion and the Motor System", *Università degli Studi di Genova*, Italy, 2013.

63. Ioannis Delis, "Modularity in Muscle Activations Underlying Voluntary Human Actions: Novel Methods for Module Identification and Evaluation", *Università degli Studi di Genova*, Italy, 2013.
64. Jacopo Zenzeri, "Coordination of Redundant Degrees of Freedom in Humans and Humanoids: Modeling and Experiments", *Università degli Studi di Genova*, Italy, 2013.
65. Abdolkarim Pahlani, "Active Cooperative Perception in Networked Robot Systems", *Instituto Superior Técnico, Universidade de Lisboa*, Portugal, 2012.
66. Luciano Rebouças de Oliveira, "Semantically Integrating Laser and Vision in Pedestrian Detection", *Universidade de Coimbra*, Portugal, 2010.
67. Nuno Miguel Pinho da Silva, "Robust Nonmetric Perception of Moving Rigid Bodies", *Instituto Superior Técnico, Universidade de Lisboa*, Portugal, 2010.
68. Davide Migliore, "Monocular Simultaneous Localization And Mapping with Bearing-Only Tracking", *Politecnico di Milano*, Italy, 2009.
69. Adolfo Martínez Usó, "Unsupervised Band Selection and Segmentation in Hyper/Multispectral Images", *Universitat Jaume I*, Castellon de la Plana, Espanha, 2008.
70. Sandra Esperanza Nope Rodríguez, "Sistema Bioinspirado de Reconocimiento e Imitación de Gestos Aplicado en Robótica", *Universidad del Valle*, Cali, Colombia, 2008.
71. Manuel Fernando Cabido Peres Lopes, "A Developmental Roadmap for Learning by Imitation in Robots", *Instituto Superior Técnico, Universidade de Lisboa*, Portugal, 2006.

13.3.2 MSc Thesis Committees

1. João Pedro Conde Ribeiro Alves Novo, "Augmentation-Based Approaches for Overcoming Low-visibility in Street Object Detection", *Instituto Superior Técnico*, 2023.
2. Miguel Alexandre Vieira Gonçalves, "Vision Transformers for Remote Sensing Image Segmentation", *Instituto Superior Técnico*, 2023.
3. Rotislav Eduardovich Andreev, "Session-Based Clothing Recommendation with Semantic and Aesthetic Information", *Instituto Superior Técnico*, 2022.
4. Pedro Gonçalo Santos Pires Pinheiro, "Artificial intelligence-based software for recognizing parkinsonian gait patterns based on wearable miniaturized sensors", *Universidade do Minho*, 2022.
5. Miguel Duarte Gonçalves, "Automatic detection of forest fires: a deep learning approach: Exploiting data augmentation for the development of early forest fire detection systems", *Instituto Superior Técnico*, 2022.
6. João Paulo Dinis Lopes, "Detection and Tracking of Non-Cooperative UAVs: a deep learning moving-object tracking approach", *Instituto Superior Técnico*, 2022.
7. Miguel Silva Amaral, "Analysis of Transformer Behaviour in Reinforcement Learning", *Instituto Superior Técnico*, 2022.
8. António Manuel de Almeida Gama Mendes, "Ugly Duckling. Outlier Detection in Dermoscopy Images using Deep Neural Networks", *Instituto Superior Técnico*, 2022.
9. Gonçalo Correia de Matos, "Automatic User Interface Generation", *Instituto Superior Técnico*, 2022.

10. Gonalo dos Santos Querido, “Learning to Perceive in Deep Model-Free Reinforcement Learning”, *Instituto Superior Tcnico*, 2022.
11. Guilherme de Melo Antunes, “Georeferencing Aerial Imagery”, *Instituto Superior Tcnico*, 2022.
12. Joo Pedro Rato Coimbra, “Enhancing WiFi Positioning Systems for IoT Asset Tracking”, *Instituto Superior Tcnico*, 2022.
13. Nivaldo S Caires, “Impedance Controller Planar Flexible Robot for Identification of Human Ankle Dynamics”, *Instituto Superior Tcnico*, 2021.
14. Alexandre Daniel Ramos Fonseca, “Sistemas No Cooperativos para Registo de Assiduidade em Ambiente de Sala de Aula”, *Universidade da Beira Interior*, 2021.
15. Andr Miguel Martins Rocha, “Aerial Sensor Networks for Early Wildfire Detection”, *Instituto Superior Tcnico*, 2021.
16. Valter Andr Ribeiro dos Santos Piedade, “Retail Store Visual Structure-From-Motion”, *Instituto Superior Tcnico*, 2021.
17. Tiago Alexandre da Silva Oliveira, “Rapid Development and Prototyping Environment for Testing of unmanned Aerial Vehicles”, *Instituto Superior Tcnico*, 2021.
18. Sofia Teodoro Bebian, “Vehicle Tracking in Urban Environment”, *Instituto Superior Tcnico*, 2021.
19. Pedro Miguel de Aguiar Coelho, “Addressing the Texture-Bias and Domain Generalization Performance of Convolutional Neural Networks for Simulation-Based Learning”, *Instituto Superior Tcnico*, 2021.
20. Francisco Antnio Drumond Freitas de Sousa, “Impedance Learning in Physical Human-Robot Interaction”, *Instituto Superior Tcnico*, 2021.
21. Antnio Jos Evaristo Farinhas, “Visual Attention with Sparse and Continuous Transformations”, *Instituto Superior Tcnico*, 2021.
22. Dinis Loureno Tavares Rodrigues, “Automated Assessment of Coronary Artery Stenosis in X-ray Angiography using Deep Neural Networks”, *Instituto Superior Tcnico*, 2021.
23. Guilherme Kaname Reis Kano, “Automatic Detection of Railway Track Obstacles using a Monocular Camera”, *Instituto Superior Tcnico*, 2020.
24. Bruno Manuel Degardin, “Weakly and Partially Supervised Learning Frameworks for Anomaly Detection”, *Universidade da Beira Interior*, 2020.
25. Andr Rosa de Sousa Porfrio Correia, “Software for a Service Robot”, *Universidade da Beira Interior*, 2020.
26. Francisco Baslio de Aguiar Gomes, “Vision based real-time obstacle avoidance for drones while following a moving target”, *Instituto Superior Tcnico*, 2020.
27. Adriana Maria Marques Fernandes, “Object Handover between Humans and Robots in Microgravity”, *Instituto Superior Tcnico*, 2020.
28. Rui Fernando Faustino Bettencourt, “Multimodal Navigation for Autonomous Service Robots”, *Instituto Superior Tcnico*, 2019.
29. Nuno Miguel Calejo Alves, “From Mockup to UI”, *Instituto Superior Tcnico*, 2019.

30. Luis Carlos Gonçalves de Castro, “Alzheimer’s Disease Diagnosis using a Variational Autoencoder”, *Instituto Superior Técnico*, 2019.
31. Jorge Filipe Rosa Alves, “Identifying 3D objects: Descriptors for pointclouds”, *Instituto Superior Técnico*, 2019.
32. João José Estêvão Nisa, “People Detection from Laser Range Finder Data”, *Instituto Superior Técnico*, 2019.
33. João Pedro Pires Queiroz, “Deep Learning applied to the Semantic Segmentation of the Landslides in Madeira Island”, *Instituto Superior Técnico*, 2019.
34. Guilherme Duarte Tavares Saraiva, “A Model Predictive Control approach on Dynamic Bike Reposition”, *Instituto Superior Técnico*, 2019.
35. Ricardo Jorge da Silva Diniz, “Applying Deep Learning to Medical Images”, *Instituto Superior Técnico*, 2019.
36. Inês de Miranda de Matos Lourenço, “Temporal Perspectives: Exploring robot’s perception of time”, *Instituto Superior Técnico*, 2018.
37. Gonçalo José Dias Pais, “OmniDRL: Robust Pedestrian Detection using Omnidirectional Cameras and Deep Reinforcement Learning”, *Instituto Superior Técnico*, 2018.
38. Emilia Anna Brzozowska, “An Optimization Based Cartesian Controller for Mobile Manipulation of Service Robots in Domestic Environments”, *Instituto Superior Técnico*, 2018.
39. João Miguel Vieira Cartucho, “Robust Object Recognition Through Symbiotic Deep Learning in Mobile Robots”, *Instituto Superior Técnico*, 2018.
40. João Manuel Sousa Quartin Borges, “Estimating head pose for assistive technologies”, *Instituto Superior Técnico*, 2018.
41. Inês Afonso Alexandre, “Agent-Aware mobile robot navigation in domestic environments”, *Instituto Superior Técnico*, 2018.
42. Ricardo Dantas da Silva Simas, “Human-Aware Task Planning Under Uncertainty in Networked Robot Systems”, *Instituto Superior Técnico*, 2017.
43. Miguel João Bordalo Fernandes, “Hydroponic Greenhouse Crop Optimization”, *Instituto Superior Técnico*, 2017.
44. Tiago José Ribeiro Ricardo, “Distributed Adaptive Control of a Water Delivery Canal”, *Instituto Superior Técnico*, 2016.
45. Tiago José Pinheiro Paulino, “3-Dimensional Soft Magnetic Tactile Sensors for the Human-Friendly Robot Vizzy”, *Instituto Superior Técnico*, 2016.
46. Pedro César Alves Guerreiro Godinho Agostinho, “Obstacle Tracking in a Dynamic Environment”, *Instituto Superior Técnico*, 2016.
47. Luís Carlos Barreira Luz, “Cooperative Perception for People Tracking and Human-Aware Navigation”, *Instituto Superior Técnico*, 2016.
48. João Pedro Almas Garcia, “Multimodal Human-Robot Interaction Using Gestures and Speech Recognition”, *Instituto Superior Técnico*, 2016.
49. Ana Sofia Frutuoso Oliveira, “Sail and Rig Shape from Single Images”, *Instituto Superior Técnico*, 2016.

50. Nuno Alexandre Antunes Martins Pessanha-Santos, "Sistema de Visão para Aterragem Automática de UAV", *Instituto Superior de Engenharia de Lisboa*, 2015.
51. Kevin do Coito Ramos, "Model and Control of a Solar Tower for Energy Production", *Instituto Superior Técnico*, 2015.
52. André Miguel Alves Farinha, "Robust and Self-Adaptable 5-Finger 4-DoA Robotic Hand", *Instituto Superior Técnico*, 2015.
53. Pedro Miguel Dias Neto, "Energy Optimization in Building Water Supply Using Economic MPC", *Instituto Superior Técnico*, 2013.
54. Mao Ye, "O Problema de Correspondência em Visão por Computadores: uma Abordagem Convexa", *Instituto Superior Técnico*, 2013.
55. Francisco Figueira Moitinho de Almeida, "Development and Implementation of a Motion Controller for an Aircraft Simulator", *Instituto Superior Técnico*, 2013.
56. Tiago Manuel Rua Matos Talhada, "Perceção 3D Utilizando uma Câmara Estéreo", *Universidade de Aveiro*, 2011.
57. Sérgio Miguel dos Santos Copeto, "Detecção de Objectos Activos em Ambientes Dinâmicos", *Instituto Superior Técnico*, 2012.
58. Mário Luís Pinto Antunes, "Agente de Visão Semântica para Robótica", *Universidade de Aveiro*, 2011.
59. André Filipe Marques da Silva, "Vision Based Pose Computation from Landmarks: application to Quadrotors", *Instituto Superior Técnico*, 2011.
60. Alina Liliana Trifan, "Desenvolvimento de um Sistema de Visão para Robôs Humanoides", *Universidade de Aveiro*, 2011.
61. Alexandre Lobo Fernandes, "Controlo de um Forno Solar para Testes com Variações Rápidas de Temperatura", *Instituto Superior Técnico*, 2010.
62. Carlos Alberto Falé Cabral, "Automatic Classification of Cognitive States", *Instituto Superior Técnico*, 2010.
63. André Romão Ribeiro, "Seguimento de Jogadores num jogo de futebol", *Instituto Superior Técnico*, 2009.
64. João Carlos da Silva Santos, "Multi-Robot Cooperative Object Localization : Decentralized Bayesian Approach", *Instituto Superior Técnico*, 2008.
65. Daniel Filipe de Almeida Martins, "Sistema de Processamento de Imagem para Aplicações Robóticas", *Universidade de Aveiro*, 2008.
66. Mauro André Moreira Rodrigues, "Unidade de Processamento e Sistema de Visão para um Robô Homanóide", *Universidade de Aveiro*, 2008.
67. Elisabete Maria Recto Pereira, "Girassol-E: Multi-objective Control in Solar and Wind Energies", *Instituto Superior Técnico*, 2008.
68. Dario António Bacellar Figueira, "Cognitive Architecture for Humanoid Social Robots – Spatial Environment Model", *Instituto Superior Técnico*, 2008.
69. Ricardo Daniel Rita Beira, "Mechanical Design of an Anthropomorphic Robot Head", *Instituto Superior Técnico*, 2008. (Pre-Bologna)

70. Ricardo Jorge Coelho da Silva Nunes, “Sistema Stereo de Reconstrução de Cenas Submersas”, *Instituto Superior Técnico*, 2007.
71. André Pires Vaz Gonçalves, “Highway Autonomous Navigation System - An Integrated approach based on low cost video cameras”, *Instituto Superior Técnico*, 2007.
72. Nuno Gonçalo Matos Pinto Ferreira, “Automatic Fusion of Three-Dimensional Surfaces”, *Instituto Superior Técnico*, 2007.
73. Manuel Ricardo de Almeida Rodrigues Marques, “Reconstrução 3D a partir de Imagens 2D com Dados Incompletos”, *Instituto Superior Técnico*, 2007. (Pre-Bologna)
74. Ricardo André da Silva Marranita, “Visual Tracking of Articulated Objects: An Application to the human Hand”, *Instituto Superior Técnico*, 2005. (Pre-Bologna)

13.4 Committees of Research and Faculty Positions

1. Member of the Evaluating Committee for the recruitment of an Assistant Professor at the Faculty of Engineering of Porto University.
2. Member of the Evaluating Committee for the renewal of the position of researcher of Dr. Egidio Falotico, Instituto di Biorobotica, Scuola Superiore Sant’Anna, Pontedera, Italy, 2020.
3. Member of the CAPE (commission for the evaluation of the experimental period of assistant professors at IST) of Prof. Daniel Lopes, 2018-2022.
4. Member of the CAPE (commission for the evaluation of the experimental period of assistant professors at IST) of Prof. Hugo Nicolau, 2016-2020.
5. Member of the CAPE (commission for the evaluation of the experimental period of assistant professors at IST) of Prof. Duarte Valério, 2015-2019.
6. Evaluator for the position of Technologist of Dr. Jesus Ortiz Sanches Lafuente, at Istituto Italiano di Tecnologia, Genoa, Italy, 2014.

13.5 Juries of Academic Competitions and Awards

1. Member of the evaluation committee of the Maria de Lourdes Pintasilgo Award, IST, 2021 - 2025.
2. Member of the evaluation committee of the FEEDZAI Scholarships Women in Science, MSc & PhD, Feedzai & IST, 2022 - 2025.
3. Member of the evaluation committee of the TagEnergy Scholarships Women in Energy, MSc, TagEnergy&IST, 2022 - 2025.
4. Member of the evaluation committee of the Celfocus Merit Award for the students of the course of Machine Learning, IST, 2024 - 2025.
5. Member of the evaluation committee of the Prof. Luís Vidigal Award, IST, 2017, 2020.
6. Member of the evaluation committee of the BitChallenge Award, DEEC, IST, 2018, 2019, 2020.